

ECE-494 — Processor Design

Data-Aware Adaptive ALU and Pipeline Restructuring for Power & Performance Optimization in a RISC-V Core (CV32E40P)

Final Project Report

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Introduction

This work is carried out in the context of the ECE-494 (Processor Design) course and focuses on the study and modification of the RISC-V core CV32E40P, targeting power and performance optimization.

Initial proposal. The initial project proposal (see the project progress presentation) centered on a *Data-Aware Adaptive ALU*: introducing logic that recognizes “trivial” operands (e.g., multiplication by 0/1, division by 1) so that the standard, multi-cycle execution in the core’s multiplier and divider can be bypassed.

Project evolution. Following the professor’s suggestion, an alternative/complementary direction is also explored: restructuring the pipeline itself (splitting the Execute stage into two stages), aiming to shorten the critical path and increase the maximum operating frequency, combined — where feasible — with the data-aware logic from the original proposal.

Purpose of this document. This document records, step by step, the methodology followed: from tool setup and the baseline RTL, through the RTL modifications, to the PPA (Power, Performance, Area) evaluation via the `OpenROAD` tool.

Chapter 1

Tool Setup and Baseline RTL

1.1 Source code of the CV32E40P core

CV32E40P is a small and efficient, 32-bit, in-order RISC-V core with a 4-stage pipeline, implementing the RV32IM[F|Zfinx]C instruction set, as well as the PULP custom extensions for higher code density, performance, and energy efficiency.

The source code was obtained from the official OpenHW Group repository:

<https://github.com/openhwgroup/cv32e40p>

```
git clone --recurse-submodules https://github.com/openhwgroup/cv32e40p.git
```

Key RTL files. Within the `rtl/` folder of the core, the modules that the study initially focused on are:

- `cv32e40p_ex_stage.sv` — the top-level wrapper of the Execute stage, responsible for instantiating the functional units (ALU, Multiplier, APU) and managing communication between the Decode (ID) and Write-Back (WB) stages.
- `cv32e40p_alu.sv` — the module implementing basic single-cycle arithmetic/logic operations, within which the divider is instantiated.
- `cv32e40p_alu_div.sv` — the divider, implemented as a 3-state FSM (IDLE, DIVIDE, FINISH).
- `cv32e40p_mult.sv` — the multiplier, which internally uses a 16×16 -bit multiplier and a 3-step FSM (STEP0, STEP1, STEP2) to implement 32-bit multiplication.

1.2 Processor architecture overview

CV32E40P is a small and efficient, 32-bit, **in-order, single-issue** RISC-V core, implementing the RV32IM[F|Zfinx]C instruction set architecture together with the PULP custom extensions (hardware loops, post-increment load/store, MAC instructions, and SIMD sub-word arithmetic) for higher code density, performance, and energy efficiency. The core is organized as a classic **4-stage pipeline**:

- **IF (Instruction Fetch)** — `cv32e40p_if_stage.sv`. Fetches instructions from memory (through a prefetch buffer/FIFO for latency hiding) and forwards them, decompressed if needed (RVC), to the Decode stage.
- **ID (Instruction Decode)** — `cv32e40p_id_stage.sv`. Decodes the instruction, reads operands from the register file, resolves data hazards (via forwarding or stalling), and dispatches control signals to the Execute stage. This is also where this project's data-flow crosses back into: the EX-stage result is consumed here by forwarding logic (Section 3.5's critical-path analysis).
- **EX (Execute)** — `cv32e40p_ex_stage.sv`, the wrapper this project modifies. Houses the ALU (`cv32e40p_alu.sv`, which in turn instantiates the divider, `cv32e40p_alu_div.sv`), the multiplier (`cv32e40p_mult.sv`), and (unused in this project's configuration) the APU/FPU interface. Performs the actual computation, and multiplexes the result from whichever functional unit was active.
- **WB (Write Back)** — implemented as part of the ID/EX stage interface rather than a fully separate physical stage in this core (the register file write happens directly from the EX-stage result), a common simplification in small, area-efficient in-order cores.

Because the core is in-order and single-issue, at most one instruction occupies each pipeline stage per cycle, and instructions are neither reordered nor issued in parallel — forwarding and stalling in the ID stage are the only mechanisms used to resolve hazards. This project's Execute-stage pipeline split (Chapter 6) effectively turns the 4-stage pipeline into a 5-stage one for instructions that use the ALU, by inserting an additional register inside what was previously a single EX stage.

1.3 Installing the OpenROAD-flow-scripts tool

For the synthesis and physical design flow (RTL-to-GDSII), `OpenROAD-flow-scripts` (ORFS) was used:

<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts>

```
git clone --recursive https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git
```

The tool was built with:

```
sudo ./setup.sh
```

The OpenROAD core executable is produced at `OpenROAD/build/src/openroad`.

1.4 Tutorial

To become familiar with the ORFS workflow, the official tutorial was followed:

<https://github.com/The-OpenROAD-Project/micro2022tutorial>

```
git clone --recursive https://github.com/The-OpenROAD-Project/micro2022tutorial.git
```

A trial run of the full flow was carried out inside the following directory (after first setting up the required tools in the /tools folder):

```
micro2022tutorial/OpenROAD-flow-scripts/flow
```

The flow was then run as follows:

```
make synth → make floorplan → make place → make cts → make route → make  
finish
```

1.5 Problem: Yosys does not accept SystemVerilog

The first step of the flow (`make synth`) is carried out by **Yosys**, which performs logic synthesis — converting RTL into a gate-level netlist, which is then used by the remaining OpenROAD tools (floorplanning, placement, CTS, routing).

The problem encountered is that open-source Yosys only offers *partial* SystemVerilog support, while the CV32E40P core makes extensive use of SystemVerilog features (`always_comb`, `always_ff`, `packages`, `typedef/enum` types, etc.) — making direct synthesis of the RTL impossible.

Solution: the sv2v tool. To address this, the `sv2v` tool was installed:

<https://github.com/zachjs/sv2v>

`sv2v` converts SystemVerilog to plain Verilog, resolving SystemVerilog-specific constructs (e.g., `enum` types from packages) into plain bit-vectors, making the output compatible with Yosys.

1.6 Converting the RTL with sv2v

The prebuilt `sv2v` binary (`sv2v-Linux`) was downloaded and extracted:

```
unzip -o sv2v-Linux.zip  
chmod +x sv2v-Linux/sv2v  
./sv2v-Linux/sv2v --version
```

which reported version `sv2v v0.0.11-0-g6082cae`.

Conversion command. The full RTL of the core — all three package files under `rtl/include/`, plus every module except `cv32e40p_fp_wrapper.sv` (no FPU is used) and `cv32e40p_register_file_latch.sv` (the flip-flop-based register file, `cv32e40p_register_file_ff.sv`, is used instead) — was converted in a single `sv2v` invocation, run from `cv32e40p/rtl/`, with `cv32e40p_top` set as the top module:

```
../../sv2v-Linux/sv2v --top=cv32e40p_top \  
include/*.sv \  
cv32e40p_core.sv \  
cv32e40p_top.sv \  
cv32e40p_if_stage.sv \  
cv32e40p_id_stage.sv \  

```

```

cv32e40p_ex_stage.sv \
cv32e40p_alu.sv \
cv32e40p_alu_div.sv \
cv32e40p_mult.sv \
cv32e40p_controller.sv \
cv32e40p_decoder.sv \
cv32e40p_compressed_decoder.sv \
cv32e40p_cs_registers.sv \
cv32e40p_load_store_unit.sv \
cv32e40p_register_file_ff.sv \
cv32e40p_prefetch_buffer.sv \
cv32e40p_prefetch_controller.sv \
cv32e40p_aligner.sv \
cv32e40p_obi_interface.sv \
cv32e40p_int_controller.sv \
cv32e40p_sleep_unit.sv \
cv32e40p_hwloop_regs.sv \
cv32e40p_fifo.sv \
cv32e40p_ff_one.sv \
cv32e40p_popcnt.sv \
> ../../flattened_design_full.v

```

Verification. The conversion completed with no errors reported. The resulting file, `flattened_design_full.v` (~10,000 lines), was checked for completeness and correctness:

- It contains **24 modules**, matching every RTL file supplied to `sv2v` — including `cv32e40p_core`, `cv32e40p_top`, `cv32e40p_if_stage`, `cv32e40p_id_stage`, `cv32e40p_ex_stage`, `cv32e40p_alu`, `cv32e40p_alu_div`, `cv32e40p_mult`, and all remaining supporting modules.
- No leftover SystemVerilog constructs (`always_comb`, `always_ff`, `typedef`, `import ...:*`) remain in the file — confirming a clean conversion to plain Verilog.

This flattened, plain-Verilog file is used as the input RTL for the Yosys synthesis step (`make synth`) of the OpenROAD flow.

Chapter 2

Baseline Logic Synthesis (Yosys)

2.1 Synthesis script: `yosys_synth.tcl`

The Yosys logic-synthesis commands were collected into a standalone script, run against the flattened RTL produced by `sv2v` and the following standard-cell library:

`yosys_synth.tcl`

`sky130_fd_sc_hd__tt_025C_1v80.lib`

```
read_verilog flattened_design_full.v
synth -top cv32e40p_top
dfflibmap -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
abc -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
clean
stat -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
write_verilog synth_netlist.v
```

Command-by-command explanation.

- `read_verilog flattened_design_full.v` — loads the plain-Verilog RTL produced by the `sv2v` conversion (Section 1.6).
- `synth -top cv32e40p_top` — runs Yosys’s built-in generic synthesis flow (hierarchy check, process/proc conversion, optimization, FSM extraction, memory mapping, generic techmapping) with `cv32e40p_top` as the top module. The result is still expressed in generic gates, not yet the target standard-cell library.
- `dfflibmap -liberty sky130_fd_sc_hd__tt_025C_1v80.lib` — maps the generic flip-flops produced by `synth` onto real flip-flop cells from the sky130 library.
- `abc -liberty sky130_fd_sc_hd__tt_025C_1v80.lib` — performs technology mapping: the external `abc` logic-synthesis tool (invoked from within Yosys) maps the combinational logic onto actual sky130 standard cells (AND2, NAND, MUX, etc.), optimizing for timing/area.
- `clean` — removes dangling wires and unused signals left over from the previous passes.

- `stat -liberty sky130_fd_sc_hd__tt_025C_1v80.lib` — prints synthesis statistics (cell counts by type, estimated area) as a first sanity check on the result.
- `write_verilog synth_netlist.v` — writes the final gate-level netlist to file, to be used as the input to the OpenROAD physical-design steps (floorplanning, placement, CTS, routing).

2.2 Problem: missing cv32e40p_clock_gate module

Running the synthesis script raised the following error while elaborating the design hierarchy:

```
2.1.6. Executing AST frontend in derive mode using pre-parsed AST for module '\
cv32e40p_if_stage'.
Parameter \COREV_PULP = 0
Parameter \PULP_OBI = 0
Parameter \PULP_SECURE = 0
Parameter \FPU = 0
Parameter \ZFINX = 0
Generating RTLIL representation for module '
    $paramod$a557e35cf00b750c4faba28985285b554efef6e4\cv32e40p_if_stage'.
Parameter \COREV_CLUSTER = 0
2.1.7. Executing AST frontend in derive mode using pre-parsed AST for module '\
cv32e40p_sleep_unit'.
Parameter \COREV_CLUSTER = 0
Generating RTLIL representation for module '$paramod\cv32e40p_sleep_unit\
COREV_CLUSTER=s32'00000000000000000000000000000000'.
ERROR: Module '\cv32e40p_clock_gate' referenced in module '\cv32e40p_sleep_unit'
in cell '\core_clock_gate_i' is not part of the design.
```

Cause. The `cv32e40p_sleep_unit` module instantiates an integrated clock-gating cell (ICG) through a module named `cv32e40p_clock_gate`, used to gate the clock during sleep mode. The OpenHW Group intentionally does *not* ship this module inside `rtl/`: it is meant to be a technology-specific wrapper around the real ICG standard cell of the target PDK, supplied by the integrator. Only a simulation-only behavioral model is provided, at `bhv/cv32e40p_sim_clock_gate.sv`, which already defines a module named `cv32e40p_clock_gate` with the correct port interface:

```
module cv32e40p_clock_gate (
    input logic clk_i,
    input logic en_i,
    input logic scan_cg_en_i,
    output logic clk_o
);
    logic clk_en;
    always_latch begin
        if (clk_i == 1'b0) clk_en <= en_i | scan_cg_en_i;
    end
    assign clk_o = clk_i & clk_en;
endmodule
```

Fix. The `sv2v` conversion command was re-run with `bhv/cv32e40p_sim_clock_gate.sv` added to the file list, so that the resulting flattened netlist includes a definition of `cv32e40p_clock_gate`:

```
../../sv2v-Linux/sv2v --top=cv32e40p_top \  
include/*.sv \  
../bhv/cv32e40p_sim_clock_gate.sv \  
cv32e40p_core.sv \  
cv32e40p_top.sv \  
... (remaining files unchanged, see Section 1.6)  
> ../../flattened_design_full.v
```

Limitation. The OpenHW Group explicitly documents this file as simulation-only (“*must not be used for ASIC synthesis*”), since it models the clock gate behaviorally with an `always_latch` block. A consequence of this, and its fix, is discussed in Section 2.4 below.

2.3 Adding a delay target for abc (matching the SDC clock period)

The initial synthesis run gave `abc` no timing target at all, so the mapping (Section 2.1) optimized purely for area rather than for a specific operating frequency. The project’s SDC file, `cv32e40p_core.sdc`, defines the intended clock:

```
set clock_period 5.0  
...  
create_clock -name clk_i -period $clock_period [get_ports clk_i]
```

i.e. a 5.0 ns period (200 MHz).

Why the SDC file itself cannot be read directly by Yosys. SDC commands such as `create_clock`, `set_input_delay`, and `get_ports` belong to the standard SDC/STA command set, which is consumed later in the flow by **OpenSTA** — Yosys has no built-in SDC reader. What Yosys’s `abc` pass *does* accept is a single, simplified delay target via the `-D` flag, specified in picoseconds. Converting the SDC’s 5.0 ns clock period gives a target of **5000 ps**, which was added to the `abc` invocation:

```
abc -D 5000 -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
```

Purpose. This makes the baseline netlist’s technology mapping timing-aware (targeting the same 200 MHz operating point as the rest of the project) instead of purely area-driven, so that this early Yosys-level result is consistent with, and roughly comparable to, the full timing sign-off that OpenSTA will later perform on the placed-and-routed design using the complete SDC file (including the per-port input/output delay constraints, which are not representable through the single `-D` value and are only applied once the design is fed into the full OpenROAD flow via `SDC_FILE`).

2.4 Technology mapping of the clock-gate latch

Running `stat` after synthesis revealed the following warning, printed twice:

```
Area for cell type $_DLATCH_N_ is unknown!
```

Cause. `dfflibmap` maps only flip-flop cells to the target liberty library; it does not map latches. The `always_latch` block inside `cv32e40p_clock_gate` (Section 2.2) is inferred by Yosys as a generic, active-low-enable latch primitive, `$_DLATCH_N_`, which was left unmapped: a generic internal cell with no corresponding physical footprint. This is harmless for the area/cell-count report (a single cell, negligible contribution) but would block the OpenROAD physical-design steps later, which require every cell in the netlist to correspond to a real standard-cell footprint.

Fix. A small Yosys techmap rule was written to replace `$_DLATCH_N_` with a real `sky130_fd_sc_hd` cell. The library’s high-density latch cell, `sky130_fd_sc_hd__dlxtp` (“*delay latch, non-inverted enable*”), is active-high — confirmed directly from its Liberty description, `latch ("IQ","IQ_N") { data_in : "D"; enable : "GATE"; }`, with no inversion on `enable` — so an inverter is added ahead of it to match the design’s active-low polarity:

```
module \$_DLATCH_N_ (input E, input D, output Q);
    wire E_inv;
    sky130_fd_sc_hd__inv_1 _inv (
        .A(E),
        .Y(E_inv)
    );
    sky130_fd_sc_hd__dlxtp_1 _TECHMAP_REPLACE_ (
        .D(D),
        .GATE(E_inv),
        .Q(Q)
    );
endmodule
```

This file, `dlatch_map.v`, is applied via a `techmap` pass inserted after `dfflibmap` and before `abc`. The complete, final synthesis script is:

```
read_verilog flattened_design_full.v
synth -top cv32e40p_top
dfflibmap -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
techmap -map dlatch_map.v
abc -D 5000 -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
clean
stat -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
write_verilog synth_netlist.v
```

Verification. Re-running synthesis with this fix removed the `$_DLATCH_N_` warning entirely; `cv32e40p_clock_gate` now contains a real `sky130_fd_sc_hd__dlxtp_1` and `sky130_fd_sc_hd__inv_1` instance. The total chip area increased from $248,211.80\text{ }\mu\text{m}^2$ to $248,230.57\text{ }\mu\text{m}^2$ — a difference of $18.77\text{ }\mu\text{m}^2$, which matches exactly the area of the

newly-mapped latch and inverter, confirming that this was the only change introduced by the fix.

Remaining limitation. `dlxtp` is a general-purpose transparent latch, not the library’s purpose-built integrated clock-gating cell (`sky130_fd_sc_hd__dlclkp/sdlclkp`), which combines the latch and enable-AND gate in a single, glitch-free-verified cell suitable for insertion into a clock tree. The current fix makes the netlist fully technology-mapped and placeable, but a proper clock-tree-safe implementation using `dlclkp/sdlclkp` is left as future work; it does not affect the modules under study in this project (ALU, multiplier, divider, EX stage).

2.5 Baseline synthesis results

Table 2.1 summarizes the per-module area breakdown from the final baseline synthesis run (200 MHz target, `sky130_fd_sc_hd__tt_025C_1v80`), as reported by Yosys’s `stat` pass. Figures marked with $\sim$ are read from Yosys’s abbreviated scientific-notation output and are approximate; all others are exact.

Module	Local area (μm^2)	% of chip	Sequential %
<code>cv32e40p_mult</code>	77,140.23	31.08%	0.20%
<code>cv32e40p_register_file</code>	$\sim 60,900$	$\sim 24.53\%$	—
<code>cv32e40p_cs_registers</code>	25,410.62	10.24%	45.72%
<code>cv32e40p_id_stage (local)</code>	23,092.15	9.30%	45.52%
<code>cv32e40p_alu (local)</code>	20,936.33	8.43%	0.00%
<code>cv32e40p_alu_div</code>	7,603.54	3.06%	35.56%
<code>cv32e40p_load_store_unit (local)</code>	5,291.32	2.13%	19.86%
<code>cv32e40p_aligner</code>	$\sim 5,210$	$\sim 2.10\%$	—
<code>cv32e40p_if_stage (local)</code>	4,803.36	1.94%	34.90%
<code>cv32e40p_obi_interface</code>	$\sim 3,610$	$\sim 1.45\%$	—
<code>cv32e40p_fifo</code>	$\sim 2,940$	$\sim 1.18\%$	—
<code>cv32e40p_prefetch_controller</code>	$\sim 2,550$	$\sim 1.03\%$	—
<code>cv32e40p_controller</code>	$\sim 2,510$	$\sim 1.01\%$	—
<code>cv32e40p_decoder</code>	1,488.93	0.60%	0.00%
<code>cv32e40p_compressed_decoder</code>	1,123.58	0.45%	0.00%
<code>cv32e40p_ex_stage (local wrapper)</code>	818.28	0.33%	21.41%
<code>cv32e40p_popcnt</code>	940.90	0.38%	—
<code>cv32e40p_int_controller</code>	899.61	0.36%	—
<code>cv32e40p_sleep_unit (local)</code>	78.83	0.03%	63.49%
<code>cv32e40p_prefetch_buffer (local)</code>	360.35	0.15%	0.00%
<code>cv32e40p_core (local, top wiring)</code>	150.14	0.06%	—
<code>cv32e40p_clock_gate</code>	31.28	0.01%	0.00%
<code>cv32e40p_top (grand total)</code>	248,230.57	100%	24.39%

Table 2.1: Baseline area breakdown by module. “Local” figures exclude nested submodules, listed as separate rows. A dash indicates the sequential-element breakdown was not captured for that module in this run.

The EX-stage subsystem dominates chip area. Summing the disjoint leaf modules that make up the Execute stage — `ex_stage` (local wrapper), `alu` (local), `alu_div`, `ff_one`, `popcnt`, and `mult` — gives a combined area of **107,744.59 μm^2** , or **43.4% of the entire core**. The multiplier alone (`cv32e40p_mult`) accounts for 31.08% of total chip area on its own, more than any other single module, including the register file. This

quantitatively motivates focusing optimization effort on the EX stage: it is, by a wide margin, the most area-costly part of the design.

Chapter 3

Manual Floorplan Initialization (OpenROAD)

3.1 Switching to a manual OpenROAD flow

An attempt to run the automated ORFS flow (`make DESIGN_CONFIG=...synth`) on the shared server installation failed with a Python error:

```
Traceback (most recent call last):
  File "/OpenROAD-flow-scripts/flow/scripts/run_command.py", line 59, in <module>
    >
    def run(argv: list[str] | None = None) -> int:
TypeError: unsupported operand type(s) for |: 'types.GenericAlias' and 'NoneType'
```

This is a Python version incompatibility: the `list[str] | None` syntax is a modern type-hint feature (PEP 604) requiring Python 3.10+, while the shared server's default `python3` is older. Since this is an environment issue on shared infrastructure outside this project's control, the decision was made to bypass the ORFS Python/Makefile wrapper entirely and drive **OpenROAD directly**, via its own Tcl interface — the same approach already used successfully for Yosys synthesis in Chapter 2.

This also means the already-verified `synth_netlist.v` (Section 2.4, with the latch-mapping fix already applied) is used directly as input, without needing to re-run synthesis through ORFS.

3.2 Floorplan initialization script: `floorplan_init.tcl`

```
read_lef <ORFS_PATH>/flow/platforms/sky130hd/lef/sky130_fd_sc_hd.tlef
read_lef <ORFS_PATH>/flow/platforms/sky130hd/lef/sky130_fd_sc_hd_merged.lef

read_liberty sky130_fd_sc_hd__tt_025C_1v80.lib

read_verilog synth_netlist.v
link_design cv32e40p_top
```

```

read_sdc cv32e40p_core.sdc

initialize_floorplan -utilization 40 -aspect_ratio 1 -core_space 2 -site unithd

make_tracks

add_global_connection -net VDD -pin_pattern {^VPWR$} -power
add_global_connection -net VSS -pin_pattern {^VGND$} -ground
set_voltage_domain -power VDD -ground VSS

define_pdn_grid -name "Core"
add_pdn_stripe -grid "Core" -followpins -layer met1 -width 0.48
add_pdn_stripe -grid "Core" -layer met4 -width 1.6 -pitch 153.6 -offset 16.32 -
    extend_to_core_ring
add_pdn_stripe -grid "Core" -layer met5 -width 1.6 -pitch 153.18 -offset 16.65 -
    extend_to_core_ring
add_pdn_connect -grid "Core" -layers {met1 met4}
add_pdn_connect -grid "Core" -layers {met4 met5}

pdngen

write_def 1_floorplan_pdn.def
write_db 1_floorplan_pdn.odb

```

Command-by-command explanation.

- **read_lef** (technology LEF) — loads the routing-layer and via geometry of the sky130 process.
- **read_lef** (standard-cell LEF, `_merged.lef`) — loads the physical footprint (size, pin locations) of every standard cell.
- **read_liberty** — the same library used for Yosys synthesis; OpenROAD also needs it for timing-aware steps later in the flow.
- **read_verilog** + **link_design** `cv32e40p_top` — loads the already-synthesized, latch-fixed gate-level netlist and binds it as the active design.
- **read_sdc** — loads the 200 MHz timing constraints.
- **initialize_floorplan** — creates the die/core area: 40% utilization, square (1:1) aspect ratio, 2µm core margin, using the `unithd` placement site (the row/grid definition for `sky130_fd_sc_hd` cells, confirmed from the platform's own default configuration).
- **make_tracks** — defines the per-layer routing track grid (no arguments: read directly from the tech LEF), required before any pin placement or routing step can legally place anything.
- **add_global_connection**, **set_voltage_domain**, **define_pdn_grid**, **add_pdn_stripe**, **add_pdn_connect**, and **pdngen** — builds the Power Distribution Network: connects each standard cell's VPWR/VGND pins to logical VDD/VSS nets, then generates physical power rails (`met1`, following the standard-cell rows) and straps (`met4/met5`), with

vias connecting the layers together. See Section 3.8 for why this was added and confirmation that it does not change any previously-reported result.

- `write_def / write_db` — saves the initialized floorplan in both a human-readable (DEF) and OpenROAD-native binary (ODB) format; the ODB file is the input to the next stage (placement).

3.3 Problem discovered later: missing Power Distribution Network

The original version of `floorplan_init.tcl` did *not* include the PDN-generation block above (`add_global_connection` through `pdngen`) — this was added only after a severe routing failure was diagnosed at the routing stage (Chapter 5, Section 5.1), which traced back to the complete absence of a power distribution network. The symptom was visible even at this early floorplan stage, in retrospect: `Number of snets: 0` in the routing-stage design summary meant zero power/ground (“special”) nets existed anywhere in the design.

Fix. The PDN-generation block shown above was added to `floorplan_init.tcl`, connecting standard-cell VPWR/VGND pins to VDD/VSS nets and building `met1` rails plus `met4/met5` straps. One issue arose while adding it: `pdngen` initially failed with

```
[ERROR PDN-0106] Width (0.0350 um) specified for layer met1 is less than minimum
width (0.1400 um).
```

because no explicit width was given for the `met1` followpins stripe, and the default computed value fell below sky130’s minimum. Adding `-width 0.48` (matching the standard sky130 power-rail width) resolved this.

Verification: results are unchanged. `floorplan_init.tcl` was re-run in full, saving output under new filenames (`1_floorplan_pdn.*`) to keep both versions available. The resulting timing report (`wns max` `−20.23 ns`, `tns max` `−30,884.59`, and the identical critical path already reported in Section 3.5) and power report (8.69 mW total, identical to Section 3.6) matched the original, PDN-less run *exactly*. This is expected: PDN generation adds physical power-routing infrastructure only — it does not alter the netlist, cell count, or logic, so it cannot change timing or power results. All timing/area/power figures reported in this document prior to this point (Chapters 2–4) therefore remain valid; only the routing stage (Chapter 5) required rework once the PDN was added.

3.4 SDC compatibility: missing `remove_from_collection`

Loading `cv32e40p_core.sdc` with `read_sdc` failed with:

```
Error: cv32e40p_core.sdc, 81 invalid command name "remove_from_collection"
```

Cause. OpenSTA (the timing engine used inside OpenROAD to parse SDC files) implements a large subset of the SDC/Tcl command set, but not the full Synopsys “collection” object API. `remove_from_collection` — used three times in the SDC, e.g. to exclude `irq_id_o*` ports from an `irq_i*` wildcard match, since `irq_id_o` also matches the prefix `irq_i*` — is one such unsupported command.

Fix. Rather than rewriting each of the three affected SDC lines individually, a single Tcl compatibility shim defining `remove_from_collection` was added at the top of the SDC file, so the rest of the file is unmodified:

```
proc remove_from_collection {coll1 coll2} {
    set result {}
    foreach p $coll1 {
        if {[lsearch -exact $coll2 $p] == -1} {
            lappend result $p
        }
    }
    return $result
}
```

This works because OpenSTA represents SDC “collections” (the results of `get_ports`, etc.) as plain Tcl lists rather than opaque objects, so ordinary list operations (`foreach`, `lsearch`) apply directly, without needing any further collection-specific API commands.

3.5 Floorplan initialization results

Running `floorplan_init.tcl` completed successfully, reporting:

```
[INFO IFP-0001] Added 289 rows of 1711 site unithd.
[INFO IFP-0100] Die BBox: ( 0.000 0.000 ) ( 791.765 791.765 ) um
[INFO IFP-0101] Core BBox: ( 2.300 2.720 ) ( 789.360 788.800 ) um
[INFO IFP-0102] Core area: 618692.125 um^2
[INFO IFP-0103] Total instances area: 248230.573 um^2
[INFO IFP-0104] Effective utilization: 0.401
[INFO IFP-0105] Number of instances: 32671
```

Consistency check. Both the **total instances area** (248,230.573 μm^2) and the **instance count** (32,671) match *exactly* the baseline synthesis results obtained independently from the standalone Yosys run (Section 2.5, Table 2.1: 248,230.57 μm^2 , 32671 cells). This confirms that `synth_netlist.v` was carried into OpenROAD without any loss, duplication, or unintended modification of cells — the floorplan step only adds physical die/core geometry around the already-correct netlist. The effective utilization of 40.1% closely matches the 40% target passed to `initialize_floorplan`, as expected (the small difference comes from row/site quantization: rows must be an integer number of the `unithd` site height).

3.6 Critical-path analysis: where the timing violation lives

The worst timing-violating path from Section 3.4 is reproduced in full below, exactly as reported by `report_checks -path_delay min_max`:

```

Startpoint: core_i/id_stage_i/_4562_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:   core_i/id_stage_i/_4655_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type:  max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4562_/CLK (sky130_fd_sc_hd__dfrtp_1)
0.42 0.42 ^ core_i/id_stage_i/_4562_/Q (sky130_fd_sc_hd__dfrtp_1)
4.22 4.64 ^ core_i/ex_stage_i/alu_i/_3675_/Y (sky130_fd_sc_hd__nor4bb_1)
2.32 6.96 v core_i/ex_stage_i/alu_i/_3811_/Y (sky130_fd_sc_hd__a21oi_1)
0.59 7.55 ^ core_i/ex_stage_i/alu_i/_3863_/Y (sky130_fd_sc_hd__nand3_1)
0.14 7.69 v core_i/ex_stage_i/alu_i/_4055_/Y (sky130_fd_sc_hd__a32oi_1)
0.40 8.10 v core_i/ex_stage_i/alu_i/_4057_/X (sky130_fd_sc_hd__maj3_1)
0.39 8.49 v core_i/ex_stage_i/alu_i/_4059_/X (sky130_fd_sc_hd__maj3_1)
0.39 8.88 v core_i/ex_stage_i/alu_i/_4061_/X (sky130_fd_sc_hd__maj3_1)
0.39 9.27 v core_i/ex_stage_i/alu_i/_4063_/X (sky130_fd_sc_hd__maj3_1)
0.39 9.66 v core_i/ex_stage_i/alu_i/_4065_/X (sky130_fd_sc_hd__maj3_1)
0.39 10.06 v core_i/ex_stage_i/alu_i/_4067_/X (sky130_fd_sc_hd__maj3_1)
0.35 10.40 v core_i/ex_stage_i/alu_i/_4069_/X (sky130_fd_sc_hd__maj3_1)
0.25 10.65 ^ core_i/ex_stage_i/alu_i/_4070_/Y (sky130_fd_sc_hd__o21ai_0)
0.16 10.81 v core_i/ex_stage_i/alu_i/_4385_/Y (sky130_fd_sc_hd__xnor2_1)
0.37 11.18 v core_i/ex_stage_i/alu_i/_4429_/X (sky130_fd_sc_hd__maj3_1)
0.39 11.57 v core_i/ex_stage_i/alu_i/_4430_/X (sky130_fd_sc_hd__maj3_1)
0.39 11.97 v core_i/ex_stage_i/alu_i/_4431_/X (sky130_fd_sc_hd__maj3_1)
0.39 12.36 v core_i/ex_stage_i/alu_i/_4433_/X (sky130_fd_sc_hd__maj3_1)
0.40 12.76 v core_i/ex_stage_i/alu_i/_4435_/X (sky130_fd_sc_hd__maj3_1)
0.26 13.01 ^ core_i/ex_stage_i/alu_i/_4437_/Y (sky130_fd_sc_hd__o2111ai_1)
0.13 13.14 v core_i/ex_stage_i/alu_i/_4439_/Y (sky130_fd_sc_hd__a21oi_1)
0.19 13.33 v core_i/ex_stage_i/alu_i/_4441_/X (sky130_fd_sc_hd__o21a_1)
0.43 13.76 ^ core_i/ex_stage_i/alu_i/_4444_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 13.92 v core_i/ex_stage_i/alu_i/_4446_/Y (sky130_fd_sc_hd__a21oi_1)
0.21 14.14 v core_i/ex_stage_i/alu_i/_4448_/X (sky130_fd_sc_hd__o21a_1)
0.59 14.73 ^ core_i/ex_stage_i/alu_i/_4451_/Y (sky130_fd_sc_hd__o311ai_0)
0.21 14.93 v core_i/ex_stage_i/alu_i/_4454_/Y (sky130_fd_sc_hd__a311oi_1)
0.37 15.30 ^ core_i/ex_stage_i/alu_i/_4456_/Y (sky130_fd_sc_hd__o31ai_1)
0.18 15.48 v core_i/ex_stage_i/alu_i/_4458_/Y (sky130_fd_sc_hd__o211ai_1)
0.22 15.70 v core_i/ex_stage_i/alu_i/_4459_/X (sky130_fd_sc_hd__and3_1)
0.30 16.00 v core_i/ex_stage_i/alu_i/_4461_/X (sky130_fd_sc_hd__o311a_1)
0.34 16.34 v core_i/ex_stage_i/alu_i/_4473_/X (sky130_fd_sc_hd__or3_1)
0.26 16.60 v core_i/ex_stage_i/alu_i/_4475_/X (sky130_fd_sc_hd__a31o_1)

```

```

1.51 18.10 ^ core_i/ex_stage_i/alu_i/_4477_/Y (sky130_fd_sc_hd__a21oi_1)
0.18 18.28 v core_i/ex_stage_i/alu_i/_4705_/Y (sky130_fd_sc_hd__nor3_1)
0.63 18.91 ^ core_i/ex_stage_i/alu_i/_4714_/Y (sky130_fd_sc_hd__o311ai_0)
0.18 19.09 v core_i/ex_stage_i/alu_i/_5512_/Y (sky130_fd_sc_hd__a311oi_1)
0.44 19.52 ^ core_i/ex_stage_i/alu_i/_5521_/Y (sky130_fd_sc_hd__o31ai_1)
0.06 19.59 v core_i/ex_stage_i/alu_i/_5754_/Y (sky130_fd_sc_hd__nor2_1)
0.36 19.95 ^ core_i/ex_stage_i/alu_i/_5755_/Y (sky130_fd_sc_hd__a31oi_1)
0.12 20.07 v core_i/ex_stage_i/alu_i/_5759_/Y (sky130_fd_sc_hd__a21oi_1)
0.19 20.26 ^ core_i/ex_stage_i/alu_i/_5780_/Y (sky130_fd_sc_hd__a41oi_1)
0.10 20.36 v core_i/ex_stage_i/alu_i/_5781_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 20.51 ^ core_i/ex_stage_i/alu_i/_5830_/Y (sky130_fd_sc_hd__a31oi_1)
0.73 21.23 v core_i/ex_stage_i/alu_i/_5831_/Y (sky130_fd_sc_hd__o21ai_0)
0.60 21.83 v core_i/ex_stage_i/alu_i/_6796_/X (sky130_fd_sc_hd__a221o_1)
0.09 21.92 ^ core_i/ex_stage_i/alu_i/_6831_/Y (sky130_fd_sc_hd__o211ai_1)
0.10 22.02 v core_i/ex_stage_i/_168_/Y (sky130_fd_sc_hd__a22oi_1)
1.82 23.84 ^ core_i/ex_stage_i/_170_/Y (sky130_fd_sc_hd__o21ai_0)
0.30 24.14 ^ core_i/id_stage_i/_2950_/X (sky130_fd_sc_hd__a22o_1)
0.07 24.21 v core_i/id_stage_i/_2951_/Y (sky130_fd_sc_hd__a21oi_1)
0.37 24.58 v core_i/id_stage_i/_3220_/X (sky130_fd_sc_hd__o221a_1)
0.08 24.66 ^ core_i/id_stage_i/_3221_/Y (sky130_fd_sc_hd__nand2_1)
0.05 24.71 v core_i/id_stage_i/_3222_/Y (sky130_fd_sc_hd__nand2_1)
0.12 24.82 ^ core_i/id_stage_i/_3223_/Y (sky130_fd_sc_hd__a21oi_1)
0.14 24.97 ^ core_i/id_stage_i/_3389_/Y (sky130_fd_sc_hd__o21bai_1)
0.09 25.06 v core_i/id_stage_i/_3753_/Y (sky130_fd_sc_hd__o21ai_0)
0.10 25.16 ^ core_i/id_stage_i/_3754_/Y (sky130_fd_sc_hd__nand2_1)
0.00 25.16 ^ core_i/id_stage_i/_4655_/D (sky130_fd_sc_hd__dftrtp_1)
    25.16 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4655_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.07 4.93 library setup time
    4.93 data required time

-----
    4.93 data required time
-25.16 data arrival time
-----

-20.23 slack (VIOLATED)

```

Where each part of the path lives in the RTL. Tracing the instance-name hierarchy (full names in the listing above) against the source files:

- **Startpoint** — a flip-flop inside `cv32e40p_id_stage.sv` (instance `id_stage_i`, the operand register that feeds the ALU). Its output (Q) delay is only 0.42 ns — negligible.
- **The bulk of the path** (from cell `_3675_` to `_6831_`, roughly 50 of the path's ~60 cells, all under instance `alu_i`) — entirely inside `cv32e40p_alu.sv`. This segment alone consumes **21.92 ns** of the 25.16 ns total (87%). The long, repeated run of `maj3_1` cells (a majority-of-3 gate, the standard building block for carry computation

in a fast adder) is the signature of a ripple/carry-chain adder — almost certainly the ALU’s arithmetic adder, used for `ADD/SUB` and the `SLT/SLTU` comparisons that depend on it.

- **Cells `_168_` and `_170_`** — the EX-stage wrapper level, `cv32e40p_ex_stage.sv` (outside `alu_i` itself, instance `ex_stage_i`): almost certainly the result-selection mux that chooses between the ALU/multiplier/APU outputs before leaving the EX stage. Adds 1.92 ns.
- **The tail** (cells `_2950_` to `_3754_`) — back inside `cv32e40p_id_stage.sv`: combinational forwarding/operand-select logic that consumes the EX-stage result (e.g. for back-to-back dependent instructions). Adds a further 3.14 ns before reaching the destination flip-flop (cell `_4655_`).

What this means for the planned pipeline split. The critical path confirms, quantitatively, that the bottleneck is exactly where the project already intended to intervene: the ALU’s adder chain inside `cv32e40p_alu.sv` (instantiated within `cv32e40p_ex_stage.sv`) accounts for the large majority of the 25.16 ns path. However, the path is *not* contained entirely within `alu_i`: a non-trivial 3.14 ns tail lives in `cv32e40p_id_stage.sv`’s forwarding logic, downstream of the EX stage. This means the new pipeline register (splitting EX into EX1/EX2, Chapter 1) should not simply bisect `alu_i`’s delay by half — the `id_stage` forwarding tail must be budgeted into whichever half of the split ends up feeding it, so that both resulting stages fit within the 5 ns (200 MHz) target once combined with their respective downstream logic. This is also flagged as a limitation of the current, floorplan-stage report: with ideal (zero-skew) clocks and no real wire delay yet, these numbers will change (typically worsen) once CTS and routing are complete — the split-point decision should be revisited against post-route timing, not finalized from this report alone.

3.7 Floorplan-stage power results

`report_power` gives the following breakdown:

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	1.19e-03	1.72e-04	2.96e-08	1.36e-03	15.7%
Combinational	3.63e-03	3.53e-03	6.28e-08	7.16e-03	82.5%
Clock	6.70e-07	1.62e-04	7.19e-12	1.63e-04	1.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	4.82e-03 55.5%	3.87e-03 44.5%	9.25e-08 0.0%	8.69e-03	100.0%

Table 3.1: Power breakdown at the floorplan stage (ideal clock, no real switching activity).

Interpretation. Combinational logic accounts for 82.5% of total power (7.16 mW out of 8.69 mW), versus only 15.7% for sequential elements and 1.9% for clock distribution. This is consistent with, and reinforces, the area finding from Section 2.5: the same combinational-heavy modules (ALU, multiplier) that dominate chip area also dominate power. Leakage power is negligible at this stage (0.0%, i.e. ~ 92.5 nW out of 8.69 mW) — dynamic power (internal + switching) is what matters here.

Important caveat. This report is generated *before* placement or routing, and — critically — *without* real switching-activity data from a testbench or simulation (no VCD/-SAIF was supplied). OpenSTA falls back to a generic default activity-factor model in this case. This means the *absolute* 8.69 mW figure should not be read as “the chip will consume this much power running real code” — it is not workload-representative. The *relative* breakdown (combinational logic dominating power, matching its dominance in area) is, however, a meaningful and expected signal, and consistent between the two independent metrics (area and power).

3.8 Global Placement (timing-driven)

The final placement script, `global_placement.tcl`:

```
# --- Load the floorplanned design ---
read_db ../1_floorplan_init.odb

# --- Liberty + SDC are not stored in the .odb, so reload them for timing-aware
      steps ---
read_liberty ../sky130_fd_sc_hd_tt_025C_1v80.lib
read_sdc ../cv32e40p_core.sdc

# --- Place I/O pins on the die boundary (required before global_placement) ---
place_pins -hor_layers met3 -ver_layers met2

# --- Wire RC estimates (required by the resizer for timing-driven placement)
      ---
set_wire_rc -signal -layer met2
set_wire_rc -clock -layer met3

# --- Global placement: spread the 32671 cells across the core rows (timing-
      aware) ---
global_placement -timing_driven -density 0.65

# --- Detailed placement: legalize cells onto the row/site grid, remove overlaps
      ---
detailed_placement

# --- Save the result ---
write_def ../2_placement.def
write_db ../2_placement.odb

# --- Stage reports (placement, IDEAL wire -- same assumption as the floorplan-
      stage report) ---
puts "==== TIMING (placement, ideal wire) ====="
report_checks -path_delay min_max
report_wns
report_tns

puts "==== POWER (placement, ideal wire) ====="
report_power
```

```
# --- Estimate wire parasitics from the real placement geometry ---
estimate_parasitics -placement

# --- Stage reports (placement, ESTIMATED wire -- realistic) ---
puts "===== TIMING (placement, estimated wire) ====="
report_checks -path_delay min_max
report_wns
report_tns

puts "===== POWER (placement, estimated wire) ====="
report_power
```

Command-by-command explanation.

- `read_db` — loads the floorplanned design (Chapter 3, Sections 3.1–3.4).
- `read_liberty` / `read_sdc` — reloaded, since neither is stored in the `.odb`.
- `place_pins -hor_layers met3 -ver_layers met2` — places all I/O pins on the die boundary, on the routing track grid (Section 3.7 below explains why this step needs `make_tracks` to have run first).
- `set_wire_rc` — defines a per-micron resistance/capacitance rule for signal (`met2`) and clock (`met3`) nets, read directly from the tech LEF. This is the input the resizer needs to evaluate wire delay; by itself it does not compute anything.
- `global_placement -timing_driven -density 0.65` — the main placement step: spreads all cells across the core rows using the Nesterov-method (RePlAce) analytic placer, additionally weighting critical (low-slack) nets more heavily so their connected cells are pulled closer together.
- `detailed_placement` — legalizes the result: snaps every cell onto a real row/site, removing overlaps, with minimum total displacement.
- `write_def` / `write_db` — saves the placed design.
- `report_checks` / `report_wns` / `report_tns` / `report_power` (first block) — stage report using the same ideal-wire (zero wire delay) assumption as the floorplan-stage report, for direct comparison.
- `estimate_parasitics -placement` — combines the `set_wire_rc` per-unit-length values with each net's actual routed-estimate length (from the real cell positions just computed) to produce a per-net wire delay estimate.
- Second report block — the same reports, now with realistic (estimated) wire delay included, for direct before/after comparison.

3.9 Placement legalization results

Detailed placement legalized all cells successfully:

```
Total cells: 40184
Diamond Move Success: 40184 (100.00%)
```

```
Diamond Move Failure: 0
Total Placement Failures: 0
```

Placement Analysis

```
-----
total displacement 93222.4 u
average displacement 2.3 u
max displacement 9.9 u
original HPWL 1083178.6 u
legalized HPWL 1170295.7 u
delta HPWL 8 %
```

Zero placement failures, with modest average displacement (2.3 μm) and an 8% HPWL increase from legalization — both indicate a healthy, non-overconstrained placement at 65% target density.

Why did the cell count jump from 32,671 to 40,184 (+23%)? This is a direct, expected consequence of the `-timing_driven` flag: internally, `global_placement` performs a *virtual repair design* pass, using the Resizer (`rsz`) tool to fix timing and electrical violations *during* placement, not just move existing cells. Concretely, it:

- **Inserts buffers** on long or high-fanout nets (visible in the new critical path below as auto-generated instances such as `place12015`, `place8414`, `wire315`);
- **Upsizes cells** on critical paths to higher-drive-strength library variants (e.g. the startpoint flip-flop’s clock-to-Q path now runs through `sky130_fd_sc_hd__nor4bb_4` instead of the original `_1` drive-strength cell; further downstream, `a21oi_2` replaces `a21oi_1`, and the recovery-check flip-flop is now `dfrtp_4` instead of `dfrtp_1`).

This added 7,513 new cells ($7,513 = 40,184 - 32,671$) beyond the original synthesized netlist.

3.10 Timing and power: before/after comparison

Stage	wns max (ns)	tns max
Floorplan (ideal wire)	−20.23	−30,884.59
Placement, timing-driven (ideal wire)	−9.83	−13,280.61
Placement, timing-driven (estimated wire)	−9.83	−13,421.99

Table 3.2: Worst (wns) and total (tns) negative slack across stages.

Stage	Total power (mW)	Combinational share
Floorplan	8.69	82.5%
Placement (ideal wire)	12.70	88.5%
Placement (estimated wire)	12.70	88.5%

Table 3.3: Power breakdown across stages.

Interpretation. Enabling `-timing_driven` placement roughly **halved wns** (from −20.23 ns to −9.83 ns), but this came almost entirely from the resizer’s buffering/up-sizing (Section 3.6), not from cell repositioning alone: the design grew by 23% in cell count and total power rose by 46% ($8.69\text{ mW} \rightarrow 12.70\text{ mW}$) as a direct cost of that

effort. **Critically, the design remains timing-violated even after this aggressive automatic repair** — this is strong evidence that the bottleneck is structural (too many logic levels in series inside `cv32e40p_alu.sv`) rather than something buffering/upsizing alone can fully resolve; the planned EX-stage pipeline split remains necessary.

Comparing ideal-wire vs. estimated-wire results *at the same placement*: **wns is unchanged** (−9.83 ns both times) — the worst path is not gated by wire delay — while **tns worsens slightly** (−13,280.61 → −13,421.99, about +1%), showing real wire delay adds a small but real cumulative penalty across other, less-critical paths. This confirms the earlier conclusion: wire delay is a real, secondary contributor, while intrinsic logic depth remains the dominant factor.

3.11 Critical path after placement (estimated wire)

The worst path after placement and `estimate_parasitics`, reproduced in full:

```
Startpoint: core_i/id_stage_i/_4563_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:   core_i/id_stage_i/_4435_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type:  max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4563_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.39 0.39 v core_i/id_stage_i/_4563_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.35 0.74 ^ core_i/ex_stage_i/alu_i/_3675_/Y (sky130_fd_sc_hd__nor4bb_4)
0.11 0.85 v core_i/ex_stage_i/alu_i/_4050_/Y (sky130_fd_sc_hd__a21oi_1)
0.25 1.11 ^ core_i/ex_stage_i/alu_i/_4053_/Y (sky130_fd_sc_hd__o31ai_1)
0.16 1.26 ^ core_i/ex_stage_i/alu_i/_place12015/X (sky130_fd_sc_hd__buf_4)
0.13 1.39 ^ core_i/ex_stage_i/alu_i/_4054_/Y (sky130_fd_sc_hd__xnor2_1)
0.11 1.50 v core_i/ex_stage_i/alu_i/_4055_/Y (sky130_fd_sc_hd__a32oi_1)
0.42 1.92 v core_i/ex_stage_i/alu_i/_4057_/X (sky130_fd_sc_hd__maj3_1)
0.41 2.32 v core_i/ex_stage_i/alu_i/_4059_/X (sky130_fd_sc_hd__maj3_1)
0.38 2.71 v core_i/ex_stage_i/alu_i/_4061_/X (sky130_fd_sc_hd__maj3_1)
0.38 3.09 v core_i/ex_stage_i/alu_i/_4063_/X (sky130_fd_sc_hd__maj3_1)
0.38 3.47 v core_i/ex_stage_i/alu_i/_4065_/X (sky130_fd_sc_hd__maj3_1)
0.38 3.85 v core_i/ex_stage_i/alu_i/_4067_/X (sky130_fd_sc_hd__maj3_1)
0.35 4.20 v core_i/ex_stage_i/alu_i/_4069_/X (sky130_fd_sc_hd__maj3_1)
0.25 4.45 ^ core_i/ex_stage_i/alu_i/_4070_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 4.59 v core_i/ex_stage_i/alu_i/_4385_/Y (sky130_fd_sc_hd__xnor2_1)
0.36 4.94 v core_i/ex_stage_i/alu_i/_4429_/X (sky130_fd_sc_hd__maj3_1)
0.38 5.32 v core_i/ex_stage_i/alu_i/_4430_/X (sky130_fd_sc_hd__maj3_1)
0.38 5.70 v core_i/ex_stage_i/alu_i/_4431_/X (sky130_fd_sc_hd__maj3_1)
0.39 6.09 v core_i/ex_stage_i/alu_i/_4433_/X (sky130_fd_sc_hd__maj3_1)
0.38 6.46 v core_i/ex_stage_i/alu_i/_4435_/X (sky130_fd_sc_hd__maj3_1)
0.23 6.69 ^ core_i/ex_stage_i/alu_i/_4437_/Y (sky130_fd_sc_hd__o2111ai_1)
0.12 6.82 v core_i/ex_stage_i/alu_i/_4439_/Y (sky130_fd_sc_hd__a21oi_1)
```

```

0.20 7.02 v core_i/ex_stage_i/alu_i/_4441_/X (sky130_fd_sc_hd__o21a_1)
0.30 7.31 ^ core_i/ex_stage_i/alu_i/_4444_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 7.49 ^ core_i/ex_stage_i/alu_i/place8414/X (sky130_fd_sc_hd__buf_4)
0.07 7.56 v core_i/ex_stage_i/alu_i/_4446_/Y (sky130_fd_sc_hd__a21oi_1)
0.20 7.75 v core_i/ex_stage_i/alu_i/_4448_/X (sky130_fd_sc_hd__o21a_1)
0.28 8.03 ^ core_i/ex_stage_i/alu_i/_4451_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 8.20 ^ core_i/ex_stage_i/alu_i/place8217/X (sky130_fd_sc_hd__buf_4)
0.10 8.30 v core_i/ex_stage_i/alu_i/_4454_/Y (sky130_fd_sc_hd__a311oi_1)
0.27 8.57 ^ core_i/ex_stage_i/alu_i/_4456_/Y (sky130_fd_sc_hd__o31ai_1)
0.14 8.71 v core_i/ex_stage_i/alu_i/_4458_/Y (sky130_fd_sc_hd__o211ai_1)
0.21 8.92 v core_i/ex_stage_i/alu_i/_4459_/X (sky130_fd_sc_hd__and3_1)
0.31 9.22 v core_i/ex_stage_i/alu_i/_4461_/X (sky130_fd_sc_hd__o311a_1)
0.15 9.37 ^ core_i/ex_stage_i/alu_i/_4472_/Y (sky130_fd_sc_hd__o21ai_0)
0.22 9.59 ^ core_i/ex_stage_i/alu_i/_4475_/X (sky130_fd_sc_hd__a31o_1)
0.11 9.70 v core_i/ex_stage_i/alu_i/_4477_/Y (sky130_fd_sc_hd__a21oi_2)
0.16 9.86 v core_i/ex_stage_i/alu_i/place7874/X (sky130_fd_sc_hd__buf_12)
0.23 10.09 ^ core_i/ex_stage_i/alu_i/_4875_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 10.21 v core_i/ex_stage_i/alu_i/_5052_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 10.44 ^ core_i/ex_stage_i/alu_i/_5187_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 10.55 v core_i/ex_stage_i/alu_i/_5188_/Y (sky130_fd_sc_hd__a21oi_1)
0.33 10.88 ^ core_i/ex_stage_i/alu_i/_5197_/Y (sky130_fd_sc_hd__o311ai_0)
0.21 11.10 v core_i/ex_stage_i/alu_i/_5202_/Y (sky130_fd_sc_hd__nand4_1)
0.19 11.28 v core_i/ex_stage_i/alu_i/place7737/X (sky130_fd_sc_hd__buf_4)
0.17 11.45 ^ core_i/ex_stage_i/alu_i/_5821_/Y (sky130_fd_sc_hd__mux2i_1)
0.10 11.55 v core_i/ex_stage_i/alu_i/_5826_/Y (sky130_fd_sc_hd__o22ai_1)
0.24 11.79 ^ core_i/ex_stage_i/alu_i/_5828_/Y (sky130_fd_sc_hd__nor4_1)
0.09 11.88 v core_i/ex_stage_i/alu_i/_5829_/Y (sky130_fd_sc_hd__a211oi_1)
0.31 12.19 ^ core_i/ex_stage_i/alu_i/_5831_/Y (sky130_fd_sc_hd__o21ai_2)
0.12 12.31 v core_i/ex_stage_i/alu_i/_6935_/Y (sky130_fd_sc_hd__a221oi_1)
0.56 12.87 v core_i/ex_stage_i/alu_i/_6965_/X (sky130_fd_sc_hd__or4_1)
0.26 13.13 ^ core_i/ex_stage_i/_147_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 13.23 v core_i/ex_stage_i/_149_/Y (sky130_fd_sc_hd__o21ai_2)
0.21 13.44 v core_i/id_stage_i/_2984_/X (sky130_fd_sc_hd__a22o_1)
0.19 13.63 ^ core_i/id_stage_i/_2985_/Y (sky130_fd_sc_hd__a21oi_1)
0.12 13.75 v core_i/id_stage_i/_3296_/Y (sky130_fd_sc_hd__o22ai_1)
0.17 13.92 ^ core_i/id_stage_i/_3297_/Y (sky130_fd_sc_hd__a21oi_1)
0.07 13.99 v core_i/id_stage_i/_3357_/Y (sky130_fd_sc_hd__nand2_1)
0.09 14.09 ^ core_i/id_stage_i/_3358_/Y (sky130_fd_sc_hd__nand2_1)
0.06 14.15 v core_i/id_stage_i/_3359_/Y (sky130_fd_sc_hd__a21oi_1)
0.26 14.41 v core_i/id_stage_i/_4088_/X (sky130_fd_sc_hd__lpflow_inputiso1p_1
)
0.29 14.70 v core_i/id_stage_i/_4089_/X (sky130_fd_sc_hd__mux2_1)
0.00 14.70 v core_i/id_stage_i/_4435_/D (sky130_fd_sc_hd__dfrrtp_1)
    14.70 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4435_/CLK (sky130_fd_sc_hd__dfrrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

```

```

-----
4.87 data required time
-14.70 data arrival time
-----

-9.83 slack (VIOLATED)

```

Comparison to the floorplan-stage critical path (Section 3.5). The path retains the *same overall structure* — `id_stage_i` → through `ex_stage_i/alu_i`’s `maj3_1` carry chain → `ex_stage_i` → back into `id_stage_i` — confirming this is a stable, structural bottleneck rather than an artifact of a particular run. The endpoint changed slightly (`_4655_` → `_4435_`), and several `place...` buffer instances now appear along the path (inserted by the timing-driven resizer, as discussed above), but the path’s data arrival time only dropped from 25.16 ns to 14.70 ns — a reduction, but still far short of the 5 ns budget, and still dominated by the same long `alu_i` adder chain.

3.12 Global placement convergence log

The final iterations of the Nesterov-method (RePlAce) solver used by `global_placement`, as printed to the GUI’s Scripting panel:

Iteration	Overflow	HPWL (um)	HPWL(%)	Penalty	Group
410	0.2005	1.114183e+06	+1.59%	1.05e-07	
420	0.1684	1.096769e+06	-1.56%	1.54e-07	
430	0.1445	1.088494e+06	-0.75%	2.27e-07	
440	0.1232	1.085206e+06	-0.30%	3.34e-07	
450	0.1059	1.082798e+06	-0.22%	4.92e-07	
454	0.0984	1.081925e+06		5.98e-07	

[INFO GPL-1001] Global placement finished at iteration 454

Interpretation. The **overflow** metric (how much cells still overlap/violate the target density) decreases monotonically from 0.2005 to 0.0984 across the shown iterations, and the solver terminates cleanly once it crosses the default convergence threshold (~ 0.1) — not due to an iteration limit or a stall. **HPWL** (total estimated wirelength) decreases in step, from 1,114,183 μm to 1,081,925 μm , confirming the placement was actively improving, not oscillating, in its final phase. This is a useful sanity check supporting the placement results reported in Section 3.9: the solution is a genuine, converged optimum for the given density target, not an artifact of an early or forced termination.

Visual confirmation. No visible placement anomalies (e.g. large unexplained gaps, cells clustered in one corner) are present, consistent with the clean legalization results (0 placement failures) and convergence log reported above.

Chapter 4

Clock Tree Synthesis (CTS)

4.1 What CTS does, and the clock structure found in this design

Clock Tree Synthesis builds a physical distribution network (a tree of clock buffers) that delivers the clock signal from its single source to every sequential element (flip-flop/latch) in the design, while trying to minimize **skew** (the difference in arrival time of the clock edge at different endpoints) and insertion delay. Before CTS, every previous report in this document used an *ideal* clock (zero network delay, zero skew, shown as **clock network delay (ideal)**); after CTS, the clock network delay becomes *propagated* (a real, non-zero value that differs per sink).

Although the SDC defines a single logical clock (`clk_i`, Section 1.1), TritonCTS identified **3 separate clock nets** to build trees for, because the design contains internal clock gating (the `cv32e40p_sleep_unit/cv32e40p_clock_gate` from Chapter 2) that creates electrically distinct gated derivatives of the same logical clock:

- `clk_i` — 1 sink (the direct, ungated top-level clock net).
- `clk_i_regs` — 4–5 sinks (a small subtree for a handful of registers).
- `core_i/clk` — 2,416 sinks (later 2,557 after dummy-load insertion) — the main clock net feeding almost the entire core, downstream of the sleep-unit’s clock gate.

4.2 CTS script: `cts.tcl`

```
# --- Load the placed design ---
read_db ../2_placement.odb

# --- Liberty + SDC are not stored in the .odb, so reload them ---
read_liberty ../sky130_fd_sc_hd__tt_025C_1v80.lib
read_sdc ../cv32e40p_core.sdc

# --- Wire RC estimates (same as placement stage) ---
set_wire_rc -signal -layer met2
set_wire_rc -clock -layer met3
```

```

# --- Build the clock tree ---
clock_tree_synthesis \
  -root_buf sky130_fd_sc_hd__clkbuf_4 \
  -buf_list "sky130_fd_sc_hd__clkbuf_1 sky130_fd_sc_hd__clkbuf_2
            sky130_fd_sc_hd__clkbuf_4 sky130_fd_sc_hd__clkbuf_8
            sky130_fd_sc_hd__clkbuf_16" \
  -sink_clustering_enable

# --- Legalize the newly-inserted clock buffers ---
detailed_placement

# --- Re-estimate parasitics: the clock network is no longer "ideal" ---
estimate_parasitics -placement

# --- Save the result ---
write_def ../3_cts.def
write_db ../3_cts.odb

# --- Stage reports (CTS) ---
puts "==== TIMING (CTS) ====="
report_checks -path_delay min_max
report_wns
report_tns
report_clock_skew

puts "==== POWER (CTS) ====="
report_power

```

4.3 CTS build log

```

The following clock buffers will be used for CTS:
    sky130_fd_sc_hd__clkbuf_1
    sky130_fd_sc_hd__clkbuf_2
    sky130_fd_sc_hd__clkbuf_4
    sky130_fd_sc_hd__clkbuf_8
    sky130_fd_sc_hd__clkbuf_16
[INFO CTS-0049] Characterization buffer is sky130_fd_sc_hd__clkbuf_16.
[INFO CTS-0007] Net "clk_i" found for clock "clk_i".
[INFO CTS-0011] Clock net "clk_i" for macros has 1 sinks.
[INFO CTS-0011] Clock net "clk_i_regs" for registers has 4 sinks.
[INFO CTS-0010] Clock net "core_i/clk" has 2416 sinks.
[INFO CTS-0008] TritonCTS found 3 clock nets.
[INFO CTS-0097] Characterization used 5 buffer(s) types.
[INFO CTS-0201] 0 blockages from hard placement blockages and placed macros will
    be used.
[INFO CTS-0027] Generating H-Tree topology for net clk_i.
[INFO CTS-0028] Total number of sinks: 1.
[INFO CTS-0060] Macro sinks will be clustered in groups of up to 4 and based on
    buffer max cap.

```

```

[INFO CTS-0030] Number of static layers: 0.
[INFO CTS-0020] Wire segment unit: 13600 dbu (13 um).
[INFO CTS-0023] Original sink region: [(391752, 315670), (391752, 315670)].
[INFO CTS-0024] Normalized sink region: [(28.8053, 23.211), (28.8053, 23.211)].
[INFO CTS-0025] Width: 0.0000.
[INFO CTS-0026] Height: 0.0000.
Level 1
    Direction: Vertical
    Sinks per sub-region: 1
    Sub-region size: 0.0000 X 0.0000
[INFO CTS-0034] Segment length (rounded): 1.
[INFO CTS-0032] Stop criterion found. Max number of sinks is 15.
[INFO CTS-0035] Number of sinks covered: 1.
[INFO CTS-0201] 0 blockages from hard placement blockages and placed macros will
    be used.
[INFO CTS-0027] Generating H-Tree topology for net clk_i_regs.
[INFO CTS-0028] Total number of sinks: 4.
[INFO CTS-0090] Sinks will be clustered based on buffer max cap.
[INFO CTS-0030] Number of static layers: 0.
[INFO CTS-0020] Wire segment unit: 13600 dbu (13 um).
[INFO CTS-0023] Original sink region: [(201505, 295180), (214625, 327820)].
[INFO CTS-0024] Normalized sink region: [(14.8165, 21.7044), (15.7812, 24.1044)
    ].
[INFO CTS-0025] Width: 0.9647.
[INFO CTS-0026] Height: 2.4000.
Level 1
    Direction: Vertical
    Sinks per sub-region: 2
    Sub-region size: 0.9647 X 1.2000
[INFO CTS-0034] Segment length (rounded): 1.
[INFO CTS-0032] Stop criterion found. Max number of sinks is 15.
[INFO CTS-0035] Number of sinks covered: 4.
[INFO CTS-0201] 0 blockages from hard placement blockages and placed macros will
    be used.
[INFO CTS-0027] Generating H-Tree topology for net core_i\clk.
[INFO CTS-0028] Total number of sinks: 2416.
[INFO CTS-0090] Sinks will be clustered based on buffer max cap.
[INFO CTS-0030] Number of static layers: 0.
[INFO CTS-0020] Wire segment unit: 13600 dbu (13 um).
[INFO CTS-0206] Best clustering solution was found from max clustering size of
    21 and max clustering diameter of 215.
[INFO CTS-0019] Total number of sinks after clustering: 151.
[INFO CTS-0024] Normalized sink region: [(1.0698, 1.31047), (55.932, 43.256)].
[INFO CTS-0025] Width: 54.8622.
[INFO CTS-0026] Height: 41.9456.
Level 1
    Direction: Horizontal
    Sinks per sub-region: 76
    Sub-region size: 27.4311 X 41.9456
[INFO CTS-0034] Segment length (rounded): 14.
Level 2

```

```

    Direction: Vertical
    Sinks per sub-region: 38
    Sub-region size: 27.4311 X 20.9728
[INFO CTS-0034] Segment length (rounded): 10.
Level 3
    Direction: Horizontal
    Sinks per sub-region: 19
    Sub-region size: 13.7156 X 20.9728
[INFO CTS-0034] Segment length (rounded): 6.
Level 4
    Direction: Vertical
    Sinks per sub-region: 10
    Sub-region size: 13.7156 X 10.4864
[INFO CTS-0034] Segment length (rounded): 6.
[INFO CTS-0032] Stop criterion found. Max number of sinks is 15.
[INFO CTS-0035] Number of sinks covered: 151.
[INFO CTS-0018] Created 2 clock buffers.
[INFO CTS-0012] Minimum number of buffers in the clock path: 2.
[INFO CTS-0013] Maximum number of buffers in the clock path: 2.
[INFO CTS-0015] Created 2 clock nets.
[INFO CTS-0016] Fanout distribution for the current clock = 1:1..
[INFO CTS-0017] Max level of the clock tree: 1.
[INFO CTS-0018] Created 3 clock buffers.
[INFO CTS-0012] Minimum number of buffers in the clock path: 2.
[INFO CTS-0013] Maximum number of buffers in the clock path: 2.
[INFO CTS-0015] Created 3 clock nets.
[INFO CTS-0016] Fanout distribution for the current clock = 2:2..
[INFO CTS-0017] Max level of the clock tree: 1.
[INFO CTS-0018] Created 172 clock buffers.
[INFO CTS-0012] Minimum number of buffers in the clock path: 4.
[INFO CTS-0013] Maximum number of buffers in the clock path: 4.
[INFO CTS-0015] Created 172 clock nets.
[INFO CTS-0016] Fanout distribution for the current clock = 5:1, 6:1, 7:4, 9:8,
    10:4, 11:3, 12:2, 13:6, 14:13, 15:22, 16:34, 17:35, 18:20, 19:12, 21:2..
[INFO CTS-0017] Max level of the clock tree: 4.
[INFO CTS-0124] Clock net "clk_i"
[INFO CTS-0125] Sinks 1
[INFO CTS-0098] Clock net "clk_i_regs"
[INFO CTS-0099] Sinks 5
[INFO CTS-0100] Leaf buffers 0
[INFO CTS-0101] Average sink wire length 25.79 um
[INFO CTS-0102] Path depth 2 - 2
[INFO CTS-0207] Dummy loads inserted 1
[INFO CTS-0098] Clock net "core_i\clk"
[INFO CTS-0099] Sinks 2557
[INFO CTS-0100] Leaf buffers 151
[INFO CTS-0101] Average sink wire length 521.07 um
[INFO CTS-0102] Path depth 3 - 4
[INFO CTS-0207] Dummy loads inserted 141
[INFO CTS-0033] Balancing latency for clock clk_i
[INFO CTS-0036] inserted 9 delay buffers

```

```
[INFO CTS-0037] Total number of delay buffers: 9
```

Summary. TritonCTS used all 5 available buffer sizes (clkbuf_1 through clkbuf_16), with clkbuf_16 as the characterization reference. It built $2 + 3 + 172 = 177$ **clock buffers** across the three clock nets, plus **9 additional delay buffers** inserted specifically to balance latency across the clk_i tree. The main clock net (core_i/clk) required a 4-level H-tree, an average sink wire length of 521.07 μm , and a path depth of 3–4 buffer levels from source to sink.

4.4 Post-CTS legalization and area

```
[INFO DPL-0006] Core area: 618692.12 um^2, Instances area: 313306.74 um^2,
  Utilization: 50.6%
[INFO DPL-0005] Diamond search max displacement: +/- 500 sites horizontally, +/-
  100 rows vertically.
[INFO DPL-1101] Legalizing using diamond search.
Movements Summary
-----
Total cells: 40513
Diamond Move Success: 40513 (100.00%)
Diamond Move Failure: 0
Rip-up and replace Success: 0 ( 0.00% of diamond failures)
Rip-up and replace Failure: 0
Total Placement Failures: 0
-----
Placement Analysis
-----
total displacement 4224.0 u
average displacement 0.1 u
max displacement 11.8 u
original HPWL 1189491.9 u
legalized HPWL 1192918.6 u
delta HPWL 0 %
```

Interpretation. Adding the clock tree grew the design from 40,184 to **40,513 cells** (+329, consistent with the 177 clock buffers + 9 delay buffers, plus a handful of dummy loads). Legalization again succeeded with 0 failures and very small displacement (average 0.1 μm). Instance area is now explicitly reported as **313,306.74 μm^2** (up from 248,230.57 μm^2 at baseline synthesis), giving a core utilization of **50.6%** — up from the 40.1% baseline and the 65% *target* density used during placement (the difference between 65% target and 50.6% reported here reflects that -density in global_placement controls local bin density during the placement algorithm, not the final overall utilization figure).

4.5 Post-CTS timing (full report) and clock skew

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_238_
```

(removal check against rising-edge clock clk_i)

Path Group: asynchronous

Path Type: min

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (propagated)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.03 0.53 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_238_/
  RESET_B (sky130_fd_sc_hd__dfrtp_1)
  0.53 data arrival time

0.00 0.00 clock clk_i (rise edge)
1.57 1.57 clock network delay (propagated)
0.00 1.57 clock reconvergence pessimism
  1.57 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_238_/CLK
    (sky130_fd_sc_hd__dfrtp_1)
0.33 1.90 library removal time
  1.90 data required time
-----
  1.90 data required time
 -0.53 data arrival time
-----
-1.37 slack (VIOLATED)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_4_

(positive level-sensitive latch clocked by clk_i')

Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_

(rising clock gating-check end-point clocked by clk_i)

Path Group: gated clock

Path Type: min

Delay Time Description

```
-----
2.50 2.50 clock clk_i' (rise edge)
1.57 4.07 clock network delay (propagated)
0.00 4.07 ^ core_i/sleep_unit_i/core_clock_gate_i/_4_/GATE (
  sky130_fd_sc_hd__dlxtp_1)
0.31 4.38 v core_i/sleep_unit_i/core_clock_gate_i/_4_/Q (
  sky130_fd_sc_hd__dlxtp_1)
0.00 4.39 v core_i/sleep_unit_i/core_clock_gate_i/_3_/B (
  sky130_fd_sc_hd__and2_0)
  4.39 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.22 2.72 clock network delay (propagated)
0.00 2.72 clock reconvergence pessimism
```

```

    2.72 v core_i/sleep_unit_i/core_clock_gate_i/_3_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.72 clock gating hold time
    2.72 data required time
-----
    2.72 data required time
   -4.39 data arrival time
-----

    1.66 slack (MET)

Startpoint: irq_i[18] (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/int_controller_i/_128_
          (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
    0.00 0.00 clock clk_i (rise edge)
    0.00 0.00 clock network delay (propagated)
    0.50 0.50 v input external delay
    0.00 0.50 v irq_i[18] (in)
    0.00 0.50 v core_i/id_stage_i/int_controller_i/_128_/D (
        sky130_fd_sc_hd__dftrtp_1)
        0.50 data arrival time

    0.00 0.00 clock clk_i (rise edge)
    1.57 1.57 clock network delay (propagated)
    0.00 1.57 clock reconvergence pessimism
        1.57 ^ core_i/id_stage_i/int_controller_i/_128_/CLK (
            sky130_fd_sc_hd__dftrtp_1)
   -0.02 1.55 library hold time
        1.55 data required time
-----
        1.55 data required time
       -0.50 data arrival time
-----

       -1.04 slack (VIOLATED)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/ex_stage_i/alu_i/alu_div_i/_1698_
          (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
    0.00 0.00 clock clk_i (rise edge)
    0.00 0.00 clock network delay (propagated)

```

```

0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.70 1.20 ^ core_i/wire315/X (sky130_fd_sc_hd__buf_16)
0.74 1.94 ^ core_i/wire314/X (sky130_fd_sc_hd__buf_16)
0.78 2.72 ^ core_i/wire313/X (sky130_fd_sc_hd__buf_16)
0.81 3.52 ^ core_i/wire311/X (sky130_fd_sc_hd__buf_16)
0.21 3.73 ^ core_i/ex_stage_i/alu_i/alu_div_i/_1698_/RESET_B (
    sky130_fd_sc_hd__dfrtp_4)
    3.73 data arrival time

5.00 5.00 clock clk_i (rise edge)
1.44 6.44 clock network delay (propagated)
0.00 6.44 clock reconvergence pessimism
    6.44 ^ core_i/ex_stage_i/alu_i/alu_div_i/_1698_/CLK (
        sky130_fd_sc_hd__dfrtp_4)
-0.15 6.29 library recovery time
    6.29 data required time

-----
    6.29 data required time
    -3.73 data arrival time
-----

    2.56 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_4_
    (positive level-sensitive latch clocked by clk_i')
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
    (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i' (rise edge)
1.57 4.07 clock network delay (propagated)
1.78 5.86 time given to startpoint
0.00 5.86 v core_i/sleep_unit_i/core_clock_gate_i/_4_/D (
    sky130_fd_sc_hd__dlxtp_1)
0.28 6.13 v core_i/sleep_unit_i/core_clock_gate_i/_4_/Q (
    sky130_fd_sc_hd__dlxtp_1)
0.00 6.13 v core_i/sleep_unit_i/core_clock_gate_i/_3_/B (
    sky130_fd_sc_hd__and2_0)
    6.13 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.20 5.20 clock network delay (propagated)
0.00 5.20 clock reconvergence pessimism
    5.20 ^ core_i/sleep_unit_i/core_clock_gate_i/_3_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.20 clock gating setup time
    5.20 data required time

```

```
-----
5.20 data required time
-6.13 data arrival time
-----
```

```
-0.93 slack (VIOLATED)
```

```
Startpoint: core_i/id_stage_i/_4563_
(rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4435_
(rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max
```

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
1.45 1.45 clock network delay (propagated)
0.00 1.45 ^ core_i/id_stage_i/_4563_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.42 1.87 v core_i/id_stage_i/_4563_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.35 2.22 ^ core_i/ex_stage_i/alu_i/_3675_/Y (sky130_fd_sc_hd__nor4bb_4)
0.11 2.33 v core_i/ex_stage_i/alu_i/_4050_/Y (sky130_fd_sc_hd__a21oi_1)
0.25 2.59 ^ core_i/ex_stage_i/alu_i/_4053_/Y (sky130_fd_sc_hd__o31ai_1)
0.16 2.75 ^ core_i/ex_stage_i/alu_i/place12015/X (sky130_fd_sc_hd__buf_4)
0.13 2.87 ^ core_i/ex_stage_i/alu_i/_4054_/Y (sky130_fd_sc_hd__xnor2_1)
0.11 2.98 v core_i/ex_stage_i/alu_i/_4055_/Y (sky130_fd_sc_hd__a32oi_1)
0.42 3.40 v core_i/ex_stage_i/alu_i/_4057_/X (sky130_fd_sc_hd__maj3_1)
0.41 3.80 v core_i/ex_stage_i/alu_i/_4059_/X (sky130_fd_sc_hd__maj3_1)
0.38 4.19 v core_i/ex_stage_i/alu_i/_4061_/X (sky130_fd_sc_hd__maj3_1)
0.38 4.57 v core_i/ex_stage_i/alu_i/_4063_/X (sky130_fd_sc_hd__maj3_1)
0.38 4.95 v core_i/ex_stage_i/alu_i/_4065_/X (sky130_fd_sc_hd__maj3_1)
0.38 5.33 v core_i/ex_stage_i/alu_i/_4067_/X (sky130_fd_sc_hd__maj3_1)
0.35 5.68 v core_i/ex_stage_i/alu_i/_4069_/X (sky130_fd_sc_hd__maj3_1)
0.25 5.93 ^ core_i/ex_stage_i/alu_i/_4070_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 6.07 v core_i/ex_stage_i/alu_i/_4385_/Y (sky130_fd_sc_hd__xnor2_1)
0.36 6.43 v core_i/ex_stage_i/alu_i/_4429_/X (sky130_fd_sc_hd__maj3_1)
0.38 6.81 v core_i/ex_stage_i/alu_i/_4430_/X (sky130_fd_sc_hd__maj3_1)
0.38 7.19 v core_i/ex_stage_i/alu_i/_4431_/X (sky130_fd_sc_hd__maj3_1)
0.39 7.57 v core_i/ex_stage_i/alu_i/_4433_/X (sky130_fd_sc_hd__maj3_1)
0.38 7.95 v core_i/ex_stage_i/alu_i/_4435_/X (sky130_fd_sc_hd__maj3_1)
0.23 8.17 ^ core_i/ex_stage_i/alu_i/_4437_/Y (sky130_fd_sc_hd__o2111ai_1)
0.12 8.30 v core_i/ex_stage_i/alu_i/_4439_/Y (sky130_fd_sc_hd__a21oi_1)
0.20 8.50 v core_i/ex_stage_i/alu_i/_4441_/X (sky130_fd_sc_hd__o21a_1)
0.30 8.80 ^ core_i/ex_stage_i/alu_i/_4444_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 8.97 ^ core_i/ex_stage_i/alu_i/place8414/X (sky130_fd_sc_hd__buf_4)
0.07 9.04 v core_i/ex_stage_i/alu_i/_4446_/Y (sky130_fd_sc_hd__a21oi_1)
0.20 9.24 v core_i/ex_stage_i/alu_i/_4448_/X (sky130_fd_sc_hd__o21a_1)
0.28 9.51 ^ core_i/ex_stage_i/alu_i/_4451_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 9.69 ^ core_i/ex_stage_i/alu_i/place8217/X (sky130_fd_sc_hd__buf_4)
0.10 9.79 v core_i/ex_stage_i/alu_i/_4454_/Y (sky130_fd_sc_hd__a311oi_1)
0.27 10.05 ^ core_i/ex_stage_i/alu_i/_4456_/Y (sky130_fd_sc_hd__o31ai_1)
```

```

0.14 10.19 v core_i/ex_stage_i/alu_i/_4458_/Y (sky130_fd_sc_hd__o211ai_1)
0.21 10.40 v core_i/ex_stage_i/alu_i/_4459_/X (sky130_fd_sc_hd__and3_1)
0.31 10.70 v core_i/ex_stage_i/alu_i/_4461_/X (sky130_fd_sc_hd__o311a_1)
0.15 10.85 ^ core_i/ex_stage_i/alu_i/_4472_/Y (sky130_fd_sc_hd__o21ai_0)
0.22 11.07 ^ core_i/ex_stage_i/alu_i/_4475_/X (sky130_fd_sc_hd__a31o_1)
0.11 11.18 v core_i/ex_stage_i/alu_i/_4477_/Y (sky130_fd_sc_hd__a21oi_2)
0.16 11.34 v core_i/ex_stage_i/alu_i/place7874/X (sky130_fd_sc_hd__buf_12)
0.23 11.57 ^ core_i/ex_stage_i/alu_i/_4875_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 11.69 v core_i/ex_stage_i/alu_i/_5052_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 11.92 ^ core_i/ex_stage_i/alu_i/_5187_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 12.04 v core_i/ex_stage_i/alu_i/_5188_/Y (sky130_fd_sc_hd__a21oi_1)
0.33 12.37 ^ core_i/ex_stage_i/alu_i/_5197_/Y (sky130_fd_sc_hd__o311ai_0)
0.21 12.58 v core_i/ex_stage_i/alu_i/_5202_/Y (sky130_fd_sc_hd__nand4_1)
0.19 12.76 v core_i/ex_stage_i/alu_i/place7737/X (sky130_fd_sc_hd__buf_4)
0.17 12.93 ^ core_i/ex_stage_i/alu_i/_5821_/Y (sky130_fd_sc_hd__mux2i_1)
0.10 13.03 v core_i/ex_stage_i/alu_i/_5826_/Y (sky130_fd_sc_hd__o22ai_1)
0.24 13.27 ^ core_i/ex_stage_i/alu_i/_5828_/Y (sky130_fd_sc_hd__nor4_1)
0.09 13.36 v core_i/ex_stage_i/alu_i/_5829_/Y (sky130_fd_sc_hd__a211oi_1)
0.31 13.67 ^ core_i/ex_stage_i/alu_i/_5831_/Y (sky130_fd_sc_hd__o21ai_2)
0.12 13.79 v core_i/ex_stage_i/alu_i/_6935_/Y (sky130_fd_sc_hd__a221oi_1)
0.56 14.36 v core_i/ex_stage_i/alu_i/_6965_/X (sky130_fd_sc_hd__or4_1)
0.26 14.61 ^ core_i/ex_stage_i/_147_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 14.71 v core_i/ex_stage_i/_149_/Y (sky130_fd_sc_hd__o21ai_2)
0.21 14.92 v core_i/id_stage_i/_2984_/X (sky130_fd_sc_hd__a22o_1)
0.19 15.11 ^ core_i/id_stage_i/_2985_/Y (sky130_fd_sc_hd__a21oi_1)
0.12 15.23 v core_i/id_stage_i/_3296_/Y (sky130_fd_sc_hd__o22ai_1)
0.17 15.40 ^ core_i/id_stage_i/_3297_/Y (sky130_fd_sc_hd__a21oi_1)
0.07 15.48 v core_i/id_stage_i/_3357_/Y (sky130_fd_sc_hd__nand2_1)
0.09 15.57 ^ core_i/id_stage_i/_3358_/Y (sky130_fd_sc_hd__nand2_1)
0.06 15.63 v core_i/id_stage_i/_3359_/Y (sky130_fd_sc_hd__a21oi_1)
0.26 15.89 v core_i/id_stage_i/_4088_/X (sky130_fd_sc_hd__lpflow_inputisolp_1
)
0.29 16.18 v core_i/id_stage_i/_4089_/X (sky130_fd_sc_hd__mux2_1)
0.00 16.18 v core_i/id_stage_i/_4435_/D (sky130_fd_sc_hd__dftrtp_1)
    16.18 data arrival time

5.00 5.00 clock clk_i (riseedge)
1.44 6.44 clock network delay (propagated)
0.00 6.44 clock reconvergence pessimism
    6.44 ^ core_i/id_stage_i/_4435_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.11 6.32 library setup time
    6.32 data required time

-----
    6.32 data required time
-16.18 data arrival time

-----
    -9.86 slack (VIOLATED)

wns max -9.86
tns max -13596.81

```

```

Clock clk_i
  1.57 source latency core_i/cs_registers_i/_4354_/CLK ^
-1.37 target latency core_i/if_stage_i/prefetch_buffer_i/fifo_i/_274_/CLK ^
-0.00 CRPR
-----
  0.19 setup skew

```

New violations introduced by propagated (real) clock skew. Three paths that were **MET** at earlier ideal-clock stages are now **VIOLATED**, purely because of real clock arrival-time differences between source and destination flip-flops:

- The `rst_ni` removal check on `instruction_obi_i`: **MET** (+0.21 ns) at floorplan → **VIOLATED** (−1.37 ns) now.
- An input-port hold check on `irq_i[18]`: not previously reported as a distinct violation; now **VIOLATED** (−1.04 ns).
- The gated-clock max (setup) check on `core_clock_gate_i`: **MET** (−0.16 ns slack, i.e. barely met) at floorplan → **VIOLATED** (−0.93 ns) now.

These are a normal and expected consequence of introducing a real clock tree: reset/async and gated-clock paths are especially sensitive to clock skew between the flip-flops involved, since (unlike the main `clk_i` data path) they are not protected by the full clock period. They would need dedicated timing exceptions or skew-group constraints to resolve properly — noted here as a known limitation rather than fixed, since it does not affect the main EX-stage bottleneck this project targets.

Setup skew report. `report_clock_skew` reports a worst-case **setup skew of 0.19 ns** between the source flip-flop (`cs_registers_i/_4354_`, 1.57 ns clock latency) and the target flip-flop (`prefetch_buffer_i/fifo_i/_274_`). This is small relative to the 5 ns clock period (3.8%), indicating the clock tree itself is reasonably well-balanced — the new violations above stem from specific asynchronous/gated-clock paths, not from generally poor clock tree quality.

Main critical path. The dominant EX-stage bottleneck persists: `wns_max` is **−9.86 ns** (essentially unchanged from −9.83 ns at the placement stage), with the same `id_stage_i` → `alu_i` carry-chain → `id_stage_i` structure as before. The data arrival time increased slightly (14.70 ns → 16.18 ns) due to real, non-zero clock network delay now being added on both the launch and capture side of the path.

4.6 Post-CTS power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.14e-02	1.80e-03	2.96e-08	2.32e-02	11.1%
Combinational	8.70e-02	8.62e-02	1.00e-07	1.73e-01	82.6%
Clock	7.34e-03	6.05e-03	2.86e-09	1.34e-02	6.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	1.16e-01 55.2%	9.41e-02 44.8%	1.33e-07 0.0%	2.10e-01	100.0%

Table 4.1: Power breakdown at the CTS stage.

An important, unresolved discrepancy. Total power jumped from 12.70 mW (placement stage) to **210.1 mW** at CTS — a $\sim 16.5\times$ increase. The clock group’s own power rose as expected (0.163 mW $\rightarrow$ 13.4 mW, since 186 real clock buffers now exist and toggle every cycle, versus zero before). However, **combinational** power alone also rose by $\sim 16\times$ (11.3 mW $\rightarrow$ 173 mW), which the +329-cell (+0.8%) growth in cell count cannot explain on its own. The most likely explanation is that OpenSTA’s default (non-workload) switching-activity model, used throughout this project in the absence of real VCD/SAIF simulation data (Section 2.6), is sensitive to whether the clock is ideal or propagated — once real clock latency and skew are introduced, the default activity assumptions may be recomputed differently for many nets, not just clock-adjacent ones. This is flagged explicitly as an **open question requiring further verification** (e.g. rerunning `report_power` at the CTS stage with the clock temporarily forced back to ideal, to isolate the cause) rather than presented as a settled conclusion; the relative combinational-dominance pattern remains consistent with earlier stages, but the absolute power figures at this stage should be treated with caution until this is investigated further.

4.7 Stage-by-stage summary and conclusions

Stage	wns max (ns)	tns max
Floorplan (ideal wire)	−20.23	−30,884.59
Placement, timing-driven (ideal wire)	−9.83	−13,280.61
Placement, timing-driven (estimated wire)	−9.83	−13,421.99
CTS (propagated clock, estimated wire)	−9.86	−13,596.81

Table 4.2: wns/tns across all stages completed so far.

Stage	Total power (mW)	Cell count	Instance area (μm^2)
Baseline synthesis	—	32,671	248,230.57
Placement	12.70	40,184	—
CTS	210.1	40,513	313,306.74

Table 4.3: Power, cell count, and instance area across stages (dashes: not separately measured at that stage).

Overall conclusions from the physical design flow (Chapters 3–4).

1. **The EX-stage bottleneck is real, large, and stable across every stage of the physical flow.** From baseline synthesis through floorplan, placement, and CTS, the worst path consistently runs through the same ~ 50 -cell `maj3_1` carry chain inside `cv32e40p_alu.sv`. Its magnitude shrank considerably once the resizer was allowed to buffer/upscale aggressively during timing-driven placement (wns $-20.23\text{ ns} \rightarrow -9.83\text{ ns}$), but never closed, and CTS did not meaningfully change it further ($-9.83\text{ ns} \rightarrow -9.86\text{ ns}$). This is strong, multi-stage evidence that the fix must be structural (the planned EX-stage pipeline split), not something the backend tools can resolve through buffering, sizing, or placement alone.
2. **Fixing timing this aggressively has a real, measured cost.** Enabling timing-driven optimization grew the design by 23% in cell count and roughly doubled (at least) total power between baseline and CTS. This is a useful, quantified trade-off

to report: better timing was bought with area and power, and even that was not enough on its own.

3. **Introducing a real clock tree exposes new, secondary violations** (reset removal, an input hold check, and the gated-clock setup check) that were invisible under the ideal-clock assumption used earlier. These are normal at this stage of a physical design flow and are noted as known, unresolved limitations rather than something this project’s scope requires fixing, since none of them lie on the EX-stage path under study.
4. **The area, cell-count, and power figures from this document form a consistent baseline** for the eventual before/after comparison once the pipeline is split into EX1/EX2 (Chapter 1): this document already establishes what “before” looks like at every stage (synthesis, floorplan, placement, CTS), so the same script sequence can be re-run on the modified RTL for a fair, stage-by-stage comparison.

4.8 Post-CTS timing repair (repair_timing)

After the CTS results in Section 4.5, the following was run on the live session (reloaded standalone afterwards as `repair_timing.tcl`, reusing `3_cts.odb`):

```
repair_timing -setup -hold

write_def 3_cts_repaired.def
write_db 3_cts_repaired.odb
```

4.8.1 Repair progress log

Repairing 2056 out of 2056 (100.00%) violating endpoints...													
Iter	Removed	Resized	Inserted	Cloned	Pin	Area	WNS	StTNS	EnTNS	Viol	Worst		
	Buffers	Gates	Buffers	Gates	Swaps				Endpts	St/EnPt			
0*	0	0	0	0	0	+0.0%	-9.826	-5988.6	-13423.6	2056	core_i/id_stage_i/_4435_/D		
2300*	2	415	36	5	206	+0.5%	-8.450	-5912.5	-11760.1	2056	core_i/id_stage_i/_4752_/D		
2300+	2	415	36	5	206	+0.5%	-8.450	-5912.5	-11760.1	2056	core_i/id_stage_i/_4752_/D		
final	2	461	36	5	219	+0.5%	-8.450	-5900.3	-11756.3	2056	core_i/id_stage_i/_4662_/D		

[INFO RSZ-0059] Removed 2 buffers.													
[INFO RSZ-0040] Inserted 36 buffers.													
[INFO RSZ-0051] Resized 461 instances: 461 up, 0 up match, 0 down, 0 VT													
[INFO RSZ-0043] Swapped pins on 219 instances.													
[INFO RSZ-0049] Cloned 5 instances.													
[WARNING RSZ-0062] Unable to repair all setup violations.													
[INFO RSZ-0033] No hold violations found.													

Summary. The resizer found **2,056 violating endpoints** (out of 2,056 checked — i.e. every endpoint it examined was in violation). To repair them it: removed 2 buffers, **resized 461 instances** (all upsized to higher-drive-strength variants), **inserted 36 new buffers**, cloned 5 instances, and swapped pins on 219 instances — a substantial additional repair effort on top of the buffering/upsizing already done during timing-driven placement (Section 3.8). This reduced **wns from -9.826 ns to -8.450 ns** (a 1.376 ns, ~14% improvement) and reduced the **EnTNS** column from $-13,423.6$ to $-11,756.3$ (~12.4% improvement, and consistent with the **tns max** value reported independently below). **Critically, the tool itself reports [WARNING RSZ-0062] Unable to repair all setup violations** — despite this effort, the violating-endpoint count did not drop from 2,056. [INFO RSZ-0033] No hold violations found — the previously-seen `irq_i[18]` hold

violation (Section 4.5) no longer appears among the checked/reported paths at this stage; this is noted as an observed change, not independently re-verified in isolation.

4.8.2 Post-repair timing report (full)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/fifo_i/_241_
(removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min
Delay Time Description

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/if_stage_i/prefetch_buffer_i/fifo_i/_241_/RESET_B (
sky130_fd_sc_hd__dfrrtp_1)
0.50 data arrival time
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
0.00 ^ core_i/if_stage_i/prefetch_buffer_i/fifo_i/_241_/CLK (
sky130_fd_sc_hd__dfrrtp_1)
0.30 0.30 library removal time
0.30 data required time

0.30 data required time
-0.50 data arrival time

0.21 slack (MET)
Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_4_
(positive level-sensitive latch clocked by clk_i')
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
(rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min
Delay Time Description

2.50 2.50 clock clk_i' (rise edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 ^ core_i/sleep_unit_i/core_clock_gate_i/_4_/GATE (
sky130_fd_sc_hd__dlxtp_1)
0.30 2.80 v core_i/sleep_unit_i/core_clock_gate_i/_4_/Q (
sky130_fd_sc_hd__dlxtp_1)

```

0.00 2.80 v core_i/sleep_unit_i/core_clock_gate_i/_3_/B (
    sky130_fd_sc_hd__and2_0)
    2.80 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
    2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
    2.50 data required time

-----
    2.50 data required time
    -2.80 data arrival time
-----

    0.30 slack (MET)

Startpoint: core_i/id_stage_i/controller_i/_632_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/controller_i/_632_
    (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/controller_i/_632_/CLK (
    sky130_fd_sc_hd__dfrtp_1)
0.30 0.30 ^ core_i/id_stage_i/controller_i/_632_/Q (sky130_fd_sc_hd__dfrtp_1)
0.09 0.39 ^ core_i/id_stage_i/controller_i/_501_/X (
    sky130_fd_sc_hd__lpflow_inputiso1p_1)
0.00 0.39 ^ core_i/id_stage_i/controller_i/_632_/D (sky130_fd_sc_hd__dfrtp_1)
    0.39 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/id_stage_i/controller_i/_632_/CLK (
        sky130_fd_sc_hd__dfrtp_1)
-0.04 -0.04 library hold time
    -0.04 data required time

-----
    -0.04 data required time
    -0.39 data arrival time
-----

    0.43 slack (MET)

```

Startpoint: rst_ni (input port clocked by clk_i)
 Endpoint: core_i/ex_stage_i/alu_i/alu_div_i/_1698_
 (recovery check against rising-edge clock clk_i)
 Path Group: asynchronous
 Path Type: max

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.70 1.20 ^ core_i/wire315/X (sky130_fd_sc_hd__buf_16)
0.74 1.94 ^ core_i/wire314/X (sky130_fd_sc_hd__buf_16)
0.78 2.72 ^ core_i/wire313/X (sky130_fd_sc_hd__buf_16)
0.81 3.52 ^ core_i/wire311/X (sky130_fd_sc_hd__buf_16)
0.21 3.73 ^ core_i/ex_stage_i/alu_i/alu_div_i/_1698_/RESET_B (
    sky130_fd_sc_hd__dfrtp_4)
    3.73 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/ex_stage_i/alu_i/alu_div_i/_1698_/CLK (
    sky130_fd_sc_hd__dfrtp_4)
-0.18 4.82 library recovery time
    4.82 data required time
-----
    4.82 data required time
    -3.73 data arrival time
-----
    1.09 slack (MET)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_4_
 (positive level-sensitive latch clocked by clk_i')
 Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
 (rising clock gating-check end-point clocked by clk_i)
 Path Group: gated clock
 Path Type: max

Delay Time Description

```
-----
2.50 2.50 clock clk_i' (rise edge)
0.00 2.50 clock network delay (ideal)
1.83 4.33 time given to startpoint
0.00 4.33 v core_i/sleep_unit_i/core_clock_gate_i/_4_/D (
    sky130_fd_sc_hd__dlxtp_1)
0.28 4.60 v core_i/sleep_unit_i/core_clock_gate_i/_4_/Q (
    sky130_fd_sc_hd__dlxtp_1)
```

```

0.00 4.60 v core_i/sleep_unit_i/core_clock_gate_i/_3_/B (
    sky130_fd_sc_hd__and2_0)
    4.60 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_3_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time

-----

    5.00 data required time
    -4.60 data arrival time

-----

    0.40 slack (MET)

Startpoint: core_i/id_stage_i/_4564_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4662_
    (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4564_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.39 0.39 v core_i/id_stage_i/_4564_/Q (sky130_fd_sc_hd__dftrtp_1)
0.35 0.74 ^ core_i/ex_stage_i/alu_i/_3675_/Y (sky130_fd_sc_hd__nor4bb_4)
0.11 0.85 v core_i/ex_stage_i/alu_i/_4050_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 1.07 ^ core_i/ex_stage_i/alu_i/_4053_/Y (sky130_fd_sc_hd__o31ai_1)
0.17 1.24 ^ core_i/ex_stage_i/alu_i/place12015/X (sky130_fd_sc_hd__buf_4)
0.12 1.37 ^ core_i/ex_stage_i/alu_i/_4054_/Y (sky130_fd_sc_hd__xnor2_2)
0.09 1.45 v core_i/ex_stage_i/alu_i/_4055_/Y (sky130_fd_sc_hd__a32oi_2)
0.35 1.80 v core_i/ex_stage_i/alu_i/_4057_/X (sky130_fd_sc_hd__maj3_2)
0.33 2.13 v core_i/ex_stage_i/alu_i/_4059_/X (sky130_fd_sc_hd__maj3_2)
0.31 2.44 v core_i/ex_stage_i/alu_i/_4061_/X (sky130_fd_sc_hd__maj3_2)
0.32 2.75 v core_i/ex_stage_i/alu_i/_4063_/X (sky130_fd_sc_hd__maj3_2)
0.32 3.07 v core_i/ex_stage_i/alu_i/_4065_/X (sky130_fd_sc_hd__maj3_2)
0.32 3.39 v core_i/ex_stage_i/alu_i/_4067_/X (sky130_fd_sc_hd__maj3_2)
0.31 3.69 v core_i/ex_stage_i/alu_i/_4069_/X (sky130_fd_sc_hd__maj3_2)
0.16 3.86 ^ core_i/ex_stage_i/alu_i/_4070_/Y (sky130_fd_sc_hd__o21ai_2)
0.10 3.96 v core_i/ex_stage_i/alu_i/_4385_/Y (sky130_fd_sc_hd__xnor2_1)
0.32 4.28 v core_i/ex_stage_i/alu_i/_4429_/X (sky130_fd_sc_hd__maj3_2)
0.31 4.59 v core_i/ex_stage_i/alu_i/_4430_/X (sky130_fd_sc_hd__maj3_2)
0.31 4.89 v core_i/ex_stage_i/alu_i/_4431_/X (sky130_fd_sc_hd__maj3_2)
0.32 5.21 v core_i/ex_stage_i/alu_i/_4433_/X (sky130_fd_sc_hd__maj3_2)
0.31 5.53 v core_i/ex_stage_i/alu_i/_4435_/X (sky130_fd_sc_hd__maj3_2)

```

```

0.23 5.75 ^ core_i/ex_stage_i/alu_i/_4437_/Y (sky130_fd_sc_hd__o2111ai_2)
0.09 5.84 v core_i/ex_stage_i/alu_i/_4439_/Y (sky130_fd_sc_hd__a21oi_2)
0.21 6.05 v core_i/ex_stage_i/alu_i/_4441_/X (sky130_fd_sc_hd__o21a_1)
0.28 6.33 ^ core_i/ex_stage_i/alu_i/_4444_/Y (sky130_fd_sc_hd__o311ai_1)
0.16 6.49 ^ core_i/ex_stage_i/alu_i/place8414/X (sky130_fd_sc_hd__buf_4)
0.05 6.54 v core_i/ex_stage_i/alu_i/_4446_/Y (sky130_fd_sc_hd__a21oi_2)
0.20 6.74 v core_i/ex_stage_i/alu_i/_4448_/X (sky130_fd_sc_hd__o21a_1)
0.26 7.00 ^ core_i/ex_stage_i/alu_i/_4451_/Y (sky130_fd_sc_hd__o311ai_1)
0.16 7.16 ^ core_i/ex_stage_i/alu_i/place8217/X (sky130_fd_sc_hd__buf_4)
0.08 7.24 v core_i/ex_stage_i/alu_i/_4454_/Y (sky130_fd_sc_hd__a311oi_2)
0.31 7.55 ^ core_i/ex_stage_i/alu_i/_4456_/Y (sky130_fd_sc_hd__o31ai_1)
0.15 7.70 v core_i/ex_stage_i/alu_i/_4458_/Y (sky130_fd_sc_hd__o211ai_1)
0.19 7.90 v core_i/ex_stage_i/alu_i/_4459_/X (sky130_fd_sc_hd__and3_1)
0.35 8.25 v core_i/ex_stage_i/alu_i/_4461_/X (sky130_fd_sc_hd__o311a_1)
0.37 8.61 v core_i/ex_stage_i/alu_i/_4473_/X (sky130_fd_sc_hd__or3_1)
0.27 8.88 v core_i/ex_stage_i/alu_i/_4475_/X (sky130_fd_sc_hd__a31o_1)
0.32 9.20 ^ core_i/ex_stage_i/alu_i/_4477_/Y (sky130_fd_sc_hd__a21oi_4)
0.18 9.38 ^ core_i/ex_stage_i/alu_i/place7871/X (sky130_fd_sc_hd__buf_6)
0.05 9.43 v core_i/ex_stage_i/alu_i/_4861_/Y (sky130_fd_sc_hd__nor3_4)
0.12 9.55 v core_i/ex_stage_i/alu_i/place7847/X (sky130_fd_sc_hd__buf_6)
0.24 9.79 ^ core_i/ex_stage_i/alu_i/_5067_/Y (sky130_fd_sc_hd__o311ai_2)
0.16 9.95 ^ core_i/ex_stage_i/alu_i/place7809/X (sky130_fd_sc_hd__buf_4)
0.04 9.99 v core_i/ex_stage_i/alu_i/_5253_/Y (sky130_fd_sc_hd__a21oi_2)
0.17 10.16 ^ core_i/ex_stage_i/alu_i/_5254_/Y (sky130_fd_sc_hd__a31oi_1)
0.09 10.25 v core_i/ex_stage_i/alu_i/_5365_/Y (sky130_fd_sc_hd__a21oi_1)
0.28 10.53 ^ core_i/ex_stage_i/alu_i/_5374_/Y (sky130_fd_sc_hd__o211ai_1)
0.14 10.67 v core_i/ex_stage_i/alu_i/_5784_/Y (sky130_fd_sc_hd__mux2i_1)
0.19 10.86 ^ core_i/ex_stage_i/alu_i/_5787_/Y (sky130_fd_sc_hd__o2bb2ai_1)
0.11 10.97 v core_i/ex_stage_i/alu_i/_5806_/Y (sky130_fd_sc_hd__o22ai_1)
0.21 11.19 ^ core_i/ex_stage_i/alu_i/_5829_/Y (sky130_fd_sc_hd__a211oi_2)
0.09 11.28 v core_i/ex_stage_i/alu_i/_5831_/Y (sky130_fd_sc_hd__o21ai_2)
0.15 11.42 v core_i/ex_stage_i/alu_i/rebuffer14958/X (sky130_fd_sc_hd__buf_6)
0.13 11.55 v core_i/ex_stage_i/alu_i/place7644/X (sky130_fd_sc_hd__buf_12)
0.12 11.67 v core_i/ex_stage_i/alu_i/place7643/X (sky130_fd_sc_hd__buf_6)
0.28 11.95 v core_i/ex_stage_i/alu_i/_6885_/X (sky130_fd_sc_hd__a221o_1)
0.10 12.05 ^ core_i/ex_stage_i/alu_i/_6933_/Y (sky130_fd_sc_hd__o2111ai_1)
0.10 12.15 v core_i/ex_stage_i/_123_/Y (sky130_fd_sc_hd__a22oi_1)
0.17 12.32 ^ core_i/ex_stage_i/_125_/Y (sky130_fd_sc_hd__o21ai_2)
0.14 12.47 ^ core_i/id_stage_i/_2969_/X (sky130_fd_sc_hd__a22o_1)
0.06 12.53 v core_i/id_stage_i/_2970_/Y (sky130_fd_sc_hd__a21oi_1)
0.20 12.73 ^ core_i/id_stage_i/_3355_/Y (sky130_fd_sc_hd__o22ai_1)
0.08 12.81 v core_i/id_stage_i/_3356_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 13.03 ^ core_i/id_stage_i/_3359_/Y (sky130_fd_sc_hd__a21oi_1)
0.17 13.21 ^ core_i/id_stage_i/_3360_/Y (sky130_fd_sc_hd__o21bai_4)
0.09 13.30 v core_i/id_stage_i/_3725_/Y (sky130_fd_sc_hd__o21ai_0)
0.09 13.38 ^ core_i/id_stage_i/_3726_/Y (sky130_fd_sc_hd__nand2_1)
0.00 13.38 ^ core_i/id_stage_i/_4662_/D (sky130_fd_sc_hd__dftrtp_1)
    13.38 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)

```

```

0.00 5.00 clock reconvergence pessimism
      5.00 ^ core_i/id_stage_i/_4662_/CLK (sky130_fd_sc_hd__dfirtp_1)
-0.07 4.93 library setup time
      4.93 data required time
-----
      4.93 data required time
     -13.38 data arrival time
-----
     -8.45 slack (VIOLATED)

wns max -8.45
tns max -11756.31
Clock clk_i
  1.26 source latency core_i/id_stage_i/controller_i/_612_/CLK ^
 -1.06 target latency core_i/if_stage_i/_928_/CLK ^
  0.00 CRPR
-----
  0.20 setup skew

```

The critical endpoint moved, but the bottleneck did not. The worst path’s endpoint shifted from cell `_4435_` (Section 4.5) to cell `_4662_`, both inside `id_stage_i` — a *different* flip-flop, fed by an overlapping but distinct portion of the same `alu_i` carry chain (many of the same cell indices reappear: `_4057_` through `_4435_`, now upsized to `_2/_4` drive-strength variants such as `maj3_2`, `a21oi_4`, and several newly-inserted buffers like `place7871`, `rebuffer14958`). This is the expected behavior of endpoint-driven repair: fixing the single worst path exposes the next-worst path in the same structurally-bottlenecked region as the new worst offender — consistent with the conclusion that the true bottleneck is the `alu_i` logic depth itself, not any single specific path.

4.8.3 Post-repair power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.18e-02	1.80e-03	2.96e-08	2.36e-02	11.3%
Combinational	8.75e-02	8.65e-02	1.01e-07	1.74e-01	83.0%
Clock	7.59e-03	4.39e-03	2.86e-09	1.20e-02	5.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	1.17e-01 55.8%	9.27e-02 44.2%	1.33e-07 0.0%	2.10e-01	100.0%

Table 4.4: Power breakdown after `repair_timing`.

Total power (210.1 mW) is essentially **unchanged** from the CTS stage (Section 4.6) despite the additional 461 resized + 36 inserted cells — the unexplained $\sim 16.5\times$ jump observed at CTS (relative to placement) neither worsens nor resolves here, and remains an open question as noted in Section 4.6.

4.9 Updated stage-by-stage summary

Stage	wns max (ns)	tns max
Floorplan (ideal wire)	−20.23	−30,884.59
Placement, timing-driven (ideal wire)	−9.83	−13,280.61
Placement, timing-driven (estimated wire)	−9.83	−13,421.99
CTS (propagated clock, estimated wire)	−9.86	−13,596.81
CTS + repair_timing	−8.45	−11,756.31

Table 4.5: wns/tns across all stages completed so far, including the post-CTS repair pass.

Updated conclusion. Even after every automatic backend technique available has now been applied — timing-driven placement, clock tree synthesis, and an explicit post-CTS `repair_timing` pass (461 resizes, 36 new buffers, 5 clones, 219 pin swaps) — the design still violates timing by **8.45 ns**, on **2,056 endpoints**, and the tool explicitly reports it could not repair all of them. Across four independent stages (floorplan, placement, CTS, and now post-repair), the wns has improved from −20.23 ns to −8.45 ns (58% reduction) purely through buffering/upsizing/cloning — a substantial but insufficient improvement. This is now very strong, multi-stage evidence that closing the remaining gap requires the planned structural change (splitting the EX stage’s `alu_i` carry chain across two pipeline registers), not further backend optimization.

Chapter 5

Routing

5.1 Fixing a constant-value net before routing: `insert_tiecells`

Running `detailed_route` directly on the routed-so-far design failed immediately:

```
[INFO DRT-0150] Reading design.  
[WARNING DRT-0120] Large net rst_ni has 150 pins which may impact routing  
performance. Consider optimization.  
[ERROR DRT-0305] Net zero_ of signal type GROUND is not routable by TritonRoute.  
Move to special nets.  
[ERROR GUI-0070] DRT-0305
```

Cause. Where the RTL ties a signal to a constant value (e.g. an unconnected input, or an explicit `1'b0`/`1'b1`), Yosys represents it in the netlist with an implicit constant net, conventionally named `zero_` (for 0) or `one_` (for 1). These are ordinary logic nets, but they get mis-typed in the physical database as `GROUND/POWER special` nets — which TritonRoute refuses to route as regular signals, since special (power/ground) nets are normally handled by the power distribution network, not by signal routing. This is a well-documented issue (see References) whose standard fix is `insert_tiecells`: replacing each constant net with a real physical **tie cell** instance (a standard cell that drives a fixed logic 0 or 1), which is an ordinary, routable signal net.

Fix, and the follow-on placement issue. `insert_tiecells` was run for both polarities, using the sky130 tie-cell master `sky130_fd_sc_hd__conb_1` (its L0 and HI pins provide the two constant values):

```
insert_tiecells sky130_fd_sc_hd__conb_1/L0 -prefix "LOGIC0_"  
insert_tiecells sky130_fd_sc_hd__conb_1/HI -prefix "LOGIC1_"
```

This alone was not sufficient: the newly-created tie-cell instances have no physical location yet, and `detailed_placement` (the legalizer) can only nudge already-approximately-placed cells onto legal sites — it cannot place a completely new, location-less instance. Attempting `global_route` at this point failed with `[ERROR GRT-0010] Instance LOGIC0_zero_ is not placed`. The fix was to run `global_placement -incremental`

first, to give the new cells an initial location (without disturbing the rest of the already-placed design), *then* legalize with `detailed_placement`.

Note on methodology. The OpenROAD-flow-scripts' own automated floorplan step performs this same tie-cell insertion *during floorplanning* (Chapter 3), before placement — the cleaner point to do it. Since this project's manual flow only discovered the issue at the routing stage (after placement, CTS, and `repair_timing` were already complete), it was patched in-place here rather than redoing the entire flow from floorplan onward; this does not affect the validity of the timing/power/area results already reported in Chapters 3–4, since tie cells carry fixed, non-switching values and do not alter the design's logic or its critical path.

5.2 Routing script: `routing.tcl`

```
# --- Reload the completed, repaired CTS design ---
read_db ../3_cts_repaired.odb

# --- Liberty + SDC are not stored in the .odb, so reload them ---
read_liberty ../sky130_fd_sc_hd_tt_025C_1v80.lib
read_sdc ../cv32e40p_core.sdc

# --- Wire RC (same as previous stages) ---
set_wire_rc -signal -layer met2
set_wire_rc -clock -layer met3

# --- Fix constant-value nets ("zero_" / "one_") that Yosys left as fake GROUND/
#      POWER
# nets instead of real signals -- TritonRoute cannot route these directly.
# insert_tiecells replaces them with real tie-cell instances.
insert_tiecells sky130_fd_sc_hd__conb_1/L0 -prefix "LOGIC0_"
insert_tiecells sky130_fd_sc_hd__conb_1/HI -prefix "LOGIC1_"

# --- Give the newly-inserted, still-unplaced tie cells an initial location ---
global_placement -incremental

# --- Legalize the newly-inserted tie cells ---
detailed_placement

# --- Global routing: coarse net topology, congestion-aware ---
# met1-met5 signal, met3-met5 clock: matches this platform's own defaults
# (MIN_ROUTING_LAYER=met1, MIN_CLK_ROUTING_LAYER=met3, MAX_ROUTING_LAYER=met5)
set_routing_layers -signal met1-met5 -clock met3-met5
global_route

# --- Re-estimate parasitics using the (more accurate) global-route topology ---
estimate_parasitics -global_routing

# --- Detailed routing: commits exact, DRC-clean wires and vias ---
detailed_route -output_drc ../5_route_drc.rpt
```

```
# --- Save the result ---
write_def ../5_route.def
write_db ../5_route.odg

# --- Reports ---
puts "==== TIMING (post-route) ====="
report_checks -path_delay min_max
report_wns
report_tns
report_clock_skew

puts "==== POWER (post-route) ====="
report_power
```

Command-by-command explanation.

- `read_db` — loads the completed, timing-repaired CTS design (Chapter 4).
- `insert_tiecells / global_placement -incremental / detailed_placement` — the fix described above.
- `set_routing_layers -signal met1-met5 -clock met3-met5` — tells the router which metal layers it may use for signal and clock nets, matching this platform's own defaults (`MIN_ROUTING_LAYER=met1`, `MIN_CLK_ROUTING_LAYER=met3`, `MAX_ROUTING_LAYER=met5`).
- `global_route` — the coarse-grained routing pass (FastRoute): decides each net's general topology and layer usage, congestion-aware, without yet committing exact geometry.
- `estimate_parasitics -global_routing` — upgrades the wire-delay estimate to use the real global-route topology (more accurate than the placement-only HPWL-based estimate used in Chapters 3–4).
- `detailed_route` — the fine-grained routing pass (TritonRoute): commits exact, DRC-clean wires and vias. `-output_drc` writes any remaining design-rule violations to a report file.
- `write_def / write_db` — saves the fully-routed design.
- Final reports — the same timing/power reporting pattern used at every previous stage, now on a fully-routed design.

Chapter 6

RTL Modification: EX-Stage Pipeline Split

6.1 Where the critical path lives in the RTL, and why

The critical-path analysis in Sections 3.5 and 3.10 traced the worst timing path structurally to `cv32e40p_alu.sv`, inside the EX stage. Reading the actual RTL confirms and refines this: the ALU implements a **36-bit partitioned adder** (a standard PULP technique letting the same adder perform one 32-bit add, two independent 16-bit SIMD adds, or four independent 8-bit SIMD adds, by selectively blocking carry propagation at bit positions 0, 9, 18, and 27):

```
// cv32e40p_alu.sv, line 168
assign adder_result_expanded = $signed(adder_in_a) + $signed(adder_in_b);
```

This single 36-bit addition is what Yosys/ABC synthesized into the long, repeated run of `maj3_1` carry-chain cells identified in Section 3.5 as consuming 87% of the critical path’s delay. Crucially, the adder’s output does not stop there: it feeds a rounding adjustment and then a 64-bit barrel shifter used for shift/rotate instructions:

```
// line 186: rounding
assign adder_round_result = adder_result + adder_round_value;

// line 255: the adder’s (rounded) output feeds the shifter as an operand
assign shift_op_a = shift_left ? operand_a_rev :
                    (shift_use_round ? adder_round_result : operand_a_i);
```

This explains the second half of the critical path (Section 3.10’s `mux2i_1/o21ai_0` cells after the adder chain): it is the barrel shifter, implemented as a tree of multiplexers, consuming the adder’s result combinatorially, in the same cycle.

Two candidate split points. Two places were considered for inserting the new pipeline register:

1. **Inside the adder, at bit 18** — the design already blocks carry propagation there for 16-bit SIMD mode (line 155: `adder_in_a[18] = 1'b0`), making it a structurally

natural place to break the carry chain. However, implementing this correctly requires manually splitting the single Verilog `+` operator into two half-additions with an explicit carry passed through the new register — a hand-written carry-propagation scheme that is easy to get subtly wrong (sign handling, the guard-bit semantics under each `vector_mode_i` case) and hard to verify with confidence in the time available.

2. **Immediately after the adder and its rounding step, before the shifter** (between line 186 and line 255) — a natural *functional* boundary: EX1 becomes “compute the arithmetic result”, EX2 becomes “shift/rotate and finalize”. No arithmetic operator is split; the register only latches already-complete signals (`adder_round_result`, plus whatever operands/control signals downstream logic needs).

Decision. Option 2 was chosen. It is lower-risk (no hand-rolled carry logic to get wrong), matches the two-phase structure already visible in the critical-path trace (a long arithmetic chunk followed by a shifter/mux chunk), and is easier to verify: the register only captures fully-formed signals rather than a partially-computed sum. The trade-off is that it does not bisect the single largest contributor (the 36-bit add) as evenly as an in-adder split might have — this is noted as a potential avenue for a more aggressive future revision if further latency reduction is needed.

6.2 Further RTL discovery: the shifter’s true reach

Reading the full 987-line file revealed that the split boundary is more consequential than it first appears: not only `shift_result` (feeding `ALU_ADD/SUB/SLL/SRL/SRA/ROR`), but also `bextins_result` (bit-insert/extract), `reverse_result` (bit-reverse), and — unexpectedly — the **divider** all depend on the shifter’s output:

```
// cv32e40p_alu_div.sv is instantiated with:
.OpB_DI (shift_left_result), // the divisor is pre-normalized through the
    shifter
```

This means the divider’s own instantiation must move into the “EX2” half along with the shifter, since one of its inputs is only available there. This was judged an acceptable, well-justified simplification: the divider already runs its own multi-cycle FSM (on the order of 32 cycles once started), so one extra cycle of input latency before it begins is negligible.

6.3 Implementation: pipeline control and signal partitioning

The core of the change is a single-bit valid register controlling a two-phase (EX1/EX2) split, re-using the ALU’s pre-existing `ready_o/ex_ready_i/enable_i` handshake (already present in the original RTL to stall the pipeline while the divider is busy):

```
logic ex1_valid_q;
logic ex1_capture; // new EX1 result is latched into the EX2 registers this
    cycle
```

```

logic ex2_consume; // EX2's result_o is accepted downstream this cycle

assign ex1_capture = enable_i & ~div_valid & ~ex1_valid_q;
assign ex2_consume = ex1_valid_q & ex_ready_i;

always_ff @(posedge clk, negedge rst_n) begin
    if (~rst_n) begin
        ex1_valid_q <= 1'b0;
    end else begin
        if (ex1_capture) ex1_valid_q <= 1'b1;
        else if (ex2_consume) ex1_valid_q <= 1'b0;
    end
end
...
assign ready_o = div_valid_q ? div_ready : ex1_valid_q;

```

Why no changes were needed in `cv32e40p_id_stage.sv` or `cv32e40p_controller.sv`. `cv32e40p_ex_stage.sv` already derives its own `ex_ready_o/ex_valid_o` (consumed by the ID stage for stalling) directly from the ALU's `ready_o` signal (wired as `alu_ready`). Since `ready_o` now correctly drops for exactly one cycle whenever a non-divide operation is in EX1 (mirroring how it already dropped for many cycles during a divide), the existing stall infrastructure propagates the extra latency automatically. No other module's RTL needed modification, and the ALU's external port list is unchanged (**verified programmatically to be byte-identical** to the original).

Signal partitioning. Roughly 25 signals cross the new EX1→EX2 register: the opcode and control fields needed to run the final case statement in EX2 (`operator_q`, `vector_mode_q`, `bmask_a_q`, `is_clpx_q`), the raw operands still needed downstream (`operand_a_q`, `operand_a_rev_q`, `operand_b_q`, `operand_c_q`), the adder's result, the shifter's already-prepared inputs (`shift_op_a_q`, `shift_amt_int_q`, `shift_left_q`, `shift_arithmetic_q`), every “cheap” EX1-computed candidate result (`cmp_result_q`, `result_minmax_q`, `clip_result_q`, `pack_result_q`, `bitop_result_q`), the bit-manipulation masks (`bmask_q`, `bmask_inv_q`), and the divider's pre-computed shift amount and sign (`div_shift_q`, `div_op_a_signed_q`, `div_valid_q`).

Design trade-off. All non-divide ALU operations now uniformly take 2 cycles instead of 1 — including simple bitwise ops (AND/OR/XOR) that do not touch the adder or shifter at all and could, in principle, bypass the new register. This was a deliberate choice for implementation simplicity and verifiability within the project's time budget, at the cost of leaving some possible performance on the table; selectively bypassing the pipeline register for operations that do not need it is noted as a natural follow-up optimization.

6.4 Verification status

The modified file was checked with Icarus Verilog (`iverilog -g2012`) against its real dependencies (`cv32e40p_pkg.sv`, `cv32e40p_popcnt.sv`, `cv32e40p_ff_one.sv`, `cv32e40p_alu_div.sv`):

- **Zero syntax/elaboration errors**, exit code 0.
- The same 63 “sorry: constant selects in always_* processes” notices appear as in the *original, unmodified* file — confirmed by running the identical check against both files — meaning these are pre-existing Icarus Verilog limitations unrelated to this change, not new problems it introduced.
- The module’s port list was confirmed **programmatically byte-identical** to the original.

Important caveat. This is a syntax/elaboration check only, **not** a functional simulation. It confirms the file is well-formed SystemVerilog that elaborates cleanly against its dependencies, but does *not* confirm the pipelined ALU produces correct results for any given instruction. Running the core’s existing testbench/riscv-tests against this modified RTL, before treating its PPA results as valid, remains necessary future work and is explicitly flagged as such rather than assumed.

6.5 Bug found during review: divider deadlock

Before running the modified RTL through `sv2v`, the pipeline-control logic was re-examined for correctness, and a real bug was found in the interaction between the new EX1/EX2 register and the divider.

The bug. The original version of `ex1_capture` explicitly excluded divide operations:

```
assign ex1_capture = enable_i & ~div_valid & ~ex1_valid_q;
```

while the divider’s start signal required `ex1_valid_q`:

```
.InVld_SI (ex1_valid_q & div_valid_q),
```

Since divide operations were never captured, `ex1_valid_q` would never become 1 for them, so `InVld_SI` would never assert — **the divider would never start, a deadlock** — and the registered operands it depends on (`operand_b_q`, `div_shift_q`, etc.) would never be updated with the actual divide operands in the first place.

The fix. Divide operations are now captured into the pipeline register like any other operation, ensuring the divider receives correctly-registered operands. Readiness is instead computed separately, gating on the divider’s own multi-cycle `div_ready` handshake once a divide is in flight, rather than on `ex1_valid_q` alone:

```
assign ex1_capture = enable_i & ~ex1_valid_q;
assign result_ready = div_valid_q ? div_ready : ex1_valid_q;
assign ex2_consume = result_ready & ex_ready_i;
...
assign ready_o = result_ready;
```

This also correctly prevents a new operation from being captured while a divide is still in flight (`ex1_capture` requires `~ex1_valid_q`, which stays high for the divide’s full multi-cycle duration).

Verification. The corrected file was re-checked with Icarus Verilog: exit code 0, the same 63 pre-existing “sorry” notices as before (no new ones introduced by the fix). The corrected RTL (Section 6.3’s listing reflects this fix) was then converted with `sv2v` and inspected: 25 modules present (matching the baseline’s module count including the `cv32e40p_clock_gate` fix from Section 2.2), no leftover SystemVerilog constructs, and the flattened pipeline-control and divider-instantiation logic matches the corrected design exactly. This is still not a substitute for functional simulation (Section 6.4’s caveat still applies in full), but it is a second, independent check that caught a real logic error before it could propagate into the physical design flow.

Chapter 7

Modified Design: Baseline Synthesis Results

7.1 Synthesis run

The corrected, pipelined `cv32e40p_alu.sv` (Chapter 6) was converted with `sv2v` (producing `flattened_design_modified.v`) and synthesized with `yosys_syntn_modified.tcl` (Section 7 uses the identical script as Section 2.4, only the input/output filenames differ). Synthesis completed without errors. The design hierarchy still lists exactly the same 25 modules as the baseline (Section 2.2), confirming no module was accidentally added, removed, or left out of the modified netlist.

7.2 Area and cell-count comparison

Metric	Baseline	Modified	Change
Total cells	32,671	33,092	+421 (+1.3%)
Chip area (μm^2)	248,230.57	264,114.56	+15,883.99 (+6.4%)
Sequential share	24.39%	27.31%	+2.9 pp

Table 7.1: Baseline vs. modified design: cell count, area, and sequential share at the Yosys synthesis stage.

Module (local area, μm^2)	Baseline	Modified	Change
<code>cv32e40p_alu</code>	20,936.33	$\sim 33,200$	+12,264 (+58.6%)
<code>cv32e40p_mult</code>	77,140.23	77,300 (7.73E+04)	$\approx 0\%$ (unchanged)
<code>cv32e40p_register_file</code>	$\sim 60,900$	64,300 (6.43E+04)	$\sim +5.6\%$
<code>cv32e40p_clock_gate</code>	31.28	27.53	-12.0%

Table 7.2: Per-module area comparison for the modules of interest (unaffected modules such as `mult` shown for reference).

Interpretation. The area increase is concentrated almost entirely where expected: `cv32e40p_alu`'s own area grew by 58.6% (+12,264 μm^2), and this single module accounts for $\sim 77\%$ of the entire design's +15,884 μm^2 growth — direct, quantitative confirmation that the new EX1/EX2 pipeline registers (Section 6.3) are the dominant cost of this modification, not some unrelated side effect elsewhere in the design. The +421 total cell-count

increase is consistent with the ~ 25 signals registered across the new pipeline boundary (accounting for their bit-widths). `cv32e40p_mult`, entirely untouched by this project's RTL changes, shows no meaningful change, as expected.

Two smaller observations, neither a cause for concern:

- `cv32e40p_register_file` (also untouched RTL) shows a small increase ($\sim 5.6\%$), most plausibly ordinary ABC technology-mapping variation from a different overall netlist context, rather than a real effect of the ALU change.
- `cv32e40p_clock_gate` (untouched RTL) happened to map to `sky130_fd_sc_hd__dlxtn_1` directly this run, instead of the `dlxtp_1+inv_1` pair seen in the baseline (Section 2.4) — the same active-low latch functionality, just a slightly more area-efficient single-cell mapping ($27.53\text{ }\mu\text{m}^2$ vs. $31.28\text{ }\mu\text{m}^2$) chosen this time by ABC.

Chapter 8

Modified Design: Floorplan (with PDN) Results

8.1 Floorplan run

`floorplan_init_modified.tcl` (Section 7.1 references the identical script structure as Section 3.7, now including the PDN block from the start — no repeat of Chapter 4’s PDN omission) was run on `synth_netlist_modified.v`. The design initialized and `pdngen` completed without error.

8.2 Timing: a completely different critical path

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/cs_registers_i/_4725_
          (removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/cs_registers_i/_4725_/RESET_B (sky130_fd_sc_hd__dfrrtp_1)
          0.50 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
          0.00 ^ core_i/cs_registers_i/_4725_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.30 0.30 library removal time
          0.30 data required time
-----
          0.30 data required time
          -0.50 data arrival time
```

```

-----
0.20 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

```

Delay Time Description

```

-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
sky130_fd_sc_hd__dlxtn_1)
0.23 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
sky130_fd_sc_hd__dlxtn_1)
0.00 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
sky130_fd_sc_hd__and2_0)
2.73 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
2.50 data required time

-----
2.50 data required time
-2.73 data arrival time
-----

```

```

0.23 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22511_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/ex_stage_i/mult_i/_22510_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

```

Delay Time Description

```

-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22511_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.34 0.34 ^ core_i/ex_stage_i/mult_i/_22511_/Q (sky130_fd_sc_hd__dftrtp_1)
0.00 0.34 ^ core_i/ex_stage_i/mult_i/_22510_/D (sky130_fd_sc_hd__dftrtp_1)

```

```

    0.34 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/ex_stage_i/mult_i/_22510_/CLK (sky130_fd_sc_hd__dfrrtp_1)
-0.05 -0.05 library hold time
    -0.05 data required time
-----
    -0.05 data required time
    -0.34 data arrival time
-----

    0.39 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/cs_registers_i/_4747_
    (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
    0.00 0.00 clock clk_i (rise edge)
    0.00 0.00 clock network delay (ideal)
    0.50 0.50 ^ input external delay
    0.00 0.50 ^ rst_ni (in)
    0.00 0.50 ^ core_i/cs_registers_i/_4747_/SET_B (sky130_fd_sc_hd__dfstp_1)
    0.50 data arrival time

    5.00 5.00 clock clk_i (rise edge)
    0.00 5.00 clock network delay (ideal)
    0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/cs_registers_i/_4747_/CLK (sky130_fd_sc_hd__dfstp_1)
    0.13 5.13 library recovery time
    5.13 data required time
-----
    5.13 data required time
    -0.50 data arrival time
-----

    4.63 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
    (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
    (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description

```

```

-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
2.33 4.83 time given to startpoint
0.00 4.83 ^ core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.31 5.14 ^ core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 5.14 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    5.14 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time

-----
    5.00 data required time
    -5.14 data arrival time

-----
    -0.14 slack (VIOLATED)

Startpoint: core_i/if_stage_i/_931_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: instr_addr_o[31] (output port clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/if_stage_i/_931_/CLK (sky130_fd_sc_hd__dftrtp_1)
7.10 7.10 ^ core_i/if_stage_i/_931_/Q (sky130_fd_sc_hd__dftrtp_1)
5.40 12.49 v core_i/id_stage_i/register_file_i/_06253_/Y (
    sky130_fd_sc_hd__clkinv_1)
0.57 13.07 ^ core_i/id_stage_i/register_file_i/_08665_/Y (
    sky130_fd_sc_hd__o21ai_0)
0.15 13.21 v core_i/id_stage_i/register_file_i/_08666_/Y (
    sky130_fd_sc_hd__o221ai_1)
0.26 13.48 ^ core_i/id_stage_i/register_file_i/_08672_/Y (
    sky130_fd_sc_hd__a211oi_1)
0.11 13.58 v core_i/id_stage_i/register_file_i/_08673_/Y (
    sky130_fd_sc_hd__a211oi_1)
0.27 13.85 v core_i/id_stage_i/_2228_/X (sky130_fd_sc_hd__and2_0)
0.23 14.08 ^ core_i/id_stage_i/_2235_/Y (sky130_fd_sc_hd__nor2_1)
0.15 14.23 v core_i/id_stage_i/_2237_/Y (sky130_fd_sc_hd__a211oi_1)

```

```

0.42 14.65 ^ core_i/id_stage_i/_2240_/Y (sky130_fd_sc_hd__o311ai_0)
0.16 14.81 v core_i/id_stage_i/_2243_/Y (sky130_fd_sc_hd__a311oi_1)
0.37 15.18 v core_i/id_stage_i/_2244_/X (sky130_fd_sc_hd__or3_1)
0.38 15.55 v core_i/id_stage_i/_2247_/X (sky130_fd_sc_hd__a311o_1)
0.40 15.95 v core_i/id_stage_i/_2250_/X (sky130_fd_sc_hd__a311o_1)
0.31 16.26 ^ core_i/id_stage_i/_2257_/Y (sky130_fd_sc_hd__o31ai_1)
0.14 16.40 v core_i/id_stage_i/_2267_/Y (sky130_fd_sc_hd__nand4_1)
0.28 16.68 ^ core_i/id_stage_i/_2272_/Y (sky130_fd_sc_hd__a31oi_1)
0.15 16.83 v core_i/id_stage_i/_2278_/Y (sky130_fd_sc_hd__nand4_1)
0.24 17.07 ^ core_i/id_stage_i/_2283_/Y (sky130_fd_sc_hd__a31oi_1)
0.14 17.21 v core_i/id_stage_i/_2288_/Y (sky130_fd_sc_hd__nand4_1)
0.30 17.51 ^ core_i/id_stage_i/_2294_/Y (sky130_fd_sc_hd__a311oi_1)
0.23 17.74 ^ core_i/id_stage_i/_2295_/X (sky130_fd_sc_hd__maj3_1)
0.07 17.81 v core_i/id_stage_i/_2296_/Y (sky130_fd_sc_hd__nor2_1)
0.17 17.98 ^ core_i/id_stage_i/_2589_/Y (sky130_fd_sc_hd__mux2i_1)
0.13 18.11 ^ core_i/id_stage_i/_2590_/Y (sky130_fd_sc_hd__xnor2_1)
0.10 18.21 v core_i/id_stage_i/_2591_/Y (sky130_fd_sc_hd__o22ai_1)
0.32 18.54 ^ core_i/if_stage_i/_795_/Y (sky130_fd_sc_hd__a222oi_1)
0.07 18.61 v core_i/if_stage_i/_797_/Y (sky130_fd_sc_hd__nand2_1)
0.10 18.71 ^ core_i/if_stage_i/_798_/Y (sky130_fd_sc_hd__a21oi_1)
0.15 18.86 v core_i/if_stage_i/_799_/Y (sky130_fd_sc_hd__o311ai_0)
0.24 19.10 ^ core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_416_/
Y (sky130_fd_sc_hd__a22oi_1)
0.08 19.18 v core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_417_/
Y (sky130_fd_sc_hd__nand2_1)
0.28 19.46 v core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_146_/X (
sky130_fd_sc_hd__mux2_1)
0.00 19.46 v instr_addr_o[31] (out)
    19.46 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
-3.00 2.00 output external delay
    2.00 data required time

-----
    2.00 data required time
   -19.46 data arrival time

-----
   -17.46 slack (VIOLATED)

wns max -17.46
tns max -30642.40

```

Metric	Baseline (Section 3.5)	Modified
wns max	−20.23 ns	−17.46 ns (13.7% better)
tns max	−30,884.59	−30,642.40
Critical path location	alu_i (adder + shifter)	register_file_i → id_stage_i → if_stage_i/prefetch_buffer → instr_addr_o[31]
Data arrival time	25.16 ns	19.46 ns

Table 8.1: Floorplan-stage timing: baseline vs. modified design.

Interpretation. The EX-stage pipeline split worked exactly as intended: `cv32e40p_alu` is no longer on the critical path at all. However, this did not close timing — it exposed a *different, pre-existing* bottleneck that was always present in the design but previously hidden behind the even-worse ALU violation. The new worst path runs through the register file’s address-decode logic and back into the instruction-fetch stage’s output port `instr_addr_o`, most plausibly part of the jump/branch-target address computation (a register value feeding a PC-mux chain that ultimately drives the next instruction fetch address). This is a classic result of single-bottleneck optimization: fixing the worst offender reveals the next one. The two gated-clock-check paths (min/max) are essentially unchanged in magnitude from the baseline (−0.14 ns vs. −0.16 ns) and are treated as the same pre-existing, unrelated issue noted in Section 3.5 — not something introduced by this change.

8.3 Power: an unexplained, flagged discrepancy

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	1.31e-03	1.64e-04	3.29e-08	1.47e-03	33.8%
Combinational	1.35e-03	1.38e-03	6.45e-08	2.73e-03	62.5%
Clock	−6.41e-07	1.61e-04	1.87e-12	1.60e-04	3.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.65e-03 60.9%	1.71e-03 39.1%	9.74e-08 0.0%	4.36e-03	100.0%

Table 8.2: Power breakdown, modified design, floorplan stage.

Total power (4.36 mW) is roughly half the baseline’s 8.69 mW (Section 3.6), despite the modified design having *more* cells (33,092 vs. 32,671) and a higher sequential share (27.31% vs. 24.39%, which would ordinarily be expected to increase, not decrease, power). This is flagged as an **open discrepancy, not a settled conclusion** — consistent with the caution already raised in Section 4.6 about default (non-workload) switching-activity estimates being sensitive to netlist structure in the absence of real VCD/SAIF data. Both figures should be treated as rough estimates until real activity-based power analysis is performed; the relative magnitude should not yet be read as a genuine power benefit of the pipeline split.

Chapter 9

Modified Design: Placement Results

9.1 Placement run and legalization

`global_placement_modified.tcl` (identical script structure to Section 3.7–3.8) was run on `1_floorplan_pdn_modified.odb`. Timing-driven global placement added 19.00% area ($56,166.55\mu\text{m}^2$), close to the baseline's +20.98%. Legalization succeeded with 0 failures:

```
Total cells: 40223
Diamond Move Success: 40223 (100.00%)
Diamond Move Failure: 0
Rip-up and replace Success: 0 ( 0.00% of diamond failures)
Rip-up and replace Failure: 0
Total Placement Failures: 0
-----
Placement Analysis
-----
total displacement 94965.4 u
average displacement 2.4 u
max displacement 9.6 u
original HPWL 1109927.9 u
legalized HPWL 1197544.4 u
delta HPWL 8 %
```

Metric	Baseline (Section 3.8)	Modified
Total cells	40,184	40,223
Timing-driven area delta	+20.98%	+19.00%
Instances area	307,562.48 μm^2	320,596.23 μm^2
Utilization	49.7%	48.7%
Total displacement / avg / max	93222.4 / 2.3 / 9.9 u	94965.4 / 2.4 / 9.6 u
HPWL delta from legalization	8%	8%

Table 9.1: Placement legalization: baseline vs. modified design.

9.2 Timing: the critical path moves again — into the untouched multiplier

9.2.1 Ideal wire

Startpoint: rst_ni (input port clocked by clk_i)	
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_266_ (removal check against rising-edge clock clk_i)	
Path Group: asynchronous	
Path Type: min	
Delay Time Description	

0.00	0.00 clock clk_i (rise edge)
0.00	0.00 clock network delay (ideal)
0.50	0.50 ^ input external delay
0.00	0.50 ^ rst_ni (in)
0.38	0.88 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.01	0.89 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_266_/RESET_B (sky130_fd_sc_hd__dfrrtp_1)
	0.89 data arrival time
0.00	0.00 clock clk_i (rise edge)
0.00	0.00 clock network delay (ideal)
0.00	0.00 clock reconvergence pessimism
	0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_266_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.43	0.43 library removal time
	0.43 data required time

	0.43 data required time
	-0.89 data arrival time

	0.45 slack (MET)
Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_ (negative level-sensitive latch clocked by clk_i)	
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_ (rising clock gating-check end-point clocked by clk_i)	
Path Group: gated clock	
Path Type: min	
Delay Time Description	

2.50	2.50 clock clk_i (fall edge)
0.00	2.50 clock network delay (ideal)
0.00	2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (sky130_fd_sc_hd__dlxtn_1)
0.23	2.73 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (sky130_fd_sc_hd__dlxtn_1)

```

0.00 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    2.73 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
    2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
    2.50 data required time

-----
    2.50 data required time
    -2.73 data arrival time
-----

    0.23 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22511_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/ex_stage_i/mult_i/_22510_
    (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22511_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.37 0.37 ^ core_i/ex_stage_i/mult_i/_22511_/Q (sky130_fd_sc_hd__dftrtp_1)
0.00 0.37 ^ core_i/ex_stage_i/mult_i/_22510_/D (sky130_fd_sc_hd__dftrtp_1)
    0.37 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/ex_stage_i/mult_i/_22510_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.06 -0.06 library hold time
    -0.06 data required time

-----
    -0.06 data required time
    -0.37 data arrival time
-----

    0.43 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/cs_registers_i/_4932_
    (recovery check against rising-edge clock clk_i)
Path Group: asynchronous

```

Path Type: max

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.38 0.88 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.55 1.43 ^ core_i/cs_registers_i/place14125/X (sky130_fd_sc_hd__buf_4)
0.72 2.15 ^ core_i/cs_registers_i/place14124/X (sky130_fd_sc_hd__buf_4)
0.02 2.17 ^ core_i/cs_registers_i/_4932_/RESET_B (sky130_fd_sc_hd__dfrrtp_1)
      2.17 data arrival time
```

```
5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
      5.00 ^ core_i/cs_registers_i/_4932_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.03 5.03 library recovery time
      5.03 data required time
```

```
-----
      5.03 data required time
     -2.17 data arrival time
```

```
-----
      2.85 slack (MET)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_

(negative level-sensitive latch clocked by clk_i)

Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_

(rising clock gating-check end-point clocked by clk_i)

Path Group: gated clock

Path Type: max

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.74 4.24 time given to startpoint
0.00 4.24 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
      sky130_fd_sc_hd__dlxtn_1)
0.22 4.46 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
      sky130_fd_sc_hd__dlxtn_1)
0.00 4.46 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
      sky130_fd_sc_hd__and2_0)
      4.46 data arrival time
```

```
5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
```

```

    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time
-----
    5.00 data required time
   -4.46 data arrival time
-----

    0.54 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22508_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4716_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22508_/CLK (sky130_fd_sc_hd__dfstp_1)
0.55 0.55 ^ core_i/ex_stage_i/mult_i/_22508_/Q (sky130_fd_sc_hd__dfstp_1)
0.14 0.70 ^ core_i/ex_stage_i/mult_i/_11259_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.16 0.85 ^ core_i/ex_stage_i/mult_i/_place12084_/X (sky130_fd_sc_hd__buf_4)
0.06 0.91 v core_i/ex_stage_i/mult_i/_12041_/Y (sky130_fd_sc_hd__nor3_1)
0.15 1.06 v core_i/ex_stage_i/mult_i/_place11563_/X (sky130_fd_sc_hd__buf_4)
0.35 1.41 ^ core_i/ex_stage_i/mult_i/_13042_/Y (sky130_fd_sc_hd__nor4_1)
0.41 1.82 v core_i/ex_stage_i/mult_i/_13085_/X (sky130_fd_sc_hd__xnor3_1)
0.52 2.34 v core_i/ex_stage_i/mult_i/_13086_/X (sky130_fd_sc_hd__or4b_1)
0.41 2.74 v core_i/ex_stage_i/mult_i/_13090_/X (sky130_fd_sc_hd__maj3_1)
0.40 3.14 v core_i/ex_stage_i/mult_i/_13094_/X (sky130_fd_sc_hd__maj3_1)
0.19 3.33 v core_i/ex_stage_i/mult_i/_13109_/X (sky130_fd_sc_hd__xor2_1)
0.22 3.55 ^ core_i/ex_stage_i/mult_i/_13111_/Y (sky130_fd_sc_hd__a31oi_1)
0.18 3.73 ^ core_i/ex_stage_i/mult_i/_13112_/Y (sky130_fd_sc_hd__xnor2_1)
0.13 3.86 v core_i/ex_stage_i/mult_i/_13123_/Y (sky130_fd_sc_hd__a21oi_1)
0.38 4.24 v core_i/ex_stage_i/mult_i/_13125_/X (sky130_fd_sc_hd__maj3_1)
0.37 4.61 v core_i/ex_stage_i/mult_i/_13126_/X (sky130_fd_sc_hd__maj3_1)
0.24 4.85 ^ core_i/ex_stage_i/mult_i/_13128_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 4.99 v core_i/ex_stage_i/mult_i/_13130_/Y (sky130_fd_sc_hd__a21oi_1)
0.25 5.24 ^ core_i/ex_stage_i/mult_i/_13132_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 5.39 v core_i/ex_stage_i/mult_i/_13134_/Y (sky130_fd_sc_hd__a21oi_1)
0.29 5.68 ^ core_i/ex_stage_i/mult_i/_13137_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 5.81 v core_i/ex_stage_i/mult_i/_13140_/Y (sky130_fd_sc_hd__a21oi_1)
0.30 6.10 ^ core_i/ex_stage_i/mult_i/_13142_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 6.24 v core_i/ex_stage_i/mult_i/_13145_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 6.46 v core_i/ex_stage_i/mult_i/_13148_/X (sky130_fd_sc_hd__o21a_1)
0.22 6.68 ^ core_i/ex_stage_i/mult_i/_13150_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 6.78 v core_i/ex_stage_i/mult_i/_13153_/Y (sky130_fd_sc_hd__a21oi_1)

```

```

0.21 6.99 ^ core_i/ex_stage_i/mult_i/_13155_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.10 v core_i/ex_stage_i/mult_i/_13157_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 7.34 ^ core_i/ex_stage_i/mult_i/_13160_/Y (sky130_fd_sc_hd__o21ai_0)
0.15 7.49 v core_i/ex_stage_i/mult_i/_13162_/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.73 ^ core_i/ex_stage_i/mult_i/_13165_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.84 v core_i/ex_stage_i/mult_i/_13167_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 8.06 ^ core_i/ex_stage_i/mult_i/_13169_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 8.17 v core_i/ex_stage_i/mult_i/_13171_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 8.40 ^ core_i/ex_stage_i/mult_i/_13174_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 8.50 v core_i/ex_stage_i/mult_i/_13176_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 8.72 ^ core_i/ex_stage_i/mult_i/_13179_/Y (sky130_fd_sc_hd__o21ai_0)
0.15 8.87 ^ core_i/ex_stage_i/mult_i/place7266/X (sky130_fd_sc_hd__buf_4)
0.14 9.01 ^ core_i/ex_stage_i/mult_i/_13241_/Y (sky130_fd_sc_hd__xnor2_1)
0.09 9.10 v core_i/ex_stage_i/mult_i/_13242_/Y (sky130_fd_sc_hd__clkinv_1)
0.46 9.57 v core_i/ex_stage_i/mult_i/_21734_/X (sky130_fd_sc_hd__mux4_2)
0.19 9.75 ^ core_i/ex_stage_i/mult_i/_21735_/Y (sky130_fd_sc_hd__mux2i_1)
0.16 9.91 ^ core_i/ex_stage_i/mult_i/_21920_/X (sky130_fd_sc_hd__a21o_1)
0.06 9.97 v core_i/ex_stage_i/mult_i/_22267_/Y (sky130_fd_sc_hd__a211oi_1)
0.52 10.49 v core_i/ex_stage_i/mult_i/_22276_/X (sky130_fd_sc_hd__or4_1)
0.26 10.75 ^ core_i/ex_stage_i/_156_/Y (sky130_fd_sc_hd__a22oi_1)
0.11 10.86 v core_i/ex_stage_i/_158_/Y (sky130_fd_sc_hd__o21ai_2)
0.19 11.05 ^ core_i/id_stage_i/_2829_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 11.15 v core_i/id_stage_i/_2830_/Y (sky130_fd_sc_hd__nand2_1)
0.18 11.33 ^ core_i/id_stage_i/_2834_/Y (sky130_fd_sc_hd__a22oi_1)
0.14 11.47 v core_i/id_stage_i/_2835_/Y (sky130_fd_sc_hd__nand3_1)
0.18 11.65 ^ core_i/id_stage_i/_3073_/Y (sky130_fd_sc_hd__o21ai_0)
0.07 11.72 v core_i/id_stage_i/_3074_/Y (sky130_fd_sc_hd__a21oi_1)
0.24 11.96 v core_i/id_stage_i/_3529_/Y (sky130_fd_sc_hd__o21bai_1)
0.29 12.25 v core_i/id_stage_i/_3554_/X (sky130_fd_sc_hd__mux2_1)
0.00 12.25 v core_i/id_stage_i/_4716_/D (sky130_fd_sc_hd__dftrtp_1)

```

```

    12.25 data arrival time

```

```

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4716_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

```

```

-----
    4.87 data required time
   -12.25 data arrival time

```

```

-----
   -7.38 slack (VIOLATED)

```

```

wns max -7.38
tns max -9553.46

```

9.2.2 Estimated wire

```

Startpoint: rst_ni (input port clocked by clk_i)

```

Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_
(removal check against rising-edge clock clk_i)

Path Group: asynchronous

Path Type: min

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.41 0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.01 0.91 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/
RESET_B (sky130_fd_sc_hd__dfirtp_1)
0.91 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/CLK
(sky130_fd_sc_hd__dfirtp_1)
0.45 0.45 library removal time
0.45 data required time

-----
0.45 data required time
-0.91 data arrival time

-----
0.47 slack (MET)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
(negative level-sensitive latch clocked by clk_i)

Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
(rising clock gating-check end-point clocked by clk_i)

Path Group: gated clock

Path Type: min

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
sky130_fd_sc_hd__dlxtn_1)
0.23 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
sky130_fd_sc_hd__dlxtn_1)
0.00 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
sky130_fd_sc_hd__and2_0)
2.73 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
```

```

0.00 2.50 clock reconvergence pessimism
      2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
          sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
      2.50 data required time
-----
      2.50 data required time
     -2.73 data arrival time
-----

      0.23 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22511_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:  core_i/ex_stage_i/mult_i/_22510_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22511_/CLK (sky130_fd_sc_hd__dfirtp_1)
0.36 0.36 ^ core_i/ex_stage_i/mult_i/_22511_/Q (sky130_fd_sc_hd__dfirtp_1)
0.00 0.36 ^ core_i/ex_stage_i/mult_i/_22510_/D (sky130_fd_sc_hd__dfirtp_1)
      0.36 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
      0.00 ^ core_i/ex_stage_i/mult_i/_22510_/CLK (sky130_fd_sc_hd__dfirtp_1)
-0.06 -0.06 library hold time
      -0.06 data required time
-----
      -0.06 data required time
     -0.36 data arrival time
-----

      0.42 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint:  core_i/cs_registers_i/_4932_
            (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)

```

```

0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.41 0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.58 1.49 ^ core_i/cs_registers_i/place14125/X (sky130_fd_sc_hd__buf_4)
0.74 2.23 ^ core_i/cs_registers_i/place14124/X (sky130_fd_sc_hd__buf_4)
0.02 2.25 ^ core_i/cs_registers_i/_4932_/RESET_B (sky130_fd_sc_hd__dftrtp_1)
    2.25 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/cs_registers_i/_4932_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.02 5.02 library recovery time
    5.02 data required time

-----
    5.02 data required time
   -2.25 data arrival time
-----

    2.76 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.69 4.19 time given to startpoint
0.00 4.19 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.22 4.40 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 4.40 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    4.40 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
    sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time

-----
    5.00 data required time
   -4.40 data arrival time

```

0.60 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22508_
(rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4716_
(rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22508_/CLK (sky130_fd_sc_hd__dfstp_1)
0.56 0.56 ^ core_i/ex_stage_i/mult_i/_22508_/Q (sky130_fd_sc_hd__dfstp_1)
0.15 0.71 ^ core_i/ex_stage_i/mult_i/_11259_/X (
sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.16 0.87 ^ core_i/ex_stage_i/mult_i/_place12084_/X (sky130_fd_sc_hd__buf_4)
0.06 0.93 v core_i/ex_stage_i/mult_i/_12041_/Y (sky130_fd_sc_hd__nor3_1)
0.15 1.08 v core_i/ex_stage_i/mult_i/_place11563_/X (sky130_fd_sc_hd__buf_4)
0.34 1.43 ^ core_i/ex_stage_i/mult_i/_13042_/Y (sky130_fd_sc_hd__nor4_1)
0.42 1.85 v core_i/ex_stage_i/mult_i/_13085_/X (sky130_fd_sc_hd__xnor3_1)
0.52 2.36 v core_i/ex_stage_i/mult_i/_13086_/X (sky130_fd_sc_hd__or4b_1)
0.41 2.77 v core_i/ex_stage_i/mult_i/_13090_/X (sky130_fd_sc_hd__maj3_1)
0.40 3.17 v core_i/ex_stage_i/mult_i/_13094_/X (sky130_fd_sc_hd__maj3_1)
0.19 3.35 v core_i/ex_stage_i/mult_i/_13109_/X (sky130_fd_sc_hd__xor2_1)
0.22 3.57 ^ core_i/ex_stage_i/mult_i/_13111_/Y (sky130_fd_sc_hd__a31oi_1)
0.18 3.75 ^ core_i/ex_stage_i/mult_i/_13112_/Y (sky130_fd_sc_hd__xnor2_1)
0.12 3.87 v core_i/ex_stage_i/mult_i/_13123_/Y (sky130_fd_sc_hd__a21oi_1)
0.39 4.26 v core_i/ex_stage_i/mult_i/_13125_/X (sky130_fd_sc_hd__maj3_1)
0.38 4.64 v core_i/ex_stage_i/mult_i/_13126_/X (sky130_fd_sc_hd__maj3_1)
0.22 4.87 ^ core_i/ex_stage_i/mult_i/_13128_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 5.00 v core_i/ex_stage_i/mult_i/_13130_/Y (sky130_fd_sc_hd__a21oi_1)
0.26 5.26 ^ core_i/ex_stage_i/mult_i/_13132_/Y (sky130_fd_sc_hd__o21ai_0)
0.15 5.41 v core_i/ex_stage_i/mult_i/_13134_/Y (sky130_fd_sc_hd__a21oi_1)
0.29 5.70 ^ core_i/ex_stage_i/mult_i/_13137_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 5.83 v core_i/ex_stage_i/mult_i/_13140_/Y (sky130_fd_sc_hd__a21oi_1)
0.28 6.11 ^ core_i/ex_stage_i/mult_i/_13142_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 6.23 v core_i/ex_stage_i/mult_i/_13145_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 6.46 v core_i/ex_stage_i/mult_i/_13148_/X (sky130_fd_sc_hd__o21a_1)
0.21 6.67 ^ core_i/ex_stage_i/mult_i/_13150_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 6.77 v core_i/ex_stage_i/mult_i/_13153_/Y (sky130_fd_sc_hd__a21oi_1)
0.21 6.98 ^ core_i/ex_stage_i/mult_i/_13155_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 7.10 v core_i/ex_stage_i/mult_i/_13157_/Y (sky130_fd_sc_hd__a21oi_1)
0.25 7.35 ^ core_i/ex_stage_i/mult_i/_13160_/Y (sky130_fd_sc_hd__o21ai_0)
0.16 7.50 v core_i/ex_stage_i/mult_i/_13162_/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.74 ^ core_i/ex_stage_i/mult_i/_13165_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.85 v core_i/ex_stage_i/mult_i/_13167_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 8.08 ^ core_i/ex_stage_i/mult_i/_13169_/Y (sky130_fd_sc_hd__o21ai_0)

```

0.12 8.20 v core_i/ex_stage_i/mult_i/_13171_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 8.43 ^ core_i/ex_stage_i/mult_i/_13174_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 8.54 v core_i/ex_stage_i/mult_i/_13176_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 8.76 ^ core_i/ex_stage_i/mult_i/_13179_/Y (sky130_fd_sc_hd__o21ai_0)
0.15 8.91 ^ core_i/ex_stage_i/mult_i/place7266/X (sky130_fd_sc_hd__buf_4)
0.14 9.05 ^ core_i/ex_stage_i/mult_i/_13241_/Y (sky130_fd_sc_hd__xnor2_1)
0.10 9.15 v core_i/ex_stage_i/mult_i/_13242_/Y (sky130_fd_sc_hd__clkinv_1)
0.46 9.61 v core_i/ex_stage_i/mult_i/_21734_/X (sky130_fd_sc_hd__mux4_2)
0.18 9.80 ^ core_i/ex_stage_i/mult_i/_21735_/Y (sky130_fd_sc_hd__mux2i_1)
0.16 9.95 ^ core_i/ex_stage_i/mult_i/_21920_/X (sky130_fd_sc_hd__a21o_1)
0.06 10.01 v core_i/ex_stage_i/mult_i/_22267_/Y (sky130_fd_sc_hd__a211oi_1)
0.52 10.54 v core_i/ex_stage_i/mult_i/_22276_/X (sky130_fd_sc_hd__or4_1)
0.27 10.80 ^ core_i/ex_stage_i/_156_/Y (sky130_fd_sc_hd__a22oi_1)
0.12 10.92 v core_i/ex_stage_i/_158_/Y (sky130_fd_sc_hd__o21ai_2)
0.18 11.10 ^ core_i/id_stage_i/_2829_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 11.21 v core_i/id_stage_i/_2830_/Y (sky130_fd_sc_hd__nand2_1)
0.19 11.40 ^ core_i/id_stage_i/_2834_/Y (sky130_fd_sc_hd__a22oi_1)
0.16 11.56 v core_i/id_stage_i/_2835_/Y (sky130_fd_sc_hd__nand3_1)
0.20 11.76 ^ core_i/id_stage_i/_3073_/Y (sky130_fd_sc_hd__o21ai_0)
0.08 11.84 v core_i/id_stage_i/_3074_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 12.07 v core_i/id_stage_i/_3529_/Y (sky130_fd_sc_hd__o21bai_1)
0.28 12.35 v core_i/id_stage_i/_3554_/X (sky130_fd_sc_hd__mux2_1)
0.00 12.35 v core_i/id_stage_i/_4716_/D (sky130_fd_sc_hd__dftrtp_1)
    12.35 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4716_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

-----

    4.87 data required time
   -12.35 data arrival time

-----

   -7.48 slack (VIOLATED)

wns max -7.48
tns max -9733.24

```

Metric	Baseline (Section 3.9–3.10)	Modified
wns max, ideal wire	−9.83 ns	−7.38 ns (25% better)
tns max, ideal wire	−13,280.61	−9,553.46
wns max, estimated wire	−9.83 ns	−7.48 ns (24% better)
tns max, estimated wire	−13,421.99	−9,733.24
Critical path location	alu_i	mult_i (untouched module) → id_stage_i

Table 9.2: Placement-stage timing: baseline vs. modified design.

Interpretation: whack-a-mole timing closure, now landing on the multiplier.

The critical path has moved a *third* time across this project’s stages: synthesis/floorplan (Chapter 3) pointed to `alu_i`; the modified design’s floorplan (Chapter 8) pointed to the register-file/jump-target path; and now, after timing-driven placement’s buffering was applied, that path closed enough that the worst path is a **completely untouched module**: `cv32e40p_mult`’s own internal carry-chain (a long run of `maj3_1` cells structurally similar to what the original ALU had, feeding into a `mux4_2/mux2i_1` result-select stage), now driving into `id_stage_i`. This is a textbook illustration of single-bottleneck optimization: closing one violation does not guarantee timing closure overall, it reveals whichever path is next-worst — here, a bottleneck this project’s RTL changes never touched. Despite this, the *overall* wns and tns numbers are meaningfully better than the baseline at the same stage (wns $-7.38/-7.48$ ns vs. -9.83 ns, roughly 24–25% improvement; tns roughly 28–29% improvement), showing the EX-stage split’s benefit persists through placement even though the specific worst path is now elsewhere.

9.3 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	1.27e-03	1.26e-04	3.29e-08	1.40e-03	21.9%
Combinational	2.05e-03	2.77e-03	1.00e-07	4.83e-03	75.6%
Clock	$-6.41e-07$	1.61e-04	1.87e-12	1.60e-04	2.5%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.32e-03 52.0%	3.06e-03 48.0%	1.33e-07 0.0%	6.38e-03	100.0%

Table 9.3: Power breakdown, modified design, placement stage (identical for both ideal- and estimated-wire reports).

Total power (6.38 mW) is again roughly half the baseline’s 12.7 mW (Section 3.9) at the same stage, continuing the pattern first observed in Chapter 8. This remains an **open, flagged discrepancy** rather than a confirmed benefit, for the same reason given in Section 8.3: without real VCD/SAIF-based switching activity, OpenSTA’s default activity estimates are known to be sensitive to netlist structure, and should not yet be read as a genuine power improvement from the pipeline split.

Chapter 10

Modified Design: Clock Tree Synthesis Results

10.1 CTS run and legalization

cts_modified.tcl was run on 2_placement_modified.odt. Clock tree synthesis and buffer legalization completed with 0 failures:

```
Total cells: 40601
Diamond Move Success: 40601 (100.00%)
Diamond Move Failure: 0
Rip-up and replace Success: 0 ( 0.00% of diamond failures)
Rip-up and replace Failure: 0
Total Placement Failures: 0
-----
Placement Analysis
-----
total displacement 4941.8 u
average displacement 0.1 u
max displacement 16.3 u
original HPWL 1219375.0 u
legalized HPWL 1223186.6 u
delta HPWL 0 %
```

Metric	Baseline (Section 4.4)	Modified
Total cells	40,513	40,601
Total / avg / max displacement	4224.0 / 0.1 / 9.9 u	4941.8 / 0.1 / 16.3 u
HPWL delta from legalization	0%	0%

Table 10.1: CTS-stage legalization: baseline vs. modified design.

10.2 Timing: the critical path returns to the register-file/PC path

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_
```

(removal check against rising-edge clock clk_i)

Path Group: asynchronous

Path Type: min

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (propagated)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.41 0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.01 0.91 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/
  RESET_B (sky130_fd_sc_hd__dftrtp_1)
  0.91 data arrival time

0.00 0.00 clock clk_i (rise edge)
1.60 1.60 clock network delay (propagated)
0.00 1.60 clock reconvergence pessimism
  1.60 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/CLK
    (sky130_fd_sc_hd__dftrtp_1)
0.47 2.07 library removal time
  2.07 data required time
-----
  2.07 data required time
-0.91 data arrival time
-----
-1.16 slack (VIOLATED)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_

(negative level-sensitive latch clocked by clk_i)

Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_

(rising clock gating-check end-point clocked by clk_i)

Path Group: gated clock

Path Type: min

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
1.55 4.05 clock network delay (propagated)
0.00 4.05 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
  sky130_fd_sc_hd__dlxtn_1)
0.31 4.36 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
  sky130_fd_sc_hd__dlxtn_1)
0.00 4.36 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
  sky130_fd_sc_hd__and2_0)
  4.36 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.22 2.72 clock network delay (propagated)
0.00 2.72 clock reconvergence pessimism
```

```

    2.72 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.72 clock gating hold time
    2.72 data required time
-----
    2.72 data required time
   -4.36 data arrival time
-----

    1.64 slack (MET)

Startpoint: irq_i[28] (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/int_controller_i/_138_
          (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
    0.00 0.00 clock clk_i (rise edge)
    0.00 0.00 clock network delay (propagated)
    0.50 0.50 v input external delay
    0.00 0.50 v irq_i[28] (in)
    0.00 0.50 v core_i/id_stage_i/int_controller_i/_138_/D (
        sky130_fd_sc_hd__dftrtp_1)
        0.50 data arrival time

    0.00 0.00 clock clk_i (rise edge)
    1.61 1.61 clock network delay (propagated)
    0.00 1.61 clock reconvergence pessimism
        1.61 ^ core_i/id_stage_i/int_controller_i/_138_/CLK (
            sky130_fd_sc_hd__dftrtp_1)
   -0.02 1.59 library hold time
        1.59 data required time
-----
        1.59 data required time
       -0.50 data arrival time
-----

       -1.08 slack (VIOLATED)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/_4561_
          (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
    0.00 0.00 clock clk_i (rise edge)
    0.00 0.00 clock network delay (propagated)

```

```

0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.41 0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.67 1.58 ^ core_i/place14152/X (sky130_fd_sc_hd__buf_4)
0.62 2.21 ^ core_i/place14151/X (sky130_fd_sc_hd__buf_4)
0.01 2.22 ^ core_i/id_stage_i/_4561_/RESET_B (sky130_fd_sc_hd__dfirtp_2)
      2.22 data arrival time

5.00 5.00 clock clk_i (rise edge)
1.53 6.53 clock network delay (propagated)
0.00 6.53 clock reconvergence pessimism
      6.53 ^ core_i/id_stage_i/_4561_/CLK (sky130_fd_sc_hd__dfirtp_2)
0.06 6.59 library recovery time
      6.59 data required time

-----
      6.59 data required time
     -2.22 data arrival time
-----

      4.37 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint:  core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
1.55 4.05 clock network delay (propagated)
1.75 5.80 time given to startpoint
0.00 5.80 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
      sky130_fd_sc_hd__dlxtn_1)
0.29 6.08 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
      sky130_fd_sc_hd__dlxtn_1)
0.00 6.09 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
      sky130_fd_sc_hd__and2_0)
      6.09 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.21 5.21 clock network delay (propagated)
0.00 5.21 clock reconvergence pessimism
      5.21 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
      sky130_fd_sc_hd__and2_0)
0.00 5.21 clock gating setup time
      5.21 data required time

-----
      5.21 data required time
     -6.09 data arrival time

```

-0.88 slack (VIOLATED)

Startpoint: core_i/if_stage_i/_930_

(rising edge-triggered flip-flop clocked by clk_i)

Endpoint: instr_addr_o[31] (output port clocked by clk_i)

Path Group: clk_i

Path Type: max

Delay Time Description

0.00 0.00 clock clk_i (rise edge)
1.52 1.52 clock network delay (propagated)
0.00 1.52 ^ core_i/if_stage_i/_930_/CLK (sky130_fd_sc_hd__dfrtp_4)
0.59 2.11 v core_i/if_stage_i/_930_/Q (sky130_fd_sc_hd__dfrtp_4)
0.45 2.56 ^ core_i/id_stage_i/register_file_i/_07894_/X (
sky130_fd_sc_hd__lpflow_isobufsrc_4)
0.18 2.74 v core_i/id_stage_i/register_file_i/_08702_/Y (
sky130_fd_sc_hd__a221oi_1)
0.36 3.10 ^ core_i/id_stage_i/register_file_i/_08705_/Y (
sky130_fd_sc_hd__o21ai_0)
0.25 3.35 v core_i/id_stage_i/register_file_i/_08711_/Y (
sky130_fd_sc_hd__a21oi_1)
0.41 3.76 ^ core_i/id_stage_i/register_file_i/_08725_/Y (
sky130_fd_sc_hd__a211oi_1)
0.12 3.88 v core_i/id_stage_i/_2230_/Y (sky130_fd_sc_hd__nand2_1)
0.23 4.11 ^ core_i/id_stage_i/_2233_/Y (sky130_fd_sc_hd__o21ai_0)
0.12 4.23 v core_i/id_stage_i/_2237_/Y (sky130_fd_sc_hd__a211oi_1)
0.40 4.63 ^ core_i/id_stage_i/_2240_/Y (sky130_fd_sc_hd__o311ai_0)
0.17 4.80 v core_i/id_stage_i/_2243_/Y (sky130_fd_sc_hd__a311oi_1)
0.39 5.18 v core_i/id_stage_i/_2244_/X (sky130_fd_sc_hd__or3_1)
0.39 5.57 v core_i/id_stage_i/_2247_/X (sky130_fd_sc_hd__a311o_1)
0.37 5.94 v core_i/id_stage_i/_2250_/X (sky130_fd_sc_hd__a311o_1)
0.25 6.18 ^ core_i/id_stage_i/_2257_/Y (sky130_fd_sc_hd__o31ai_1)
0.18 6.36 ^ core_i/id_stage_i/place8134/X (sky130_fd_sc_hd__buf_4)
0.09 6.45 v core_i/id_stage_i/_2267_/Y (sky130_fd_sc_hd__nand4_1)
0.18 6.63 ^ core_i/id_stage_i/_2272_/Y (sky130_fd_sc_hd__a31oi_1)
0.13 6.77 v core_i/id_stage_i/_2278_/Y (sky130_fd_sc_hd__nand4_1)
0.32 7.09 ^ core_i/id_stage_i/_2283_/Y (sky130_fd_sc_hd__a31oi_1)
0.17 7.26 v core_i/id_stage_i/_2288_/Y (sky130_fd_sc_hd__nand4_1)
0.25 7.51 ^ core_i/id_stage_i/_2294_/Y (sky130_fd_sc_hd__a311oi_1)
0.16 7.66 ^ core_i/id_stage_i/place7763/X (sky130_fd_sc_hd__buf_4)
0.15 7.81 ^ core_i/id_stage_i/_2295_/X (sky130_fd_sc_hd__maj3_1)
0.05 7.87 v core_i/id_stage_i/_2296_/Y (sky130_fd_sc_hd__nor2_1)
0.16 8.03 ^ core_i/id_stage_i/_2589_/Y (sky130_fd_sc_hd__mux2i_1)
0.14 8.17 ^ core_i/id_stage_i/_2590_/Y (sky130_fd_sc_hd__xnor2_1)
0.20 8.37 v core_i/id_stage_i/_2591_/Y (sky130_fd_sc_hd__o22ai_1)
0.35 8.73 ^ core_i/if_stage_i/_795_/Y (sky130_fd_sc_hd__a222oi_1)
0.10 8.83 v core_i/if_stage_i/_797_/Y (sky130_fd_sc_hd__nand2_1)
0.19 9.02 ^ core_i/if_stage_i/_798_/Y (sky130_fd_sc_hd__a21oi_1)

```

0.34 9.35 v core_i/if_stage_i/_799_/Y (sky130_fd_sc_hd__o311ai_0)
0.32 9.68 ^ core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_416_/Y
(sky130_fd_sc_hd__a22oi_1)
0.11 9.78 v core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_417_/Y
(sky130_fd_sc_hd__nand2_1)
0.36 10.15 v core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_146_/X (
sky130_fd_sc_hd__mux2_1)
0.00 10.15 v instr_addr_o[31] (out)
    10.15 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (propagated)
0.00 5.00 clock reconvergence pessimism
-3.00 2.00 output external delay
    2.00 data required time

-----
    2.00 data required time
   -10.15 data arrival time

-----

   -8.15 slack (VIOLATED)

wns max -8.15
tns max -9999.46
Clock clk_i
    1.61 source latency core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i
/_477_/CLK ^
-1.46 target latency core_i/sleep_unit_i/_6_/CLK ^
    0.00 CRPR

-----
    0.15 setup skew

```

Metric	Baseline (Section 4.5)	Modified
wns max	-9.86 ns	-8.15 ns (17% better)
tns max	-13,596.81	-9,999.46 (26% better)
Setup skew	0.19 ns	0.15 ns
Critical path location	alu_i	register_file_i → id_stage_i → if_stage_i/prefetch_buffer → instr_addr_o[31]

Table 10.2: CTS-stage timing: baseline vs. modified design.

The critical path moves a *fourth* time. After propagated clock latency and skew were introduced, the worst path moved away from the multiplier (Chapter 9) and back to essentially the same register-file/PC-generation path first seen at the modified design’s floorplan stage (Chapter 8) — now with a somewhat shorter data arrival time (10.15 ns vs. 19.46 ns there, since real placement and buffering have since been applied). This confirms that, for this modified design, the `alu_i/mult_i/register-file` paths are all now much closer in magnitude to one another than they were in the baseline (where `alu_i` dominated by a wide margin) — exactly the expected signature of successfully removing a

single dominant bottleneck: several previously-secondary paths now compete for “worst,” and small changes between stages (ideal vs. propagated clock, resizer decisions) can tip which one currently holds that position. The same three secondary violations seen in the baseline’s CTS stage (Section 4.5) reappear here with very similar magnitudes (removal check -1.16 ns vs. -1.37 ns baseline; hold check -1.08 ns vs. -1.04 ns; gated-clock max -0.88 ns vs. -0.93 ns baseline), reinforcing that these are a pre-existing, clock-tree-related characteristic largely independent of the EX-stage split.

10.3 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.64e-02	2.60e-03	3.29e-08	2.90e-02	21.1%
Combinational	4.40e-02	4.90e-02	1.00e-07	9.30e-02	67.7%
Clock	8.23e-03	7.10e-03	3.20e-09	1.53e-02	11.2%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	7.86e-02 57.2%	5.87e-02 42.8%	1.36e-07 0.0%	1.37e-01	100.0%

Table 10.3: Power breakdown, modified design, CTS stage.

Total power (137 mW) is again lower than the baseline’s 210 mW at the same stage (Section 4.6), here by about 35% rather than the roughly 50% reduction seen at the floorplan and placement stages (Chapters 8–9) — the gap narrows somewhat once the real clock tree (177+ buffers) is added, since clock power is comparatively similar in both designs. This remains part of the same **open, flagged discrepancy** noted since Chapter 8 rather than a settled conclusion.

Chapter 11

Modified Design: Post-CTS Timing Repair Results

11.1 A note on a path typo

The first `write_def` in this run failed (Failed to open `.../orfs_outt/3_cts_repaired.def.1`) due to a typo in the output directory name (`orfs_outt` instead of `orfs_out`). This was corrected in the same live session by re-running `write_def/write_db` with the correct path, which succeeded — no repair work was lost, and the reports below are from the correctly-saved database.

11.2 Repair progress log

```
Repair move sequence: UnbufferMove SizeUpMove SwapPinsMove BufferMove CloneMove SplitLoadMove
[INFO RSZ-0094] Found 2117 endpoints with setup violations.
[INFO RSZ-0099] Repairing 2117 out of 2117 (100.00%) violating endpoints...
  Iter | Removed | Resized | Inserted | Cloned | Pin | Area | WNS | StTNS | EnTNS | Viol | Worst
        | Buffers | Gates | Buffers | Gates | Swaps | | | | Endpts | St/EnPt
-----
  0*   | 0 | 0 | 0 | 0 | 0 | 0 | +0.0% | -7.485 | -5948.3 | -9749.1 | 2117 | core_i/id_stage_i/_4716_/D
1200* | 5 | 315 | 78 | 1 | 225 | +0.5% | -5.775 | -5320.9 | -7828.4 | 2117 | core_i/id_stage_i/_4716_/D
1200+ | 5 | 315 | 78 | 1 | 225 | +0.5% | -5.775 | -5320.9 | -7828.4 | 2117 | core_i/id_stage_i/_4716_/D
final | 5 | 370 | 78 | 1 | 247 | +0.6% | -5.774 | -5287.1 | -7749.0 | 2115 | core_i/id_stage_i/_4716_/D
-----
[INFO RSZ-0059] Removed 5 buffers.
[INFO RSZ-0045] Inserted 78 buffers, 1 to split loads.
[INFO RSZ-0051] Resized 370 instances: 370 up, 0 up match, 0 down, 0 VT
[INFO RSZ-0043] Swapped pins on 247 instances.
[INFO RSZ-0049] Cloned 1 instances.
[WARNING RSZ-0062] Unable to repair all setup violations.
[INFO RSZ-0033] No hold violations found.
```

Metric	Baseline (Section 4.7)	Modified
Violating endpoints found	2,056	2,117
Resized / inserted buffers / cloned / pin swaps	461 / 36 / 5 / 219	370 / 78 / 1 / 247
wns: start → final	−9.826 → −8.450 ns (14%)	−7.485 → −5.774 ns (23%)
Endpoints remaining violated	2,056 (unchanged)	2,115 (2 fewer)

Table 11.1: Post-CTS repair effort: baseline vs. modified design.

The modified design’s repair achieved a *larger relative* wns improvement (23% vs. 14%) with a different mix of moves (more buffer insertions, fewer resizes) — consistent with a design whose worst paths are now shorter and more numerous rather than one dominated by a single very long chain. As in the baseline, the tool explicitly reports [WARNING

RSZ-0062] Unable to repair all setup violations, and [INFO RSZ-0033] No hold violations found.

11.3 Full post-repair timing report

Startpoint: rst_ni (input port clocked by clk_i)	
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_	
(removal check against rising-edge clock clk_i)	
Path Group: asynchronous	
Path Type: min	
Delay Time Description	

0.00	0.00 clock clk_i (rise edge)
0.00	0.00 clock network delay (ideal)
0.50	0.50 ^ input external delay
0.00	0.50 ^ rst_ni (in)
0.41	0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.01	0.91 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/ RESET_B (sky130_fd_sc_hd__dfrtp_1)
	0.91 data arrival time
0.00	0.00 clock clk_i (rise edge)
0.00	0.00 clock network delay (ideal)
0.00	0.00 clock reconvergence pessimism
	0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_291_/CLK (sky130_fd_sc_hd__dfrtp_1)
0.45	0.45 library removal time
	0.45 data required time

	0.45 data required time
	-0.91 data arrival time

	0.47 slack (MET)
Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_	
(negative level-sensitive latch clocked by clk_i)	
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_	
(rising clock gating-check end-point clocked by clk_i)	
Path Group: gated clock	
Path Type: min	
Delay Time Description	

2.50	2.50 clock clk_i (fall edge)
0.00	2.50 clock network delay (ideal)
0.00	2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (sky130_fd_sc_hd__dlxtn_1)
0.29	2.79 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (sky130_fd_sc_hd__dlxtn_1)

```

0.00 2.80 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    2.80 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
    2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
    2.50 data required time

-----
    2.50 data required time
    -2.80 data arrival time
-----

    0.30 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22511_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/ex_stage_i/mult_i/_22510_
    (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22511_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.36 0.36 ^ core_i/ex_stage_i/mult_i/_22511_/Q (sky130_fd_sc_hd__dftrtp_1)
0.00 0.36 ^ core_i/ex_stage_i/mult_i/_22510_/D (sky130_fd_sc_hd__dftrtp_1)
    0.36 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/ex_stage_i/mult_i/_22510_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.06 -0.06 library hold time
    -0.06 data required time

-----
    -0.06 data required time
    -0.36 data arrival time
-----

    0.42 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/cs_registers_i/_4932_
    (recovery check against rising-edge clock clk_i)
Path Group: asynchronous

```

Path Type: max

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.41 0.91 ^ core_i/wire273/X (sky130_fd_sc_hd__buf_16)
0.58 1.49 ^ core_i/cs_registers_i/place14125/X (sky130_fd_sc_hd__buf_4)
0.74 2.23 ^ core_i/cs_registers_i/place14124/X (sky130_fd_sc_hd__buf_4)
0.02 2.25 ^ core_i/cs_registers_i/_4932_/RESET_B (sky130_fd_sc_hd__dfrrtp_1)
      2.25 data arrival time
```

```
5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
      5.00 ^ core_i/cs_registers_i/_4932_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.02 5.02 library recovery time
      5.02 data required time
```

```
-----
      5.02 data required time
     -2.25 data arrival time
```

```
-----
      2.77 slack (MET)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_

(negative level-sensitive latch clocked by clk_i)

Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_

(rising clock gating-check end-point clocked by clk_i)

Path Group: gated clock

Path Type: max

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.69 4.19 time given to startpoint
0.00 4.19 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
      sky130_fd_sc_hd__dlxtn_1)
0.29 4.48 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
      sky130_fd_sc_hd__dlxtn_1)
0.00 4.48 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
      sky130_fd_sc_hd__and2_0)
      4.48 data arrival time
```

```
5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
```

```

    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time
-----
    5.00 data required time
   -4.48 data arrival time
-----

    0.52 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22508_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:  core_i/id_stage_i/_4716_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22508_/CLK (sky130_fd_sc_hd__dfstp_2)
0.57 0.57 ^ core_i/ex_stage_i/mult_i/_22508_/Q (sky130_fd_sc_hd__dfstp_2)
0.15 0.72 ^ core_i/ex_stage_i/mult_i/_11259_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.17 0.89 ^ core_i/ex_stage_i/mult_i/_place12084_/X (sky130_fd_sc_hd__buf_4)
0.08 0.97 v core_i/ex_stage_i/mult_i/_11439_/Y (sky130_fd_sc_hd__nor3_4)
0.27 1.24 ^ core_i/ex_stage_i/mult_i/_12990_/Y (sky130_fd_sc_hd__nor4_4)
0.41 1.65 v core_i/ex_stage_i/mult_i/_13019_/X (sky130_fd_sc_hd__xnor3_1)
0.25 1.90 ^ core_i/ex_stage_i/mult_i/_13024_/Y (sky130_fd_sc_hd__xnor2_2)
0.11 2.01 v core_i/ex_stage_i/mult_i/_13046_/Y (sky130_fd_sc_hd__xnor2_1)
0.33 2.35 v core_i/ex_stage_i/mult_i/_13050_/X (sky130_fd_sc_hd__maj3_2)
0.37 2.71 v core_i/ex_stage_i/mult_i/_13053_/X (sky130_fd_sc_hd__xor3_1)
0.15 2.86 v core_i/ex_stage_i/mult_i/_place9013_/X (sky130_fd_sc_hd__buf_4)
0.11 2.97 v core_i/ex_stage_i/mult_i/_13055_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_4)
0.38 3.36 v core_i/ex_stage_i/mult_i/_13057_/X (sky130_fd_sc_hd__xnor3_1)
0.17 3.53 v core_i/ex_stage_i/mult_i/_13080_/X (sky130_fd_sc_hd__xor2_2)
0.16 3.68 v core_i/ex_stage_i/mult_i/_13099_/Y (sky130_fd_sc_hd__xnor2_1)
0.33 4.01 v core_i/ex_stage_i/mult_i/_13125_/X (sky130_fd_sc_hd__maj3_2)
0.33 4.34 v core_i/ex_stage_i/mult_i/_13126_/X (sky130_fd_sc_hd__maj3_2)
0.15 4.49 ^ core_i/ex_stage_i/mult_i/_13128_/Y (sky130_fd_sc_hd__o21ai_2)
0.09 4.59 v core_i/ex_stage_i/mult_i/_13130_/Y (sky130_fd_sc_hd__a21oi_2)
0.17 4.75 ^ core_i/ex_stage_i/mult_i/_13132_/Y (sky130_fd_sc_hd__o21ai_2)
0.10 4.85 v core_i/ex_stage_i/mult_i/_13134_/Y (sky130_fd_sc_hd__a21oi_2)
0.17 5.02 ^ core_i/ex_stage_i/mult_i/_13137_/Y (sky130_fd_sc_hd__o21ai_2)
0.09 5.11 v core_i/ex_stage_i/mult_i/_13140_/Y (sky130_fd_sc_hd__a21oi_2)
0.16 5.28 ^ core_i/ex_stage_i/mult_i/_13142_/Y (sky130_fd_sc_hd__o21ai_2)
0.08 5.36 v core_i/ex_stage_i/mult_i/_13145_/Y (sky130_fd_sc_hd__a21oi_2)
0.19 5.55 v core_i/ex_stage_i/mult_i/_13148_/X (sky130_fd_sc_hd__o21a_1)

```

```

0.15 5.70 ^ core_i/ex_stage_i/mult_i/_13150_/Y (sky130_fd_sc_hd__o21ai_1)
0.10 5.80 v core_i/ex_stage_i/mult_i/_13153_/Y (sky130_fd_sc_hd__a21oi_1)
0.16 5.96 ^ core_i/ex_stage_i/mult_i/_13155_/Y (sky130_fd_sc_hd__o21ai_1)
0.11 6.07 v core_i/ex_stage_i/mult_i/_13157_/Y (sky130_fd_sc_hd__a21oi_1)
0.21 6.28 ^ core_i/ex_stage_i/mult_i/_13160_/Y (sky130_fd_sc_hd__o21ai_1)
0.11 6.39 v core_i/ex_stage_i/mult_i/_13162_/Y (sky130_fd_sc_hd__a21oi_2)
0.16 6.55 ^ core_i/ex_stage_i/mult_i/_13165_/Y (sky130_fd_sc_hd__o21ai_1)
0.10 6.66 v core_i/ex_stage_i/mult_i/_13167_/Y (sky130_fd_sc_hd__a21oi_1)
0.20 6.85 ^ core_i/ex_stage_i/mult_i/_13169_/Y (sky130_fd_sc_hd__o21ai_1)
0.09 6.94 v core_i/ex_stage_i/mult_i/_13171_/Y (sky130_fd_sc_hd__a21oi_2)
0.15 7.10 ^ core_i/ex_stage_i/mult_i/_13174_/Y (sky130_fd_sc_hd__o21ai_1)
0.10 7.20 v core_i/ex_stage_i/mult_i/_13176_/Y (sky130_fd_sc_hd__a21oi_1)
0.15 7.35 ^ core_i/ex_stage_i/mult_i/_13179_/Y (sky130_fd_sc_hd__o21ai_1)
0.11 7.46 v core_i/ex_stage_i/mult_i/_13181_/Y (sky130_fd_sc_hd__a21oi_1)
0.18 7.63 v core_i/ex_stage_i/mult_i/_13226_/X (sky130_fd_sc_hd__xor2_1)
0.48 8.11 v core_i/ex_stage_i/mult_i/_21734_/X (sky130_fd_sc_hd__mux4_2)
0.18 8.30 ^ core_i/ex_stage_i/mult_i/_21735_/Y (sky130_fd_sc_hd__mux2i_1)
0.15 8.45 ^ core_i/ex_stage_i/mult_i/_21920_/X (sky130_fd_sc_hd__a21o_1)
0.06 8.51 v core_i/ex_stage_i/mult_i/_22267_/Y (sky130_fd_sc_hd__a211oi_1)
0.49 9.00 v core_i/ex_stage_i/mult_i/_22276_/X (sky130_fd_sc_hd__or4_1)
0.21 9.21 ^ core_i/ex_stage_i/_156_/Y (sky130_fd_sc_hd__a22oi_2)
0.11 9.31 v core_i/ex_stage_i/_158_/Y (sky130_fd_sc_hd__o21ai_2)
0.16 9.47 ^ core_i/id_stage_i/_2829_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 9.58 v core_i/id_stage_i/_2830_/Y (sky130_fd_sc_hd__nand2_1)
0.18 9.75 ^ core_i/id_stage_i/_2834_/Y (sky130_fd_sc_hd__a22oi_1)
0.17 9.92 v core_i/id_stage_i/_2835_/Y (sky130_fd_sc_hd__nand3_1)
0.16 10.08 ^ core_i/id_stage_i/_3073_/Y (sky130_fd_sc_hd__o21ai_1)
0.08 10.15 v core_i/id_stage_i/_3074_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 10.38 v core_i/id_stage_i/_3529_/Y (sky130_fd_sc_hd__o21bai_4)
0.27 10.65 v core_i/id_stage_i/_3554_/X (sky130_fd_sc_hd__mux2_4)
0.00 10.65 v core_i/id_stage_i/_4716_/D (sky130_fd_sc_hd__dftrtp_1)

```

10.65 data arrival time

```

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
-0.13 4.87 library setup time
4.87 data required time

```

```

-----
4.87 data required time
-10.65 data arrival time
-----

```

-5.77 slack (VIOLATED)

```

wns max -5.77
tns max -7749.03
Clock clk_i
1.26 source latency core_i/id_stage_i/controller_i/_608_/CLK ^
-1.13 target latency core_i/id_stage_i/_4386_/CLK ^
0.00 CRPR

```

0.13 setup skew

Metric	Baseline (Section 4.7–4.8)	Modified
wns max	−8.45 ns	−5.77 ns (32% better)
tns max	−11,756.31	−7,749.03 (34% better)
Setup skew	0.20 ns	0.13 ns
Critical path location	alu_i	mult_i (untouched module) → id_stage_i

Table 11.2: Post-repair timing: baseline vs. modified design.

Critical path settles on the multiplier. After repair, the worst path is again through `cv32e40p_mult` (same as the placement stage, Chapter 9) rather than the register-file/PC path seen at CTS (Chapter 10) — the repair pass’s buffering evidently helped the register-file path more than the multiplier’s, echoing the “several comparable secondary paths” pattern already noted in Chapter 10. The three secondary violations (removal, hold, gated-clock) are absent from this report’s worst-6 listing in the same pattern as previous stages, consistent with earlier findings that these are a separate, pre-existing characteristic.

11.4 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.68e-02	2.61e-03	3.29e-08	2.94e-02	21.4%
Combinational	4.46e-02	4.95e-02	1.01e-07	9.42e-02	68.6%
Clock	8.53e-03	5.18e-03	3.20e-09	1.37e-02	10.0%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	8.00e-02 58.3%	5.73e-02 41.7%	1.37e-07 0.0%	1.37e-01	100.0%

Table 11.3: Power breakdown, modified design, post-repair.

Total power (137 mW) is essentially unchanged from the CTS stage (Chapter 10), matching the same pattern seen in the baseline (Section 4.7), where `repair_timing` similarly left total power unchanged. It remains roughly 35% below the baseline’s 210 mW at the equivalent stage — part of the same open discrepancy flagged since Chapter 8.

11.5 Updated cumulative comparison, all stages

Stage	Baseline wns	Modified wns	Baseline tns	Modified tns
Floorplan	−20.23 ns	−17.46 ns	−30,884.59	−30,642.40
Placement (estimated)	−9.83 ns	−7.48 ns	−13,421.99	−9,733.24
CTS	−9.86 ns	−8.15 ns	−13,596.81	−9,999.46
CTS + repair_timing	−8.45 ns	−5.77 ns	−11,756.31	−7,749.03

Table 11.4: wns/tns across every stage completed for both designs.

Stage	Baseline area	Modified area	Change
Synthesis (chip area)	248,230.57 μm^2	264,114.56 μm^2	+6.4%
Placement (instances area)	307,562.48 μm^2	320,596.23 μm^2	+4.2%

Table 11.5: Area across stages, both designs. The CTS-stage row is intentionally omitted here (the modified design’s instances area was not captured at that stage); a post-global-route area comparison, using `report_design_area` for both designs, follows in the routing chapter instead.

Stage	Baseline power	Modified power	Change
Floorplan	8.69 mW	4.36 mW	−50%
Placement (estimated)	12.70 mW	6.38 mW	−50%
CTS	210.1 mW	137.0 mW	−35%
CTS + repair_timing	210.1 mW	137.0 mW	−35%

Table 11.6: Total power across stages, both designs. As discussed in Chapters 8–11, this reduction is an **open, unconfirmed discrepancy** rather than a validated result — see the caveat below.

Reading the area and power tables together with the timing table. The area table shows the modified design’s real, measured physical cost of the EX-stage split: +6.4% chip area at synthesis, narrowing to +4.2% by placement (the baseline’s own timing-driven resizer adds relatively more buffering to compensate for its more severe violations, partly closing the gap). The power table’s large, consistent reduction (35–50% at every stage) is **not treated as a confirmed power benefit of this design**: as flagged repeatedly since Chapter 8, these are default (non-workload) OpenSTA switching-activity estimates in the absence of real VCD/SAIF data, and they are known to be sensitive to netlist and clock-tree structure — a real activity-based power analysis would be needed before drawing conclusions from this particular number. The timing improvement (previous table) is the most load-bearing result of the three, since wns/tns are derived directly from the library’s real timing arcs and the actual (estimated or extracted) parasitics, not from an activity assumption.

Overall conclusion for the modified design. At every single stage of the physical design flow, the modified (EX-stage-split) design’s wns and tns are meaningfully better than the baseline’s — improvements ranging from roughly 14% (floorplan) to over 30% (post-repair) on wns, and similar or larger improvements on tns. This is despite the fact that `cv32e40p_alu` stopped being the critical path after the very first stage: the split’s benefit is visible in the *aggregate* timing metrics throughout the flow, even as the specific worst-path location moved between the register file/PC-generation logic and the (untouched) multiplier at different stages. The cost of this improvement is twofold, and the two costs are measured very differently: (1) a **physically measured** area cost, +6.4% chip area at synthesis (Chapter 7), concentrated almost entirely in `cv32e40p_alu` itself and narrowing to +4.2% by placement; and (2) an **architectural, not-yet-measured** cost — every non-divide ALU operation now takes 2 clock cycles instead of 1 (Chapter 6.3), which increases CPI (cycles per instruction) for ALU-heavy code in a way this physical-design flow cannot quantify on its own. Whether the net trade-off (better fmax, more area, worse CPI) is worthwhile depends on the target workload’s mix of ALU operations vs. other instruction types — a question outside the scope of the physical-design comparison presented here, but a natural next step (e.g. via cycle-accurate simulation of representative benchmarks) if this project is continued.

Chapter 12

Post-Global-Route Results

12.1 Note on scope and methodology

Both the baseline’s and the modified design’s `detailed_route` runs hung without converging in reasonable time (Chapter 5 already documented the baseline’s severe routing congestion — tens of thousands of DRC violations after the first optimization iteration). Rather than wait indefinitely, `global_route` was run (without `detailed_route`) for both designs, followed by `report_checks`, `report_power`, and `report_design_area`. This is explicitly a **post-global-route, pre-detailed-route** data point — routing topology and estimated parasitics (via `estimate_parasitics -global_routing`) are more accurate than the placement-stage HPWL-based estimate, but this is still not signoff-quality (no extracted RC from real wire geometry, and DRC-clean routing was never achieved). It is presented as an additional, honestly-labelled comparison point, not a final result, and the same methodology is applied to both designs for a fair comparison.

12.2 Baseline: timing

```
Startpoint: core_i/id_stage_i/_4564_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4478_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4564_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.40 0.40 v core_i/id_stage_i/_4564_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.39 0.78 ^ core_i/ex_stage_i/alu_i/_3675_/Y (sky130_fd_sc_hd__nor4bb_4)
0.11 0.89 v core_i/ex_stage_i/alu_i/_4050_/Y (sky130_fd_sc_hd__a21oi_1)
0.21 1.10 ^ core_i/ex_stage_i/alu_i/_4053_/Y (sky130_fd_sc_hd__o31ai_1)
0.17 1.27 ^ core_i/ex_stage_i/alu_i/place12015/X (sky130_fd_sc_hd__buf_4)
0.12 1.39 ^ core_i/ex_stage_i/alu_i/_4054_/Y (sky130_fd_sc_hd__xnor2_2)
```

```

0.09 1.48 v core_i/ex_stage_i/alu_i/_4055_/Y (sky130_fd_sc_hd__a32oi_2)
0.36 1.84 v core_i/ex_stage_i/alu_i/_4057_/X (sky130_fd_sc_hd__maj3_2)
0.33 2.17 v core_i/ex_stage_i/alu_i/_4059_/X (sky130_fd_sc_hd__maj3_2)
0.31 2.49 v core_i/ex_stage_i/alu_i/_4061_/X (sky130_fd_sc_hd__maj3_2)
0.32 2.81 v core_i/ex_stage_i/alu_i/_4063_/X (sky130_fd_sc_hd__maj3_2)
0.32 3.13 v core_i/ex_stage_i/alu_i/_4065_/X (sky130_fd_sc_hd__maj3_2)
0.32 3.45 v core_i/ex_stage_i/alu_i/_4067_/X (sky130_fd_sc_hd__maj3_2)
0.32 3.77 v core_i/ex_stage_i/alu_i/_4069_/X (sky130_fd_sc_hd__maj3_2)
0.17 3.93 ^ core_i/ex_stage_i/alu_i/_4070_/Y (sky130_fd_sc_hd__o21ai_2)
0.13 4.06 ^ core_i/ex_stage_i/alu_i/place9006/X (sky130_fd_sc_hd__buf_4)
0.19 4.25 ^ core_i/ex_stage_i/alu_i/_4072_/X (sky130_fd_sc_hd__maj3_1)
0.16 4.41 ^ core_i/ex_stage_i/alu_i/_4075_/X (sky130_fd_sc_hd__a21o_1)
0.12 4.52 ^ core_i/ex_stage_i/alu_i/place8855/X (sky130_fd_sc_hd__buf_4)
0.07 4.59 v core_i/ex_stage_i/alu_i/_4079_/Y (sky130_fd_sc_hd__a211oi_1)
0.42 5.01 ^ core_i/ex_stage_i/alu_i/_4082_/Y (sky130_fd_sc_hd__o311ai_0)
0.19 5.21 v core_i/ex_stage_i/alu_i/_4085_/Y (sky130_fd_sc_hd__a311oi_1)
0.34 5.54 ^ core_i/ex_stage_i/alu_i/_4088_/Y (sky130_fd_sc_hd__o311ai_0)
0.19 5.73 ^ core_i/ex_stage_i/alu_i/place8614/X (sky130_fd_sc_hd__buf_4)
0.13 5.86 v core_i/ex_stage_i/alu_i/_4091_/Y (sky130_fd_sc_hd__a311oi_1)
0.45 6.31 ^ core_i/ex_stage_i/alu_i/_4094_/Y (sky130_fd_sc_hd__o311ai_0)
0.21 6.51 v core_i/ex_stage_i/alu_i/_4098_/Y (sky130_fd_sc_hd__a311oi_1)
0.21 6.72 ^ core_i/ex_stage_i/alu_i/_4100_/Y (sky130_fd_sc_hd__o31ai_1)
0.19 6.91 v core_i/ex_stage_i/alu_i/_4101_/Y (sky130_fd_sc_hd__a21boi_0)
0.30 7.21 ^ core_i/ex_stage_i/alu_i/_4271_/X (sky130_fd_sc_hd__xor2_1)
0.25 7.46 ^ core_i/ex_stage_i/alu_i/_4273_/X (sky130_fd_sc_hd__xor2_1)
0.07 7.53 v core_i/ex_stage_i/alu_i/_4454_/Y (sky130_fd_sc_hd__a311oi_2)
0.34 7.87 ^ core_i/ex_stage_i/alu_i/_4456_/Y (sky130_fd_sc_hd__o31ai_1)
0.17 8.04 v core_i/ex_stage_i/alu_i/_4458_/Y (sky130_fd_sc_hd__o211ai_1)
0.20 8.25 v core_i/ex_stage_i/alu_i/_4459_/X (sky130_fd_sc_hd__and3_1)
0.36 8.60 v core_i/ex_stage_i/alu_i/_4461_/X (sky130_fd_sc_hd__o311a_1)
0.13 8.73 ^ core_i/ex_stage_i/alu_i/_4472_/Y (sky130_fd_sc_hd__o21ai_1)
0.25 8.99 ^ core_i/ex_stage_i/alu_i/_4475_/X (sky130_fd_sc_hd__a31o_1)
0.15 9.13 v core_i/ex_stage_i/alu_i/_4477_/Y (sky130_fd_sc_hd__a21oi_4)
0.19 9.32 v clone14972/X (sky130_fd_sc_hd__clkbuf_16)
0.17 9.49 ^ core_i/ex_stage_i/alu_i/_4868_/Y (sky130_fd_sc_hd__o21ai_1)
0.11 9.60 v core_i/ex_stage_i/alu_i/_5050_/Y (sky130_fd_sc_hd__a21oi_1)
0.18 9.78 ^ core_i/ex_stage_i/alu_i/_5184_/Y (sky130_fd_sc_hd__o21ai_1)
0.11 9.90 v core_i/ex_stage_i/alu_i/_5189_/Y (sky130_fd_sc_hd__a21oi_1)
0.36 10.25 ^ core_i/ex_stage_i/alu_i/_5197_/Y (sky130_fd_sc_hd__o311ai_1)
0.37 10.62 v core_i/ex_stage_i/alu_i/_5202_/Y (sky130_fd_sc_hd__nand4_1)
0.26 10.88 v core_i/ex_stage_i/alu_i/place7737/X (sky130_fd_sc_hd__buf_4)
0.21 11.10 ^ core_i/ex_stage_i/alu_i/_5821_/Y (sky130_fd_sc_hd__mux2i_1)
0.13 11.23 v core_i/ex_stage_i/alu_i/_5826_/Y (sky130_fd_sc_hd__o22ai_1)
0.32 11.54 ^ core_i/ex_stage_i/alu_i/_5828_/Y (sky130_fd_sc_hd__nor4_2)
0.07 11.62 v core_i/ex_stage_i/alu_i/_5829_/Y (sky130_fd_sc_hd__a211oi_2)
0.15 11.77 ^ core_i/ex_stage_i/alu_i/_5831_/Y (sky130_fd_sc_hd__o21ai_2)
0.16 11.93 ^ core_i/ex_stage_i/alu_i/rebuffer14958/X (sky130_fd_sc_hd__buf_6)
0.15 12.09 ^ core_i/ex_stage_i/alu_i/place7645/X (sky130_fd_sc_hd__buf_8)
0.08 12.17 v core_i/ex_stage_i/alu_i/_6967_/Y (sky130_fd_sc_hd__a221oi_1)
0.65 12.82 v core_i/ex_stage_i/alu_i/_6995_/X (sky130_fd_sc_hd__or4b_1)
0.21 13.02 ^ core_i/ex_stage_i/_176_/Y (sky130_fd_sc_hd__a22oi_2)

```

```

0.17 13.19 v core_i/ex_stage_i/_178_/Y (sky130_fd_sc_hd__o21ai_2)
0.26 13.45 v core_i/id_stage_i/_3872_/X (sky130_fd_sc_hd__a22o_1)
0.27 13.71 ^ core_i/id_stage_i/_3873_/Y (sky130_fd_sc_hd__a21oi_1)
0.08 13.79 v core_i/id_stage_i/_3874_/Y (sky130_fd_sc_hd__nor2_1)
0.25 14.04 ^ core_i/id_stage_i/_3881_/Y (sky130_fd_sc_hd__a211oi_1)
0.23 14.27 v core_i/id_stage_i/_3882_/Y (sky130_fd_sc_hd__o21ai_0)
0.47 14.74 ^ core_i/id_stage_i/_3883_/Y (sky130_fd_sc_hd__a21boi_1)
0.13 14.88 v core_i/id_stage_i/_3887_/Y (sky130_fd_sc_hd__o21ai_1)
0.00 14.88 v core_i/id_stage_i/_4478_/D (sky130_fd_sc_hd__dftrtp_1)
    14.88 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
-0.15 4.85 library setup time
    4.85 data required time
-----
    4.85 data required time
   -14.88 data arrival time
-----
   -10.03 slack (VIOLATED)

wns max -10.03
tns max -13541.69
Clock clk_i
    1.63 source latency core_i/cs_registers_i/_4624_/CLK ^
   -1.35 target latency core_i/id_stage_i/register_file_i/_13237_/CLK ^
    0.00 CRPR
-----
    0.28 setup skew

```

Metric	Baseline, post-global-route
wns max	-10.03 ns
tns max	-13,541.69
Setup skew	0.28 ns
Critical path location	alu_i (unchanged from every prior baseline stage)

Table 12.1: Baseline post-global-route timing summary.

12.3 Baseline: power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.18e-02	1.93e-03	2.96e-08	2.37e-02	10.7%
Combinational	8.44e-02	9.84e-02	1.01e-07	1.83e-01	82.6%
Clock	7.70e-03	7.16e-03	2.86e-09	1.49e-02	6.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	1.14e-01 51.5%	1.07e-01 48.5%	1.33e-07 0.0%	2.21e-01	100.0%

Table 12.2: Baseline post-global-route power breakdown (221 mW total).

12.4 Baseline: area

Design area 314907 um² 51% utilization.

This `report_design_area` figure (314,907 μm^2 , 51% utilization) is a direct, explicit area report at this stage, filling the gap left in Section 11.6 (where the modified design's CTS-stage instances area was not captured). The modified design's equivalent post-global-route figures follow once available, for a like-for-like comparison at this stage.

12.5 Modified design: timing

The excerpt below picks up mid-report (the gated-clock max path already in progress); the earlier paths (removal check, gated-clock min, hold check) from this run were not captured and are not reproduced here rather than reconstructed.

Delay Time Description

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.94 4.44 time given to startpoint
0.00 4.44 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (sky130_fd_sc_hd__dlxtn_1)
0.28 4.72 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (sky130_fd_sc_hd__dlxtn_1)
0.00 4.72 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (sky130_fd_sc_hd__and2_0)
4.72 data arrival time
5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
5.00 data required time

5.00 data required time
-4.72 data arrival time

0.28 slack (MET)
Startpoint: core_i/ex_stage_i/mult_i/_22508_ (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4716_ (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max
Delay Time Description

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22508_/CLK (sky130_fd_sc_hd__dfstp_2)
0.38 0.38 v core_i/ex_stage_i/mult_i/_22508_/Q (sky130_fd_sc_hd__dfstp_2)
0.14 0.52 v core_i/ex_stage_i/mult_i/_11259_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.17 0.68 v core_i/ex_stage_i/mult_i/place12084/X (sky130_fd_sc_hd__buf_4)
0.14 0.82 v core_i/ex_stage_i/mult_i/rebuffer14161/X (sky130_fd_sc_hd__buf_4)
0.24 1.06 ^ core_i/ex_stage_i/mult_i/_11261_/Y (sky130_fd_sc_hd__nor2_2)
0.19 1.25 ^ core_i/ex_stage_i/mult_i/place11577/X (sky130_fd_sc_hd__buf_4)
0.31 1.56 ^ core_i/ex_stage_i/mult_i/_11387_/Y (sky130_fd_sc_hd__mux2i_2)
0.19 1.75 ^ core_i/ex_stage_i/mult_i/place11129/X (sky130_fd_sc_hd__buf_4)
0.17 1.92 ^ core_i/ex_stage_i/mult_i/place11128/X (sky130_fd_sc_hd__buf_12)
0.08 2.00 v core_i/ex_stage_i/mult_i/_12829_/Y (sky130_fd_sc_hd__nor2_1)
0.19 2.20 v core_i/ex_stage_i/mult_i/_12832_/X (sky130_fd_sc_hd__a21o_1)
0.20 2.40 ^ core_i/ex_stage_i/mult_i/_12838_/Y (sky130_fd_sc_hd__xnor2_1)
0.25 2.65 ^ core_i/ex_stage_i/mult_i/_12886_/X (sky130_fd_sc_hd__xor2_1)
0.40 3.05 v core_i/ex_stage_i/mult_i/_12933_/X (sky130_fd_sc_hd__xor3_1)
0.21 3.25 ^ core_i/ex_stage_i/mult_i/_12934_/Y (sky130_fd_sc_hd__xnor2_1)
0.13 3.38 v core_i/ex_stage_i/mult_i/_12935_/Y (sky130_fd_sc_hd__xnor2_1)
0.14 3.52 v core_i/ex_stage_i/mult_i/place8681/X (sky130_fd_sc_hd__buf_4)
0.38 3.90 v core_i/ex_stage_i/mult_i/_12936_/X (sky130_fd_sc_hd__maj3_1)
0.19 4.08 v core_i/ex_stage_i/mult_i/_12939_/X (sky130_fd_sc_hd__xor2_1)
0.26 4.34 ^ core_i/ex_stage_i/mult_i/_12979_/Y (sky130_fd_sc_hd__xnor2_1)
0.28 4.62 ^ core_i/ex_stage_i/mult_i/_13012_/X (sky130_fd_sc_hd__xor2_1)
0.21 4.82 ^ core_i/ex_stage_i/mult_i/_13131_/Y (sky130_fd_sc_hd__xnor2_1)
0.12 4.95 v core_i/ex_stage_i/mult_i/_13132_/Y (sky130_fd_sc_hd__o21ai_2)
0.20 5.14 ^ core_i/ex_stage_i/mult_i/_13134_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 5.24 v core_i/ex_stage_i/mult_i/_13137_/Y (sky130_fd_sc_hd__o21ai_2)
0.19 5.43 ^ core_i/ex_stage_i/mult_i/_13140_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 5.53 v core_i/ex_stage_i/mult_i/_13142_/Y (sky130_fd_sc_hd__o21ai_2)
0.18 5.70 ^ core_i/ex_stage_i/mult_i/_13145_/Y (sky130_fd_sc_hd__a21oi_2)
0.17 5.87 ^ core_i/ex_stage_i/mult_i/_13148_/X (sky130_fd_sc_hd__o21a_1)
0.09 5.96 v core_i/ex_stage_i/mult_i/_13150_/Y (sky130_fd_sc_hd__o21ai_1)
0.21 6.17 ^ core_i/ex_stage_i/mult_i/_13153_/Y (sky130_fd_sc_hd__a21oi_1)
0.12 6.29 v core_i/ex_stage_i/mult_i/_13155_/Y (sky130_fd_sc_hd__o21ai_1)
0.27 6.56 ^ core_i/ex_stage_i/mult_i/_13157_/Y (sky130_fd_sc_hd__a21oi_1)
0.14 6.70 v core_i/ex_stage_i/mult_i/_13160_/Y (sky130_fd_sc_hd__o21ai_1)
0.20 6.90 ^ core_i/ex_stage_i/mult_i/_13162_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 7.00 v core_i/ex_stage_i/mult_i/_13165_/Y (sky130_fd_sc_hd__o21ai_1)
0.21 7.21 ^ core_i/ex_stage_i/mult_i/_13167_/Y (sky130_fd_sc_hd__a21oi_1)
0.11 7.32 v core_i/ex_stage_i/mult_i/_13169_/Y (sky130_fd_sc_hd__o21ai_1)
0.17 7.50 ^ core_i/ex_stage_i/mult_i/_13171_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 7.59 v core_i/ex_stage_i/mult_i/_13174_/Y (sky130_fd_sc_hd__o21ai_1)
0.21 7.80 ^ core_i/ex_stage_i/mult_i/_13176_/Y (sky130_fd_sc_hd__a21oi_1)
0.11 7.91 v core_i/ex_stage_i/mult_i/_13179_/Y (sky130_fd_sc_hd__o21ai_1)
0.15 8.06 v core_i/ex_stage_i/mult_i/place7266/X (sky130_fd_sc_hd__buf_4)
0.22 8.28 ^ core_i/ex_stage_i/mult_i/_13241_/Y (sky130_fd_sc_hd__xnor2_2)
0.04 8.32 v core_i/ex_stage_i/mult_i/_13242_/Y (sky130_fd_sc_hd__inv_2)
0.48 8.81 v core_i/ex_stage_i/mult_i/_21734_/X (sky130_fd_sc_hd__mux4_2)
0.25 9.05 ^ core_i/ex_stage_i/mult_i/_21735_/Y (sky130_fd_sc_hd__mux2i_1)

```

```

0.15 9.21 ^ core_i/ex_stage_i/mult_i/rebuffer14174/X (sky130_fd_sc_hd__buf_4)
0.09 9.29 ^ core_i/ex_stage_i/mult_i/rebuffer14156/X (sky130_fd_sc_hd__buf_4)
0.09 9.38 ^ core_i/ex_stage_i/mult_i/place7233/X (sky130_fd_sc_hd__buf_4)
0.13 9.51 ^ core_i/ex_stage_i/mult_i/_21736_/X (sky130_fd_sc_hd__o21a_1)
0.05 9.55 v core_i/ex_stage_i/mult_i/_21737_/Y (sky130_fd_sc_hd__nor2_1)
0.40 9.96 ^ core_i/ex_stage_i/mult_i/_21738_/Y (sky130_fd_sc_hd__o21ai_0)
0.27 10.23 v core_i/ex_stage_i/mult_i/_21753_/Y (sky130_fd_sc_hd__nand4_1)
0.29 10.51 ^ core_i/ex_stage_i/_084_/Y (sky130_fd_sc_hd__a22oi_2)
0.13 10.64 v core_i/ex_stage_i/_086_/Y (sky130_fd_sc_hd__o21ai_2)
0.24 10.88 v core_i/id_stage_i/_3525_/X (sky130_fd_sc_hd__a22o_1)
0.26 11.14 ^ core_i/id_stage_i/_3526_/Y (sky130_fd_sc_hd__a21oi_1)
0.13 11.27 v core_i/id_stage_i/_3527_/Y (sky130_fd_sc_hd__o22ai_1)
0.37 11.64 ^ core_i/id_stage_i/_3528_/Y (sky130_fd_sc_hd__nor3_1)
0.13 11.76 v core_i/id_stage_i/_3529_/Y (sky130_fd_sc_hd__o21bai_4)
0.28 12.05 v core_i/id_stage_i/_3554_/X (sky130_fd_sc_hd__mux2_4)
0.00 12.05 v core_i/id_stage_i/_4716_/D (sky130_fd_sc_hd__dftrtp_1)
    12.05 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4716_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

-----
    4.87 data required time
   -12.05 data arrival time

-----
   -7.17 slack (VIOLATED)

wns max -7.17
tns max -9560.16
Clock clk_i
  1.43 source latency core_i/id_stage_i/controller_i/_608_/CLK ^
 -1.68 target latency core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/
   _302_/CLK ^
  0.00 CRPR
-----
 -0.25 setup skew

```

Metric	Baseline, post-global-route	Modified, post-global-route
wns max	-10.03 ns	-7.17 ns (28.5% better)
tns max	-13,541.69	-9,560.16 (29.4% better)
Setup skew	0.28 ns	-0.25 ns
Critical path location	alu_i	mult_i → id_stage_i (same as CTS+repair, Chapter 11)

Table 12.3: Post-global-route timing: baseline vs. modified design.

The modified design’s critical path at this stage is the same `mult_i`→`id_stage_i` path

already seen after CTS+repair (Chapter 11), consistent with the multiplier-focused split (Section 6.3’s follow-on work) remaining the dominant bottleneck once the ALU stopped being critical. The wns/tns improvement over the baseline (28–29%) is in the same range as every other stage in this report.

12.6 Modified design: power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.69e-02	2.96e-03	3.29e-08	2.98e-02	19.8%
Combinational	4.47e-02	5.93e-02	1.01e-07	1.04e-01	68.9%
Clock	8.65e-03	8.38e-03	3.20e-09	1.70e-02	11.3%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	8.02e-02 53.2%	7.06e-02 46.8%	1.37e-07 0.0%	1.51e-01	100.0%

Table 12.4: Modified design post-global-route power breakdown (151mW total).

Total power (151mW) is about 32% below the baseline’s 221mW at this same stage, continuing the same open, unconfirmed discrepancy discussed since Chapter 8 — not treated as a validated result.

12.7 Modified design: area

Design area 329091 μm^2 50% utilization.

Metric	Baseline	Modified	Change
Design area	314,907 μm^2	329,091 μm^2	+4.5%
Utilization	51%	50%	—

Table 12.5: Post-global-route area: baseline vs. modified design.

The +4.5% area growth here reflects the combined cost of *both* RTL modifications (the ALU split, Chapter 6, and the multiplier split, Section 6.3’s follow-on), and is close to the +4.2% seen at the placement stage for the ALU-only modification (Section 11.6) — consistent with the multiplier split adding a comparatively small amount of extra area on top.

12.8 Summary: post-global-route, both designs

Metric	Baseline	Modified	Change
wns max	−10.03 ns	−7.17 ns	28.5% better
tns max	−13,541.69	−9,560.16	29.4% better
Total power	221 mW	151 mW	−32% (unconfirmed, see above)
Design area	314,907 μm^2	329,091 μm^2	+4.5%

Table 12.6: Complete post-global-route comparison, both designs, same methodology applied to each.

This final data point is consistent with every earlier stage in this report: across floorplan, placement, CTS, CTS+repair, and now post-global-route, the modified design’s timing

metrics are meaningfully and consistently better than the baseline's (roughly 14–32% depending on stage and metric), at a measured area cost in the mid-single-digit percent range, with the power figures throughout flagged as an open, unconfirmed discrepancy rather than a validated benefit.

Chapter 13

Final Modified Design (EX-Split + Multiplier Split): Floorplan Results

13.1 Floorplan run

With both RTL changes now in place (the EX-stage ALU split, Chapter 6, and the multiplier split, Section 6.3's follow-on), the design was re-flattened with `sv2v` and re-synthesized (`yosys_synth_modified.tcl`, module count confirmed unchanged at 25), then `floorplan_init_modified.tcl` (identical script structure to Section 7.1/Chapter 8, PDN block included from the start) was run on the resulting `synth_netlist_modified.v`. The design initialized and `pdngen` completed without error, producing the following timing and power reports.

13.2 Floorplan-stage timing results

```
Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint:   core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type:  max
```

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
2.38 4.88 time given to startpoint
0.00 4.88 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.22 5.10 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 5.10 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    5.10 data arrival time
```

```

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time
-----
    5.00 data required time
    -5.10 data arrival time
-----
    -0.10 slack (VIOLATED)

Startpoint: core_i/if_stage_i/_931_
    (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: instr_addr_o[31] (output port clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/if_stage_i/_931_/CLK (sky130_fd_sc_hd__dfrtp_1)
7.10 7.10 ^ core_i/if_stage_i/_931_/Q (sky130_fd_sc_hd__dfrtp_1)
5.40 12.49 v core_i/id_stage_i/register_file_i/_06253_/Y (
    sky130_fd_sc_hd__clkinv_1)
0.57 13.07 ^ core_i/id_stage_i/register_file_i/_08665_/Y (
    sky130_fd_sc_hd__o21ai_0)
0.15 13.21 v core_i/id_stage_i/register_file_i/_08666_/Y (
    sky130_fd_sc_hd__o221ai_1)
0.26 13.48 ^ core_i/id_stage_i/register_file_i/_08672_/Y (
    sky130_fd_sc_hd__a211oi_1)
0.11 13.58 v core_i/id_stage_i/register_file_i/_08673_/Y (
    sky130_fd_sc_hd__a211oi_1)
0.27 13.85 v core_i/id_stage_i/_2228_/X (sky130_fd_sc_hd__and2_0)
0.23 14.08 ^ core_i/id_stage_i/_2235_/Y (sky130_fd_sc_hd__nor2_1)
0.15 14.23 v core_i/id_stage_i/_2237_/Y (sky130_fd_sc_hd__a211oi_1)
0.42 14.65 ^ core_i/id_stage_i/_2240_/Y (sky130_fd_sc_hd__o311ai_0)
0.16 14.81 v core_i/id_stage_i/_2243_/Y (sky130_fd_sc_hd__a311oi_1)
0.37 15.18 v core_i/id_stage_i/_2244_/X (sky130_fd_sc_hd__or3_1)
0.38 15.55 v core_i/id_stage_i/_2247_/X (sky130_fd_sc_hd__a311o_1)
0.40 15.95 v core_i/id_stage_i/_2250_/X (sky130_fd_sc_hd__a311o_1)
0.31 16.26 ^ core_i/id_stage_i/_2257_/Y (sky130_fd_sc_hd__o31ai_1)
0.14 16.40 v core_i/id_stage_i/_2267_/Y (sky130_fd_sc_hd__nand4_1)
0.28 16.68 ^ core_i/id_stage_i/_2272_/Y (sky130_fd_sc_hd__a31oi_1)
0.15 16.83 v core_i/id_stage_i/_2278_/Y (sky130_fd_sc_hd__nand4_1)
0.24 17.07 ^ core_i/id_stage_i/_2283_/Y (sky130_fd_sc_hd__a31oi_1)
0.14 17.21 v core_i/id_stage_i/_2288_/Y (sky130_fd_sc_hd__nand4_1)
0.30 17.51 ^ core_i/id_stage_i/_2294_/Y (sky130_fd_sc_hd__a311oi_1)

```

```

0.23 17.74 ^ core_i/id_stage_i/_2295_/X (sky130_fd_sc_hd__maj3_1)
0.07 17.81 v core_i/id_stage_i/_2296_/Y (sky130_fd_sc_hd__nor2_1)
0.17 17.98 ^ core_i/id_stage_i/_2589_/Y (sky130_fd_sc_hd__mux2i_1)
0.13 18.11 ^ core_i/id_stage_i/_2590_/Y (sky130_fd_sc_hd__xnor2_1)
0.10 18.21 v core_i/id_stage_i/_2591_/Y (sky130_fd_sc_hd__o22ai_1)
0.32 18.54 ^ core_i/if_stage_i/_795_/Y (sky130_fd_sc_hd__a22oi_1)
0.07 18.61 v core_i/if_stage_i/_797_/Y (sky130_fd_sc_hd__nand2_1)
0.10 18.71 ^ core_i/if_stage_i/_798_/Y (sky130_fd_sc_hd__a21oi_1)
0.15 18.86 v core_i/if_stage_i/_799_/Y (sky130_fd_sc_hd__o311ai_0)
0.24 19.10 ^ core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_416_/
    Y (sky130_fd_sc_hd__a22oi_1)
0.08 19.18 v core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_417_/
    Y (sky130_fd_sc_hd__nand2_1)
0.28 19.46 v core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_146_/X (
    sky130_fd_sc_hd__mux2_1)
0.00 19.46 v instr_addr_o[31] (out)
    19.46 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
-3.00 2.00 output external delay
    2.00 data required time

-----
    2.00 data required time
   -19.46 data arrival time

-----
   -17.46 slack (VIOLATED)

wns max -17.46
tns max -27926.43

```

Critical path is unchanged from the ALU-only modified design. The worst path is, cell-for-cell, the *same* path already identified in Section 8.2 (register_file_i's address-decode logic → id_stage_i → if_stage_i/prefetch_buffer_i → instr_addr_o[31]), with an identical 19.46 ns data arrival time and identical **wns max** of −17.46 ns. This is the expected result: this path runs entirely through the register file and instruction-fetch logic and never touches cv32e40p_alu or cv32e40p_mult, so adding the multiplier split on top of the already-applied ALU split cannot change it. The gated-clock check (−0.10 ns) also remains in the same small range as the baseline (−0.16 ns, Section 3.5) and ALU-only-modified (−0.14 ns, Section 8.2) designs, consistent with it being a stable, pre-existing, unrelated issue rather than something introduced by either RTL change.

tns, however, *did* improve further: −27,926.43 here versus −30,642.40 for the ALU-only modified design (Section 8.2) and −30,884.59 for the baseline (Section 3.5) — a further ~8.9% reduction in total negative slack on top of the ALU split's own improvement, attributable to the multiplier split closing (or narrowing) a number of secondary violating paths that run through cv32e40p_mult, even though none of them was the single worst path at this stage.

Metric	Baseline	ALU-only modified	ALU +
wns max	−20.23 ns	−17.46 ns	−17.4
tns max	−30,884.59	−30,642.40	−27,926.43 (8.9% better
Critical path location	alu_i (adder + shifter)	register_file_i → instr_addr_o[31]	same as AL

Table 13.1: Floorplan-stage timing: baseline vs. ALU-only modified vs. ALU + multiplier modified.

13.3 Floorplan-stage power results

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	1.20e-03	4.60e-05	3.50e-08	1.25e-03	57.3%
Combinational	3.57e-04	4.11e-04	6.12e-08	7.68e-04	35.3%
Clock	−6.44e-07	1.61e-04	1.87e-12	1.60e-04	7.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	1.56e-03	6.18e-04	9.62e-08	2.17e-03	100.0%
	71.6%	28.4%	0.0%		

Table 13.2: Power breakdown, ALU + multiplier modified design, floorplan stage.

Total power (2.17 mW) is again lower than both the baseline (8.69 mW, Section 3.6) and the ALU-only modified design (4.36 mW, Section 8.3). Notably, **Sequential is now the largest power group (57.3%)**, ahead of Combinational (35.3%) — reversing the pattern seen at every prior stage. As with every other power figure in this report, this is measured without real switching-activity data (Section 2.6); the number is an OpenSTA default-model estimate, not a validated result.

Chapter 14

Final Modified Design (EX-Split + Multiplier Split): Placement Results

14.1 Placement run and legalization

global_placement_modified.tcl (identical script structure to Sections 3.7–3.8/Chapter 9) was run on 1_floorplan_pdn_modified.odb (the floorplan produced in the previous chapter). Legalization succeeded:

```
Placement Analysis
-----
total displacement 95833.0 u
average displacement 2.4 u
max displacement 13.8 u
original HPWL 1114942.5 u
legalized HPWL 1201870.8 u
delta HPWL 8 %
```

14.2 Placement-stage timing results (ideal wire)

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_
          (removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_/
          RESET_B (sky130_fd_sc_hd__dfrtp_1)
          0.50 data arrival time
```

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
      0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_/CLK
            (sky130_fd_sc_hd__dftrtp_1)
0.30 0.30 library removal time
      0.30 data required time
-----
      0.30 data required time
     -0.50 data arrival time
-----
      0.20 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint:  core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
      sky130_fd_sc_hd__dlxtn_1)
0.23 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
      sky130_fd_sc_hd__dlxtn_1)
0.00 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
      sky130_fd_sc_hd__and2_0)
      2.73 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
      2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
      sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
      2.50 data required time
-----
      2.50 data required time
     -2.73 data arrival time
-----
      0.23 slack (MET)

Startpoint: core_i/sleep_unit_i/_5_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:  core_i/id_stage_i/controller_i/_615_
            (rising edge-triggered flip-flop clocked by clk_i)

```

Path Group: clk_i

Path Type: min

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/sleep_unit_i/_5_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.35 0.35 ^ core_i/sleep_unit_i/_5_/Q (sky130_fd_sc_hd__dftrtp_1)
0.06 0.41 v core_i/id_stage_i/controller_i/_465_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.00 0.41 v core_i/id_stage_i/controller_i/_615_/D (sky130_fd_sc_hd__dfstp_1)
    0.41 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/id_stage_i/controller_i/_615_/CLK (
        sky130_fd_sc_hd__dfstp_1)
-0.02 -0.02 library hold time
    -0.02 data required time
-----
    -0.02 data required time
    -0.41 data arrival time
-----
    0.43 slack (MET)
```

Startpoint: rst_ni (input port clocked by clk_i)

Endpoint: core_i/id_stage_i/register_file_i/_12560_
(recovery check against rising-edge clock clk_i)

Path Group: asynchronous

Path Type: max

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.76 1.26 ^ core_i/wire299/X (sky130_fd_sc_hd__buf_16)
0.83 2.09 ^ core_i/wire297/X (sky130_fd_sc_hd__buf_16)
0.90 2.99 ^ core_i/wire296/X (sky130_fd_sc_hd__buf_16)
0.80 3.79 ^ core_i/wire295/X (sky130_fd_sc_hd__buf_16)
0.93 4.71 ^ core_i/id_stage_i/register_file_i/wire294/X (
    sky130_fd_sc_hd__buf_16)
1.04 5.75 ^ core_i/id_stage_i/register_file_i/wire292/X (
    sky130_fd_sc_hd__buf_16)
0.94 6.70 ^ core_i/id_stage_i/register_file_i/wire291/X (
    sky130_fd_sc_hd__buf_16)
```

```

0.06 6.76 ^ core_i/id_stage_i/register_file_i/_12560_/RESET_B (
    sky130_fd_sc_hd__dfrtp_1)
    6.76 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/register_file_i/_12560_/CLK (
        sky130_fd_sc_hd__dfrtp_1)
-0.07 4.93 library recovery time
    4.93 data required time
-----
    4.93 data required time
    -6.76 data arrival time
-----
    -1.83 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
    (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
    (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.73 4.23 time given to startpoint
0.00 4.23 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.22 4.45 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 4.45 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    4.45 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time
-----
    5.00 data required time
    -4.45 data arrival time
-----
    0.55 slack (MET)

```

Startpoint: core_i/ex_stage_i/mult_i/_22724_
(rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4712_
(rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22724_/CLK (sky130_fd_sc_hd__dfstp_1)
0.56 0.56 ^ core_i/ex_stage_i/mult_i/_22724_/Q (sky130_fd_sc_hd__dfstp_1)
0.08 0.64 v core_i/ex_stage_i/mult_i/_14034_/Y (sky130_fd_sc_hd__nor3_1)
0.14 0.78 v core_i/ex_stage_i/mult_i/_place12623/X (sky130_fd_sc_hd__buf_4)
0.35 1.13 ^ core_i/ex_stage_i/mult_i/_14038_/Y (sky130_fd_sc_hd__nor2_1)
0.33 1.46 ^ core_i/ex_stage_i/mult_i/_14164_/Y (sky130_fd_sc_hd__mux2i_2)
0.30 1.76 ^ core_i/ex_stage_i/mult_i/_place11678/X (sky130_fd_sc_hd__buf_4)
0.09 1.85 v core_i/ex_stage_i/mult_i/_15709_/Y (sky130_fd_sc_hd__nor2_1)
0.19 2.04 v core_i/ex_stage_i/mult_i/_15712_/X (sky130_fd_sc_hd__a21oi_1)
0.21 2.24 ^ core_i/ex_stage_i/mult_i/_15718_/Y (sky130_fd_sc_hd__xnor2_1)
0.20 2.44 ^ core_i/ex_stage_i/mult_i/_15757_/Y (sky130_fd_sc_hd__xnor2_1)
0.40 2.84 v core_i/ex_stage_i/mult_i/_15794_/X (sky130_fd_sc_hd__xor3_1)
0.38 3.22 v core_i/ex_stage_i/mult_i/_15815_/X (sky130_fd_sc_hd__xnor3_1)
0.17 3.39 v core_i/ex_stage_i/mult_i/_15816_/X (sky130_fd_sc_hd__xor2_1)
0.30 3.69 ^ core_i/ex_stage_i/mult_i/_15837_/Y (sky130_fd_sc_hd__xnor2_1)
0.30 4.00 ^ core_i/ex_stage_i/mult_i/_15879_/X (sky130_fd_sc_hd__xor3_1)
0.09 4.08 v core_i/ex_stage_i/mult_i/_15880_/Y (sky130_fd_sc_hd__a22oi_1)
0.23 4.32 ^ core_i/ex_stage_i/mult_i/_15882_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 4.45 v core_i/ex_stage_i/mult_i/_15885_/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.69 ^ core_i/ex_stage_i/mult_i/_15888_/Y (sky130_fd_sc_hd__o21ai_0)
0.16 4.85 v core_i/ex_stage_i/mult_i/_15891_/Y (sky130_fd_sc_hd__a21boi_0)
0.19 5.04 ^ core_i/ex_stage_i/mult_i/_15893_/Y (sky130_fd_sc_hd__o21bai_1)
0.12 5.15 v core_i/ex_stage_i/mult_i/_15895_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 5.37 ^ core_i/ex_stage_i/mult_i/_15897_/Y (sky130_fd_sc_hd__o21ai_0)
0.16 5.53 v core_i/ex_stage_i/mult_i/_15900_/Y (sky130_fd_sc_hd__a21boi_0)
0.23 5.76 ^ core_i/ex_stage_i/mult_i/_15902_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 5.87 v core_i/ex_stage_i/mult_i/_15905_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 6.10 ^ core_i/ex_stage_i/mult_i/_15907_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 6.21 v core_i/ex_stage_i/mult_i/_15910_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 6.43 ^ core_i/ex_stage_i/mult_i/_15911_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 6.56 v core_i/ex_stage_i/mult_i/_15913_/Y (sky130_fd_sc_hd__a21oi_1)
0.39 6.95 v core_i/ex_stage_i/mult_i/_15915_/X (sky130_fd_sc_hd__maj3_1)
0.37 7.32 v core_i/ex_stage_i/mult_i/_15917_/X (sky130_fd_sc_hd__maj3_1)
0.22 7.55 ^ core_i/ex_stage_i/mult_i/_15920_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.66 v core_i/ex_stage_i/mult_i/_15923_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 7.88 ^ core_i/ex_stage_i/mult_i/_15926_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.99 v core_i/ex_stage_i/mult_i/_15928_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 8.22 ^ core_i/ex_stage_i/mult_i/_15931_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 8.34 v core_i/ex_stage_i/mult_i/_15934_/Y (sky130_fd_sc_hd__a211oi_1)
```

```

0.34 8.68 v core_i/ex_stage_i/mult_i/_15936_/X (sky130_fd_sc_hd__or3_1)
0.09 8.77 ^ core_i/ex_stage_i/mult_i/_15937_/Y (sky130_fd_sc_hd__nand2_1)
0.17 8.95 ^ core_i/ex_stage_i/mult_i/_15984_/X (sky130_fd_sc_hd__xor2_1)
0.10 9.05 v core_i/ex_stage_i/mult_i/_16061_/Y (sky130_fd_sc_hd__mux2i_1)
0.30 9.36 v core_i/ex_stage_i/mult_i/_22038_/X (sky130_fd_sc_hd__mux2_1)
0.30 9.65 v core_i/ex_stage_i/mult_i/_22124_/X (sky130_fd_sc_hd__mux2_1)
0.30 9.96 v core_i/ex_stage_i/mult_i/_22298_/X (sky130_fd_sc_hd__mux2_1)
0.26 10.22 ^ core_i/ex_stage_i/mult_i/_22611_/Y (sky130_fd_sc_hd__o211ai_1)
0.14 10.36 v core_i/ex_stage_i/mult_i/_22620_/Y (sky130_fd_sc_hd__nand3_1)
0.24 10.59 ^ core_i/ex_stage_i/_168_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 10.69 v core_i/ex_stage_i/_170_/Y (sky130_fd_sc_hd__o21ai_2)
0.21 10.90 v core_i/id_stage_i/_2949_/X (sky130_fd_sc_hd__a22o_1)
0.15 11.06 ^ core_i/id_stage_i/_2950_/Y (sky130_fd_sc_hd__a21oi_1)
0.06 11.12 v core_i/id_stage_i/_2951_/Y (sky130_fd_sc_hd__nor2_1)
0.41 11.53 ^ core_i/id_stage_i/_2959_/Y (sky130_fd_sc_hd__a211oi_1)
0.10 11.64 v core_i/id_stage_i/_3143_/Y (sky130_fd_sc_hd__nand2_1)
0.08 11.72 ^ core_i/id_stage_i/_3144_/Y (sky130_fd_sc_hd__nand2_1)
0.04 11.76 v core_i/id_stage_i/_3145_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 12.00 v core_i/id_stage_i/_3580_/Y (sky130_fd_sc_hd__o21bai_1)
0.28 12.28 v core_i/id_stage_i/_3581_/X (sky130_fd_sc_hd__mux2_1)
0.00 12.28 v core_i/id_stage_i/_4712_/D (sky130_fd_sc_hd__dftrtp_1)
    12.28 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4712_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

-----
    4.87 data required time
-12.28 data arrival time

-----
    -7.41 slack (VIOLATED)

wns max -7.41
tns max -10651.90

```

14.2.1 Estimated wire

Re-running with estimated (post-placement) wire delay gives the following:

```

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_245_
          (removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min

```

Delay Time Description

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/ifu_stage_i/prefetch_buffer_i/instruction_obi_i/_245_/
    RESET_B (sky130_fd_sc_hd__dftrtp_1)
    0.50 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/ifu_stage_i/prefetch_buffer_i/instruction_obi_i/_245_/CLK
        (sky130_fd_sc_hd__dftrtp_1)
0.30 0.30 library removal time
    0.30 data required time
-----
    0.30 data required time
    -0.50 data arrival time
-----
    0.20 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
    (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
    (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
    sky130_fd_sc_hd__dlxtn_1)
0.23 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    2.73 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
    2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
        sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
    2.50 data required time
-----
    2.50 data required time
    -2.73 data arrival time

```

```

-----
0.23 slack (MET)

Startpoint: core_i/id_stage_i/controller_i/_628_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/controller_i/_628_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/controller_i/_628_/CLK (
            sky130_fd_sc_hd__dfrrtp_1)
0.31 0.31 ^ core_i/id_stage_i/controller_i/_628_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.09 0.40 ^ core_i/id_stage_i/controller_i/_464_/X (
            sky130_fd_sc_hd__lpflow_inputiso1p_1)
0.00 0.40 ^ core_i/id_stage_i/controller_i/_628_/D (sky130_fd_sc_hd__dfrrtp_1)
0.40 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
0.00 ^ core_i/id_stage_i/controller_i/_628_/CLK (
            sky130_fd_sc_hd__dfrrtp_1)
-0.04 -0.04 library hold time
-0.04 data required time
-----
-0.04 data required time
-0.40 data arrival time
-----

0.43 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/register_file_i/_12551_
            (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.74 1.24 ^ core_i/wire299/X (sky130_fd_sc_hd__buf_16)
0.86 2.09 ^ core_i/wire297/X (sky130_fd_sc_hd__buf_16)

```

```

0.87 2.96 ^ core_i/wire296/X (sky130_fd_sc_hd__buf_16)
0.81 3.77 ^ core_i/wire295/X (sky130_fd_sc_hd__buf_16)
0.96 4.73 ^ core_i/id_stage_i/register_file_i/wire294/X (
    sky130_fd_sc_hd__buf_16)
1.03 5.77 ^ core_i/id_stage_i/register_file_i/wire292/X (
    sky130_fd_sc_hd__buf_16)
0.97 6.73 ^ core_i/id_stage_i/register_file_i/wire291/X (
    sky130_fd_sc_hd__buf_16)
0.07 6.80 ^ core_i/id_stage_i/register_file_i/_12551_/RESET_B (
    sky130_fd_sc_hd__dfrtp_1)
    6.80 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/register_file_i/_12551_/CLK (
    sky130_fd_sc_hd__dfrtp_1)
-0.07 4.93 library recovery time
    4.93 data required time

-----
    4.93 data required time
    -6.80 data arrival time
-----

-1.87 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
    (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
    (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.91 4.41 time given to startpoint
0.00 4.41 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.22 4.63 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 4.63 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    4.63 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
    sky130_fd_sc_hd__and2_0)

```

```

0.00 5.00 clock gating setup time
      5.00 data required time
-----
      5.00 data required time
     -4.63 data arrival time
-----
      0.37 slack (MET)

Startpoint: core_i/ex_stage_i/mult_i/_22724_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:  core_i/id_stage_i/_4712_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/ex_stage_i/mult_i/_22724_/CLK (sky130_fd_sc_hd__dfstp_1)
0.56 0.56 ^ core_i/ex_stage_i/mult_i/_22724_/Q (sky130_fd_sc_hd__dfstp_1)
0.08 0.64 v core_i/ex_stage_i/mult_i/_14034_/Y (sky130_fd_sc_hd__nor3_1)
0.14 0.79 v core_i/ex_stage_i/mult_i/_place12623/X (sky130_fd_sc_hd__buf_4)
0.36 1.14 ^ core_i/ex_stage_i/mult_i/_14038_/Y (sky130_fd_sc_hd__nor2_1)
0.32 1.47 ^ core_i/ex_stage_i/mult_i/_14164_/Y (sky130_fd_sc_hd__mux2i_2)
0.30 1.77 ^ core_i/ex_stage_i/mult_i/_place11678/X (sky130_fd_sc_hd__buf_4)
0.10 1.87 v core_i/ex_stage_i/mult_i/_15709_/Y (sky130_fd_sc_hd__nor2_1)
0.19 2.05 v core_i/ex_stage_i/mult_i/_15712_/X (sky130_fd_sc_hd__a21oi_1)
0.20 2.25 ^ core_i/ex_stage_i/mult_i/_15718_/Y (sky130_fd_sc_hd__xnor2_1)
0.19 2.44 ^ core_i/ex_stage_i/mult_i/_15757_/Y (sky130_fd_sc_hd__xnor2_1)
0.40 2.84 v core_i/ex_stage_i/mult_i/_15794_/X (sky130_fd_sc_hd__xor3_1)
0.37 3.21 v core_i/ex_stage_i/mult_i/_15815_/X (sky130_fd_sc_hd__xnor3_1)
0.17 3.38 v core_i/ex_stage_i/mult_i/_15816_/X (sky130_fd_sc_hd__xor2_1)
0.29 3.67 ^ core_i/ex_stage_i/mult_i/_15837_/Y (sky130_fd_sc_hd__xnor2_1)
0.30 3.97 ^ core_i/ex_stage_i/mult_i/_15879_/X (sky130_fd_sc_hd__xor3_1)
0.09 4.06 v core_i/ex_stage_i/mult_i/_15880_/Y (sky130_fd_sc_hd__a22oi_1)
0.25 4.31 ^ core_i/ex_stage_i/mult_i/_15882_/Y (sky130_fd_sc_hd__o21ai_0)
0.14 4.45 v core_i/ex_stage_i/mult_i/_15885_/Y (sky130_fd_sc_hd__a21oi_1)
0.25 4.70 ^ core_i/ex_stage_i/mult_i/_15888_/Y (sky130_fd_sc_hd__o21ai_0)
0.16 4.86 v core_i/ex_stage_i/mult_i/_15891_/Y (sky130_fd_sc_hd__a21boi_0)
0.20 5.06 ^ core_i/ex_stage_i/mult_i/_15893_/Y (sky130_fd_sc_hd__o21bai_1)
0.12 5.18 v core_i/ex_stage_i/mult_i/_15895_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 5.40 ^ core_i/ex_stage_i/mult_i/_15897_/Y (sky130_fd_sc_hd__o21ai_0)
0.17 5.57 v core_i/ex_stage_i/mult_i/_15900_/Y (sky130_fd_sc_hd__a21boi_0)
0.25 5.82 ^ core_i/ex_stage_i/mult_i/_15902_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 5.93 v core_i/ex_stage_i/mult_i/_15905_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 6.16 ^ core_i/ex_stage_i/mult_i/_15907_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 6.27 v core_i/ex_stage_i/mult_i/_15910_/Y (sky130_fd_sc_hd__a21oi_1)
0.21 6.49 ^ core_i/ex_stage_i/mult_i/_15911_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 6.62 v core_i/ex_stage_i/mult_i/_15913_/Y (sky130_fd_sc_hd__a21oi_1)

```

```

0.38 7.00 v core_i/ex_stage_i/mult_i/_15915_/X (sky130_fd_sc_hd__maj3_1)
0.39 7.39 v core_i/ex_stage_i/mult_i/_15917_/X (sky130_fd_sc_hd__maj3_1)
0.24 7.63 ^ core_i/ex_stage_i/mult_i/_15920_/Y (sky130_fd_sc_hd__o21ai_0)
0.11 7.74 v core_i/ex_stage_i/mult_i/_15923_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 7.96 ^ core_i/ex_stage_i/mult_i/_15926_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 8.10 v core_i/ex_stage_i/mult_i/_15928_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 8.33 ^ core_i/ex_stage_i/mult_i/_15931_/Y (sky130_fd_sc_hd__o21ai_0)
0.13 8.46 v core_i/ex_stage_i/mult_i/_15934_/Y (sky130_fd_sc_hd__a211oi_1)
0.34 8.80 v core_i/ex_stage_i/mult_i/_15936_/X (sky130_fd_sc_hd__or3_1)
0.09 8.89 ^ core_i/ex_stage_i/mult_i/_15937_/Y (sky130_fd_sc_hd__nand2_1)
0.17 9.06 ^ core_i/ex_stage_i/mult_i/_15984_/X (sky130_fd_sc_hd__xor2_1)
0.10 9.16 v core_i/ex_stage_i/mult_i/_16061_/Y (sky130_fd_sc_hd__mux2i_1)
0.31 9.47 v core_i/ex_stage_i/mult_i/_22038_/X (sky130_fd_sc_hd__mux2_1)
0.30 9.77 v core_i/ex_stage_i/mult_i/_22124_/X (sky130_fd_sc_hd__mux2_1)
0.30 10.07 v core_i/ex_stage_i/mult_i/_22298_/X (sky130_fd_sc_hd__mux2_1)
0.26 10.33 ^ core_i/ex_stage_i/mult_i/_22611_/Y (sky130_fd_sc_hd__o211ai_1)
0.16 10.49 v core_i/ex_stage_i/mult_i/_22620_/Y (sky130_fd_sc_hd__nand3_1)
0.24 10.73 ^ core_i/ex_stage_i/_168_/Y (sky130_fd_sc_hd__a22oi_1)
0.10 10.83 v core_i/ex_stage_i/_170_/Y (sky130_fd_sc_hd__o21ai_2)
0.21 11.04 v core_i/id_stage_i/_2949_/X (sky130_fd_sc_hd__a22o_1)
0.16 11.20 ^ core_i/id_stage_i/_2950_/Y (sky130_fd_sc_hd__a21oi_1)
0.06 11.26 v core_i/id_stage_i/_2951_/Y (sky130_fd_sc_hd__nor2_1)
0.46 11.72 ^ core_i/id_stage_i/_2959_/Y (sky130_fd_sc_hd__a2111oi_1)
0.10 11.82 v core_i/id_stage_i/_3143_/Y (sky130_fd_sc_hd__nand2_1)
0.08 11.90 ^ core_i/id_stage_i/_3144_/Y (sky130_fd_sc_hd__nand2_1)
0.05 11.94 v core_i/id_stage_i/_3145_/Y (sky130_fd_sc_hd__a21oi_1)
0.23 12.18 v core_i/id_stage_i/_3580_/Y (sky130_fd_sc_hd__o21bai_1)
0.28 12.46 v core_i/id_stage_i/_3581_/X (sky130_fd_sc_hd__mux2_1)
0.00 12.46 v core_i/id_stage_i/_4712_/D (sky130_fd_sc_hd__dftrtp_1)
    12.46 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4712_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.13 4.87 library setup time
    4.87 data required time

-----
    4.87 data required time
   -12.46 data arrival time
-----

   -7.59 slack (VIOLATED)

wns max -7.59
tns max -10979.75

```

Compared to the ideal-wire run above, **wns** moves only slightly ($-7.41\text{ ns} \rightarrow -7.59\text{ ns}$), the critical path is the same **mult_i**→**id_stage_i** path (arrival time $12.28\text{ ns} \rightarrow 12.46\text{ ns}$), and the secondary violations (the **register_file_i** recovery/removal checks and the gated-clock checks) are essentially unchanged in magnitude.

The critical path is structurally the same as the ALU-only modified design (Section 9.2) — and this is expected. The endpoint changed only cosmetically (`_4716_` $\rightarrow$ `_4712_`), the internal `mult_i` cell chain is the same ~ 50 -cell `maj3_1`-carry-then-mux-tree structure, and the arrival time is nearly identical (12.25 ns there vs. 12.28 ns here). This is consistent with the RTL change actually applied to `cv32e40p_mult.sv`: only the `MUL_MAC32/MUL_MSU32` 32×32 MAC path was split into EX1/EX2 (Section 6.3); `MUL_H`'s existing FSM-based multi-cycle path was left untouched. This particular critical path most plausibly runs through the untouched `MUL_H` logic (or through combinational result-selection shared by both paths), so it is not expected to shrink as a direct result of the multiplier split.

wns and tns both moved slightly against the ALU-only modified design. `wns` went from -7.38 ns to -7.41 ns (ideal) and -7.48 ns to -7.59 ns (estimated); `tns` went from $-9,553.46$ to $-10,651.90$ (ideal, $\sim 11.5\%$ increase) and from $-9,733.24$ to $-10,979.75$ (estimated, $\sim 12.8\%$ increase).

Metric	Baseline	ALU-only modified	ALU + mult modified
wns max, ideal wire	-9.83 ns	-7.38 ns	-7.41 ns
tns max, ideal wire	$-13,280.61$	$-9,553.46$	$-10,651.90$
wns max, estimated wire	-9.83 ns	-7.48 ns	-7.59 ns
tns max, estimated wire	$-13,421.99$	$-9,733.24$	$-10,979.75$
Critical path location	<code>alu_i</code>	<code>mult_i</code> $\rightarrow$ <code>id_stage_i</code>	same as ALU-only modified

Table 14.1: Placement-stage timing: baseline vs. ALU-only modified vs. ALU + multiplier modified.

Despite the small regression relative to the ALU-only modified design, both `wns` and `tns` for the final (ALU+mult) modified design remain substantially better than the baseline at this stage (`wns` ~ 24 – 25% better, `tns` ~ 18 – 20% better), so the overall conclusion of Chapter 9 — that the pipeline split meaningfully improves timing over the baseline — still holds; the multiplier split's own, additional contribution at the placement stage is small and, on this specific metric, slightly negative rather than positive.

14.3 Placement-stage power results

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	$1.19\text{e-}03$	$5.60\text{e-}05$	$3.49\text{e-}08$	$1.25\text{e-}03$	41.7%
Combinational	$5.08\text{e-}04$	$1.07\text{e-}03$	$9.78\text{e-}08$	$1.58\text{e-}03$	52.9%
Clock	$-6.45\text{e-}07$	$1.61\text{e-}04$	$1.87\text{e-}12$	$1.60\text{e-}04$	5.4%
Macro	$0.00\text{e+}00$	$0.00\text{e+}00$	$0.00\text{e+}00$	$0.00\text{e+}00$	0.0%
Pad	$0.00\text{e+}00$	$0.00\text{e+}00$	$0.00\text{e+}00$	$0.00\text{e+}00$	0.0%
Total	$1.70\text{e-}03$ 56.8%	$1.29\text{e-}03$ 43.1%	$1.33\text{e-}07$ 0.0%	2.99e-03	100.0%

Table 14.2: Power breakdown, ALU + multiplier modified design, placement stage (identical for ideal- and estimated-wire reports).

Total power (2.99 mW) is again below the baseline (12.70 mW, Section 3.9) and the ALU-only modified design (6.38 mW, Section 9.3); as with every power figure in this report, this is an OpenSTA default-activity estimate, not a validated result (Section 2.6).

Chapter 15

Final Modified Design (EX-Split + Multiplier Split): Clock Tree Synthesis Results

15.1 CTS timing results

cts_modified.tcl (identical script structure to Chapter 4/Chapter 10) was run on the placed design from the previous chapter. With propagated clock latency and skew now included:

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_305_
          (removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (propagated)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_305_/
RESET_B (sky130_fd_sc_hd__dftrtp_2)
0.50 data arrival time

0.00 0.00 clock clk_i (rise edge)
1.48 1.48 clock network delay (propagated)
0.00 1.48 clock reconvergence pessimism
1.48 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_305_/CLK
(sky130_fd_sc_hd__dftrtp_2)
0.32 1.80 library removal time
1.80 data required time
-----
1.80 data required time
-0.50 data arrival time
```

```

-----
-1.30 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

```

Delay Time Description

```

-----
2.50 2.50 clock clk_i (fall edge)
1.65 4.15 clock network delay (propagated)
0.00 4.15 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
    sky130_fd_sc_hd__dlxtn_1)
0.32 4.46 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 4.47 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    4.47 data arrival time

```

```

2.50 2.50 clock clk_i (fall edge)
0.23 2.73 clock network delay (propagated)
0.00 2.73 clock reconvergence pessimism
    2.73 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
    sky130_fd_sc_hd__and2_0)
0.00 2.73 clock gating hold time
    2.73 data required time

```

```

-----
    2.73 data required time
    -4.47 data arrival time

```

```

-----
1.74 slack (MET)

```

```

Startpoint: data_rdata_i[27] (input port clocked by clk_i)
Endpoint: core_i/load_store_unit_i/_1412_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

```

Delay Time Description

```

-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (propagated)
0.50 0.50 ^ input external delay
0.00 0.50 ^ data_rdata_i[27] (in)
0.06 0.56 v core_i/load_store_unit_i/_1241_/Y (sky130_fd_sc_hd__o21ai_0)
0.06 0.62 ^ core_i/load_store_unit_i/_1242_/Y (sky130_fd_sc_hd__nand2_1)

```

```

0.00 0.62 ^ core_i/load_store_unit_i/_1412_/D (sky130_fd_sc_hd__dfirtp_1)
      0.62 data arrival time

0.00 0.00 clock clk_i (rise edge)
1.64 1.64 clock network delay (propagated)
0.00 1.64 clock reconvergence pessimism
      1.64 ^ core_i/load_store_unit_i/_1412_/CLK (sky130_fd_sc_hd__dfirtp_1)
-0.03 1.61 library hold time
      1.61 data required time
-----
      1.61 data required time
      -0.62 data arrival time
-----
      -0.99 slack (VIOLATED)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/register_file_i/_12667_
          (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (propagated)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.74 1.24 ^ core_i/wire299/X (sky130_fd_sc_hd__buf_16)
0.86 2.09 ^ core_i/wire297/X (sky130_fd_sc_hd__buf_16)
0.87 2.96 ^ core_i/wire296/X (sky130_fd_sc_hd__buf_16)
0.81 3.77 ^ core_i/wire295/X (sky130_fd_sc_hd__buf_16)
0.96 4.73 ^ core_i/id_stage_i/register_file_i/wire294/X (
      sky130_fd_sc_hd__buf_16)
1.03 5.76 ^ core_i/id_stage_i/register_file_i/wire292/X (
      sky130_fd_sc_hd__buf_16)
0.97 6.73 ^ core_i/id_stage_i/register_file_i/wire291/X (
      sky130_fd_sc_hd__buf_16)
0.04 6.77 ^ core_i/id_stage_i/register_file_i/_12667_/RESET_B (
      sky130_fd_sc_hd__dfirtp_1)
      6.77 data arrival time

5.00 5.00 clock clk_i (rise edge)
1.56 6.56 clock network delay (propagated)
0.00 6.56 clock reconvergence pessimism
      6.56 ^ core_i/id_stage_i/register_file_i/_12667_/CLK (
      sky130_fd_sc_hd__dfirtp_1)
-0.05 6.52 library recovery time
      6.52 data required time
-----
      6.52 data required time

```

```

-6.77 data arrival time
-----
-0.25 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

```

Delay Time Description

```

-----
2.50 2.50 clock clk_i (fall edge)
1.65 4.15 clock network delay (propagated)
1.85 6.00 time given to startpoint
0.00 6.00 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.30 6.30 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 6.30 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    6.30 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.21 5.21 clock network delay (propagated)
0.00 5.21 clock reconvergence pessimism
    5.21 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
    sky130_fd_sc_hd__and2_0)
0.00 5.21 clock gating setup time
    5.21 data required time
-----
    5.21 data required time
    -6.30 data arrival time
-----
-1.09 slack (VIOLATED)

```

```

Startpoint: core_i/if_stage_i/_929_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: instr_addr_o[31] (output port clocked by clk_i)
Path Group: clk_i
Path Type: max

```

Delay Time Description

```

-----
0.00 0.00 clock clk_i (rise edge)
1.66 1.66 clock network delay (propagated)
0.00 1.66 ^ core_i/if_stage_i/_929_/CLK (sky130_fd_sc_hd__dfrtp_4)
0.47 2.13 ^ core_i/if_stage_i/_929_/Q (sky130_fd_sc_hd__dfrtp_4)

```

```

0.24 2.37 ^ core_i/id_stage_i/place12899/X (sky130_fd_sc_hd__buf_4)
0.30 2.67 ^ core_i/id_stage_i/register_file_i/_07894_/X (
    sky130_fd_sc_hd__lpflow_isobufsrc_4)
0.30 2.97 ^ core_i/id_stage_i/register_file_i/place12290/X (
    sky130_fd_sc_hd__buf_4)
0.32 3.29 ^ core_i/id_stage_i/register_file_i/place12289/X (
    sky130_fd_sc_hd__buf_4)
0.12 3.41 v core_i/id_stage_i/register_file_i/_08607_/Y (
    sky130_fd_sc_hd__o21ai_0)
0.22 3.63 ^ core_i/id_stage_i/register_file_i/_08610_/Y (
    sky130_fd_sc_hd__o21ai_0)
0.14 3.77 v core_i/id_stage_i/register_file_i/_08616_/Y (
    sky130_fd_sc_hd__a21oi_1)
0.38 4.15 ^ core_i/id_stage_i/register_file_i/_08617_/Y (
    sky130_fd_sc_hd__a211oi_1)
0.19 4.34 ^ core_i/id_stage_i/place10274/X (sky130_fd_sc_hd__buf_12)
0.09 4.43 v core_i/id_stage_i/_2224_/Y (sky130_fd_sc_hd__xnor2_1)
0.32 4.74 ^ core_i/id_stage_i/_2240_/Y (sky130_fd_sc_hd__o311ai_0)
0.18 4.92 ^ core_i/id_stage_i/place9315/X (sky130_fd_sc_hd__buf_4)
0.09 5.01 v core_i/id_stage_i/_2243_/Y (sky130_fd_sc_hd__a311oi_1)
0.33 5.33 v core_i/id_stage_i/_2244_/X (sky130_fd_sc_hd__or3_1)
0.43 5.77 v core_i/id_stage_i/_2247_/X (sky130_fd_sc_hd__a311o_1)
0.40 6.16 v core_i/id_stage_i/_2250_/X (sky130_fd_sc_hd__a311o_1)
0.34 6.50 ^ core_i/id_stage_i/_2257_/Y (sky130_fd_sc_hd__o31ai_1)
0.16 6.66 v core_i/id_stage_i/_2267_/Y (sky130_fd_sc_hd__nand4_1)
0.20 6.86 ^ core_i/id_stage_i/_2272_/Y (sky130_fd_sc_hd__a31oi_1)
0.13 6.99 v core_i/id_stage_i/_2278_/Y (sky130_fd_sc_hd__nand4_1)
0.19 7.18 ^ core_i/id_stage_i/_2283_/Y (sky130_fd_sc_hd__a31oi_1)
0.12 7.31 v core_i/id_stage_i/_2288_/Y (sky130_fd_sc_hd__nand4_1)
0.30 7.61 ^ core_i/id_stage_i/_2294_/Y (sky130_fd_sc_hd__a311oi_1)
0.23 7.84 ^ core_i/id_stage_i/_2295_/X (sky130_fd_sc_hd__maj3_1)
0.08 7.92 v core_i/id_stage_i/_2296_/Y (sky130_fd_sc_hd__nor2_1)
0.18 8.10 ^ core_i/id_stage_i/_2589_/Y (sky130_fd_sc_hd__mux2i_1)
0.13 8.23 ^ core_i/id_stage_i/_2590_/Y (sky130_fd_sc_hd__xnor2_1)
0.14 8.37 ^ core_i/id_stage_i/place7862/X (sky130_fd_sc_hd__buf_4)
0.13 8.50 v core_i/id_stage_i/_2591_/Y (sky130_fd_sc_hd__o22ai_1)
0.34 8.85 ^ core_i/if_stage_i/_795_/Y (sky130_fd_sc_hd__a222oi_1)
0.08 8.92 v core_i/if_stage_i/_797_/Y (sky130_fd_sc_hd__nand2_1)
0.20 9.13 ^ core_i/if_stage_i/_798_/Y (sky130_fd_sc_hd__a21oi_1)
0.31 9.44 v core_i/if_stage_i/_799_/Y (sky130_fd_sc_hd__o311ai_0)
0.30 9.73 ^ core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_416_/Y
    (sky130_fd_sc_hd__a22oi_1)
0.14 9.87 v core_i/if_stage_i/prefetch_buffer_i/prefetch_controller_i/_417_/Y
    (sky130_fd_sc_hd__nand2_1)
0.34 10.21 v core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_146_/X (
    sky130_fd_sc_hd__mux2_1)
0.00 10.21 v instr_addr_o[31] (out)
    10.21 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (propagated)

```

```

0.00 5.00 clock reconvergence pessimism
-3.00 2.00 output external delay
    2.00 data required time
-----
    2.00 data required time
   -10.21 data arrival time
-----
   -8.21 slack (VIOLATED)

wns max -8.21
tns max -10238.69
Clock clk_i
    1.68 source latency core_i/id_stage_i/register_file_i/_12611_/CLK ^
   -1.46 target latency core_i/if_stage_i/aligner_i/_1044_/CLK ^
   -0.00 CRPR
-----
    0.22 setup skew

```

Metric	Baseline (Section 4.5)	ALU-only modified (Section 10.2)
wns max	-9.86 ns	-8.15 ns
tns max	-13,596.81	-9,999.46
Setup skew	0.19 ns	0.15 ns

Metric	ALU + mult modified
wns max	-8.21 ns (17% better than baseline)
tns max	-10,238.69 (25% better than baseline)
Setup skew	0.22 ns

Table 15.1: CTS-stage timing: baseline vs. ALU-only modified vs. ALU + multiplier modified.

The critical path returns to the register-file/PC-generation path, as it did for the ALU-only modified design. With propagated clock latency now included, the worst path moves back to the same `register_file_i` → `id_stage_i` → `if_stage_i/prefetch_buffer` → `instr_addr_o[31]` path already seen at the floorplan stage (Chapter 13) and at the ALU-only modified design’s CTS stage (Section 10.2), with an almost identical data arrival time (10.15 ns there vs. 10.21 ns here). This reconfirms, for the third stage running, that this path is unrelated to either RTL change and behaves as a stable reference point across all three designs.

wns (−8.21 ns) and **tns** (−10,238.69) are both slightly worse than the ALU-only modified design’s CTS numbers (−8.15 ns / −9,999.46), continuing the same trend already observed at the placement stage (Chapter 14) — both remain, however, substantially better than the baseline (17% and 25% respectively). The setup skew (0.22 ns) is likewise marginally higher than both the baseline (0.19 ns) and the ALU-only modified design (0.15 ns), though still small relative to the 5 ns clock period (<4.4%).

15.2 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.48e-02	1.08e-03	3.49e-08	2.59e-02	49.4%
Combinational	4.38e-03	5.62e-03	9.78e-08	1.00e-02	19.0%
Clock	9.24e-03	7.34e-03	3.55e-09	1.66e-02	31.6%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.85e-02 73.3%	1.40e-02 26.7%	1.36e-07 0.0%	5.25e-02	100.0%

Table 15.2: Power breakdown, ALU + multiplier modified design, CTS stage.

Total power (52.5 mW) is well below both the baseline (210.1 mW, Section 4.6) and the ALU-only modified design (137 mW, Section 11.3); as with every power figure in this report, this is an OpenSTA default-activity estimate, not a validated result (Section 2.6).

Chapter 16

Final Modified Design (EX-Split + Multiplier Split): Post-CTS Timing Repair Results

16.1 Full post-repair timing report

```
Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_245_
          (removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_245_/
          RESET_B (sky130_fd_sc_hd__dftrtp_1)
          0.50 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
          0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_245_/CLK
          (sky130_fd_sc_hd__dftrtp_1)
0.30 0.30 library removal time
          0.30 data required time
-----
          0.30 data required time
          -0.50 data arrival time
-----
          0.20 slack (MET)
```

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
(negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
(rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

Delay Time Description

```
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
sky130_fd_sc_hd__dlxtn_1)
0.31 2.81 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
sky130_fd_sc_hd__dlxtn_1)
0.00 2.81 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
sky130_fd_sc_hd__and2_0)
2.81 data arrival time
```

```
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
2.50 data required time
```

```
-----
2.50 data required time
-2.81 data arrival time
-----
```

```
0.31 slack (MET)
```

Startpoint: core_i/sleep_unit_i/_5_
(rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/controller_i/_615_
(rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description

```
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/sleep_unit_i/_5_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.36 0.36 ^ core_i/sleep_unit_i/_5_/Q (sky130_fd_sc_hd__dftrtp_1)
0.06 0.41 v core_i/id_stage_i/controller_i/_465_/X (
sky130_fd_sc_hd__lpflow_isobufsrc_1)
0.00 0.41 v core_i/id_stage_i/controller_i/_615_/D (sky130_fd_sc_hd__dfstp_1)
0.41 data arrival time
```

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/id_stage_i/controller_i/_615_/CLK (
        sky130_fd_sc_hd__dfstp_1)
-0.02 -0.02 library hold time
    -0.02 data required time
-----
    -0.02 data required time
    -0.41 data arrival time
-----
    0.43 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/register_file_i/_12551_
    (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.74 1.24 ^ core_i/wire299/X (sky130_fd_sc_hd__buf_16)
0.86 2.09 ^ core_i/wire297/X (sky130_fd_sc_hd__buf_16)
0.87 2.96 ^ core_i/wire296/X (sky130_fd_sc_hd__buf_16)
0.81 3.77 ^ core_i/wire295/X (sky130_fd_sc_hd__buf_16)
0.96 4.73 ^ core_i/id_stage_i/register_file_i/wire294/X (
    sky130_fd_sc_hd__buf_16)
1.03 5.76 ^ core_i/id_stage_i/register_file_i/wire292/X (
    sky130_fd_sc_hd__buf_16)
0.97 6.73 ^ core_i/id_stage_i/register_file_i/wire291/X (
    sky130_fd_sc_hd__buf_16)
0.07 6.80 ^ core_i/id_stage_i/register_file_i/_12551_/RESET_B (
    sky130_fd_sc_hd__dfrrtp_1)
    6.80 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/register_file_i/_12551_/CLK (
        sky130_fd_sc_hd__dfrrtp_1)
-0.08 4.92 library recovery time
    4.92 data required time
-----
    4.92 data required time
    -6.80 data arrival time
-----

```

```

-1.88 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
1.91 4.41 time given to startpoint
0.00 4.41 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
    sky130_fd_sc_hd__dlxtn_1)
0.30 4.71 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
    sky130_fd_sc_hd__dlxtn_1)
0.00 4.71 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
    sky130_fd_sc_hd__and2_0)
    4.71 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
    sky130_fd_sc_hd__and2_0)
0.00 5.00 clock gating setup time
    5.00 data required time
-----
    5.00 data required time
    -4.71 data arrival time
-----

0.29 slack (MET)

Startpoint: core_i/id_stage_i/_4388_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4759_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4388_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.38 0.38 v core_i/id_stage_i/_4388_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.14 0.52 v core_i/ex_stage_i/mult_i/place14410/X (sky130_fd_sc_hd__buf_6)

```

```

0.12 0.64 v core_i/ex_stage_i/mult_i/place14409/X (sky130_fd_sc_hd__buf_6)
0.10 0.74 ^ core_i/ex_stage_i/mult_i/_16544_/Y (sky130_fd_sc_hd__nand3b_1)
0.12 0.86 ^ core_i/ex_stage_i/mult_i/place12568/X (sky130_fd_sc_hd__buf_6)
0.06 0.92 v core_i/ex_stage_i/mult_i/_18898_/Y (sky130_fd_sc_hd__nand3_4)
0.11 1.03 v core_i/ex_stage_i/mult_i/place12003/X (sky130_fd_sc_hd__buf_6)
0.32 1.35 v core_i/ex_stage_i/mult_i/_18987_/X (sky130_fd_sc_hd__maj3_2)
0.14 1.49 v core_i/ex_stage_i/mult_i/_18989_/Y (sky130_fd_sc_hd__xnor2_2)
0.24 1.73 ^ core_i/ex_stage_i/mult_i/_18990_/Y (sky130_fd_sc_hd__xnor2_1)
0.51 2.24 v core_i/ex_stage_i/mult_i/_18991_/X (sky130_fd_sc_hd__xnor3_2)
0.18 2.41 v core_i/ex_stage_i/mult_i/_19066_/X (sky130_fd_sc_hd__xor2_2)
0.38 2.80 v core_i/ex_stage_i/mult_i/_19067_/X (sky130_fd_sc_hd__xor3_1)
0.33 3.13 v core_i/ex_stage_i/mult_i/_19068_/X (sky130_fd_sc_hd__maj3_2)
0.41 3.53 v core_i/ex_stage_i/mult_i/_19080_/X (sky130_fd_sc_hd__xnor3_1)
0.20 3.73 ^ core_i/ex_stage_i/mult_i/_19138_/Y (sky130_fd_sc_hd__xnor2_2)
0.22 3.95 ^ core_i/ex_stage_i/mult_i/_19141_/X (sky130_fd_sc_hd__xor2_1)
0.23 4.18 ^ core_i/ex_stage_i/mult_i/_19190_/X (sky130_fd_sc_hd__xor2_1)
0.16 4.34 ^ core_i/ex_stage_i/mult_i/_19328_/Y (sky130_fd_sc_hd__xnor2_1)
0.19 4.53 ^ core_i/ex_stage_i/mult_i/_19329_/X (sky130_fd_sc_hd__o21a_1)
0.21 4.73 ^ core_i/ex_stage_i/mult_i/_19331_/X (sky130_fd_sc_hd__maj3_2)
0.20 4.94 ^ core_i/ex_stage_i/mult_i/_19334_/X (sky130_fd_sc_hd__maj3_2)
0.06 4.99 v core_i/ex_stage_i/mult_i/_19336_/Y (sky130_fd_sc_hd__o21ai_2)
0.19 5.18 v core_i/ex_stage_i/mult_i/_19338_/X (sky130_fd_sc_hd__a21o_1)
0.22 5.40 v core_i/ex_stage_i/mult_i/_19341_/X (sky130_fd_sc_hd__a21o_1)
0.18 5.58 ^ core_i/ex_stage_i/mult_i/_19343_/Y (sky130_fd_sc_hd__a21oi_2)
0.08 5.66 v core_i/ex_stage_i/mult_i/_19346_/Y (sky130_fd_sc_hd__o21ai_2)
0.30 5.96 v core_i/ex_stage_i/mult_i/_19348_/X (sky130_fd_sc_hd__maj3_2)
0.07 6.03 ^ core_i/ex_stage_i/mult_i/_19350_/Y (sky130_fd_sc_hd__nand2b_1)
0.07 6.10 v core_i/ex_stage_i/mult_i/_19351_/Y (sky130_fd_sc_hd__nand2_1)
0.15 6.25 ^ core_i/ex_stage_i/mult_i/_19353_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 6.35 v core_i/ex_stage_i/mult_i/_19355_/Y (sky130_fd_sc_hd__o21bai_1)
0.33 6.68 v core_i/ex_stage_i/mult_i/_19357_/X (sky130_fd_sc_hd__maj3_2)
0.32 6.99 v core_i/ex_stage_i/mult_i/_19359_/X (sky130_fd_sc_hd__maj3_2)
0.19 7.19 ^ core_i/ex_stage_i/mult_i/_19361_/Y (sky130_fd_sc_hd__o211ai_1)
0.10 7.29 v core_i/ex_stage_i/mult_i/_19363_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 7.51 ^ core_i/ex_stage_i/mult_i/_19366_/Y (sky130_fd_sc_hd__o21ai_1)
0.10 7.61 v core_i/ex_stage_i/mult_i/_19368_/Y (sky130_fd_sc_hd__a21oi_2)
0.18 7.79 ^ core_i/ex_stage_i/mult_i/_19370_/Y (sky130_fd_sc_hd__o21ai_1)
0.10 7.89 v core_i/ex_stage_i/mult_i/_19470_/Y (sky130_fd_sc_hd__a21oi_1)
0.17 8.05 ^ core_i/ex_stage_i/mult_i/_19473_/Y (sky130_fd_sc_hd__nor2_1)
0.14 8.19 ^ core_i/ex_stage_i/mult_i/place7627/X (sky130_fd_sc_hd__buf_4)
0.10 8.30 ^ core_i/ex_stage_i/mult_i/_19565_/Y (sky130_fd_sc_hd__xnor2_2)
0.06 8.35 v core_i/ex_stage_i/mult_i/_22026_/Y (sky130_fd_sc_hd__nor2_1)
0.16 8.51 ^ core_i/ex_stage_i/mult_i/_22027_/Y (sky130_fd_sc_hd__a21oi_1)
0.12 8.63 v core_i/ex_stage_i/mult_i/_22028_/Y (sky130_fd_sc_hd__mux2i_1)
0.33 8.96 ^ core_i/ex_stage_i/mult_i/_22366_/Y (sky130_fd_sc_hd__a221oi_2)
0.08 9.04 v core_i/ex_stage_i/mult_i/_22367_/Y (sky130_fd_sc_hd__nor2_1)
0.23 9.27 ^ core_i/ex_stage_i/mult_i/_22372_/Y (sky130_fd_sc_hd__a211oi_1)
0.14 9.41 v core_i/ex_stage_i/mult_i/_22373_/Y (sky130_fd_sc_hd__o21ai_1)
0.23 9.63 ^ core_i/ex_stage_i/_129_/Y (sky130_fd_sc_hd__a22oi_1)
0.11 9.74 v core_i/ex_stage_i/_131_/Y (sky130_fd_sc_hd__o21ai_2)
0.22 9.97 v core_i/id_stage_i/_3048_/X (sky130_fd_sc_hd__a22o_1)

```

```

0.14 10.10 ^ core_i/id_stage_i/_3049_/Y (sky130_fd_sc_hd__a21oi_2)
0.10 10.20 v core_i/id_stage_i/_3274_/Y (sky130_fd_sc_hd__o22ai_2)
0.18 10.38 ^ core_i/id_stage_i/_3275_/Y (sky130_fd_sc_hd__a21oi_2)
0.09 10.47 v core_i/id_stage_i/_3280_/Y (sky130_fd_sc_hd__a21oi_2)
0.27 10.74 v core_i/id_stage_i/_3281_/X (sky130_fd_sc_hd__lpflow_inputiso1p_1
)
0.26 11.01 v core_i/id_stage_i/_4363_/X (sky130_fd_sc_hd__mux2_4)
0.00 11.01 v core_i/id_stage_i/_4759_/D (sky130_fd_sc_hd__dfrrtp_1)
    11.01 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4759_/CLK (sky130_fd_sc_hd__dfrrtp_1)
-0.13 4.87 library setup time
    4.87 data required time
-----
    4.87 data required time
   -11.01 data arrival time
-----
   -6.13 slack (VIOLATED)

wns max -6.13
tns max -9269.08
Clock clk_i
    1.13 source latency core_i/cs_registers_i/_4640_/CLK ^
   -1.36 target latency core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N ^
    0.00 CRPR
-----
   -0.23 setup skew

```

Metric	Baseline (Section 4.7–4.8)	ALU-only modified (Section 11.3)
wns max	–8.45 ns	–5.77 ns
tns max	–11,756.31	–7,749.03
Setup skew	0.20 ns	0.13 ns

Metric	ALU + mult modified
wns max	–6.13 ns (27% better than baseline)
tns max	–9,269.08 (21% better than baseline)
Setup skew	–0.23 ns (sign flipped)

Table 16.1: Post-CTS-repair timing: baseline vs. ALU-only modified vs. ALU + multiplier modified.

The critical path moves into mult_i itself. Unlike the CTS-stage report (Chapter 15, register-file/PC-generation path), the worst path after repair now runs through a long maj3_2/xnor/xor-chain inside ex_stage_i/mult_i (cells _18987_ through _22373_), structurally similar in kind to the mult_i-internal path already identified as the ALU-only modified design’s own post-repair critical path (Section 11.3, also inside

`mult_i`).

What can be said from the numbers alone: both **wns** (-6.13 ns vs. -5.77 ns) and **tns** ($-9,269.08$ vs. $-7,749.03$, a $\sim 19.6\%$ increase) are worse than the ALU-only modified design at this stage. Across all stages so far: floorplan ($+8.9\%$ better tns), placement (~ 11.5 – 12.8% worse), CTS ($\sim 2.4\%$ worse), and post-repair ($\sim 19.6\%$ worse). The setup skew also flipped sign (-0.23 ns vs. $+0.13$ ns for the ALU-only modified design). The final design remains meaningfully better than the un-split baseline at this stage (27% wns, 21% tns).

16.2 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.53e-02	1.08e-03	3.49e-08	2.64e-02	51.4%
Combinational	4.39e-03	5.63e-03	9.86e-08	1.00e-02	19.5%
Clock	9.56e-03	5.39e-03	3.55e-09	1.49e-02	29.1%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.93e-02 76.4%	1.21e-02 23.6%	1.37e-07 0.0%	5.14e-02	100.0%

Table 16.2: Power breakdown, ALU + multiplier modified design, post-CTS-repair stage.

Total power (51.4 mW) remains well below both the baseline (Section 4.7 area) and the ALU-only modified design; as with every power figure in this report, this is an OpenSTA default-activity estimate, not a validated result (Section 2.6).

Chapter 17

Final Modified Design (EX-Split + Multiplier Split): Post-Global-Route Results

17.1 Note on scope and methodology

As with the ALU-only modified design (Chapter 12), `detailed_route` did not converge for this design either, so `global_route` results (with `estimate_parasitics -global_routing`) are reported here, following the same post-global-route, pre-detailed-route methodology described in Chapter 12's opening section.

17.2 Timing

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_
(removal check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: min
Delay Time Description

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.00 0.50 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_/ RESET_B (sky130_fd_sc_hd__dftrtp_1)
0.50 data arrival time
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
0.00 ^ core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/_261_/CLK (sky130_fd_sc_hd__dftrtp_1)

```

0.30 0.30 library removal time
0.30 data required time
-----
0.30 data required time
-0.50 data arrival time
-----
0.20 slack (MET)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint: core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: min

Delay Time Description
-----
2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 v core_i/sleep_unit_i/core_clock_gate_i/_3_/GATE_N (
sky130_fd_sc_hd__dlxtn_1)
0.28 2.78 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
sky130_fd_sc_hd__dlxtn_1)
0.00 2.78 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
sky130_fd_sc_hd__and2_0)
2.78 data arrival time

2.50 2.50 clock clk_i (fall edge)
0.00 2.50 clock network delay (ideal)
0.00 2.50 clock reconvergence pessimism
2.50 v core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
sky130_fd_sc_hd__and2_0)
0.00 2.50 clock gating hold time
2.50 data required time
-----
2.50 data required time
-2.78 data arrival time
-----
0.28 slack (MET)

Startpoint: core_i/id_stage_i/_4443_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint: core_i/id_stage_i/_4443_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: min

Delay Time Description
-----

```

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4443_/CLK (sky130_fd_sc_hd__dfirtp_1)
0.29 0.29 ^ core_i/id_stage_i/_4443_/Q (sky130_fd_sc_hd__dfirtp_1)
0.11 0.40 ^ core_i/id_stage_i/_3245_/X (sky130_fd_sc_hd__mux2_1)
0.00 0.40 ^ core_i/id_stage_i/_4443_/D (sky130_fd_sc_hd__dfirtp_1)
    0.40 data arrival time

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism
    0.00 ^ core_i/id_stage_i/_4443_/CLK (sky130_fd_sc_hd__dfirtp_1)
-0.04 -0.04 library hold time
    -0.04 data required time
-----
    -0.04 data required time
    -0.40 data arrival time
-----
    0.44 slack (MET)

Startpoint: rst_ni (input port clocked by clk_i)
Endpoint: core_i/id_stage_i/register_file_i/_12560_
    (recovery check against rising-edge clock clk_i)
Path Group: asynchronous
Path Type: max

Delay Time Description
-----
0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.50 0.50 ^ input external delay
0.00 0.50 ^ rst_ni (in)
0.64 1.14 ^ core_i/wire299/X (sky130_fd_sc_hd__buf_16)
0.88 2.01 ^ core_i/wire297/X (sky130_fd_sc_hd__buf_16)
0.88 2.89 ^ core_i/wire296/X (sky130_fd_sc_hd__buf_16)
0.87 3.76 ^ core_i/wire295/X (sky130_fd_sc_hd__buf_16)
1.11 4.87 ^ core_i/id_stage_i/register_file_i/wire294/X (
    sky130_fd_sc_hd__buf_16)
1.03 5.90 ^ core_i/id_stage_i/register_file_i/wire292/X (
    sky130_fd_sc_hd__buf_16)
1.04 6.95 ^ core_i/id_stage_i/register_file_i/wire291/X (
    sky130_fd_sc_hd__buf_16)
0.14 7.09 ^ core_i/id_stage_i/register_file_i/_12560_/RESET_B (
    sky130_fd_sc_hd__dfirtp_1)
    7.09 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism

```

```

    5.00 ^ core_i/id_stage_i/register_file_i/_12560_/CLK (
        sky130_fd_sc_hd__dfrtp_1)
-0.11 4.89 library recovery time
    4.89 data required time
-----
    4.89 data required time
   -7.09 data arrival time
-----
   -2.19 slack (VIOLATED)

Startpoint: core_i/sleep_unit_i/core_clock_gate_i/_3_
            (negative level-sensitive latch clocked by clk_i)
Endpoint:  core_i/sleep_unit_i/core_clock_gate_i/_2_
            (rising clock gating-check end-point clocked by clk_i)
Path Group: gated clock
Path Type: max

Delay Time Description
-----
    2.50 2.50 clock clk_i (fall edge)
    0.00 2.50 clock network delay (ideal)
    2.36 4.86 time given to startpoint
    0.00 4.86 v core_i/sleep_unit_i/core_clock_gate_i/_3_/D (
        sky130_fd_sc_hd__dlxtn_1)
    0.27 5.13 v core_i/sleep_unit_i/core_clock_gate_i/_3_/Q (
        sky130_fd_sc_hd__dlxtn_1)
    0.00 5.13 v core_i/sleep_unit_i/core_clock_gate_i/_2_/B (
        sky130_fd_sc_hd__and2_0)
        5.13 data arrival time

    5.00 5.00 clock clk_i (rise edge)
    0.00 5.00 clock network delay (ideal)
    0.00 5.00 clock reconvergence pessimism
        5.00 ^ core_i/sleep_unit_i/core_clock_gate_i/_2_/A (
            sky130_fd_sc_hd__and2_0)
    0.00 5.00 clock gating setup time
        5.00 data required time
-----
    5.00 data required time
   -5.13 data arrival time
-----
   -0.13 slack (VIOLATED)

Startpoint: core_i/id_stage_i/_4450_
            (rising edge-triggered flip-flop clocked by clk_i)
Endpoint:  core_i/id_stage_i/_4482_
            (rising edge-triggered flip-flop clocked by clk_i)
Path Group: clk_i
Path Type: max

```

Delay Time Description

```

0.00 0.00 clock clk_i (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core_i/id_stage_i/_4450_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.40 0.40 ^ core_i/id_stage_i/_4450_/Q (sky130_fd_sc_hd__dfrrtp_1)
0.26 0.66 ^ core_i/ex_stage_i/mult_i/_16076_/X (sky130_fd_sc_hd__and2_4)
0.45 1.11 ^ core_i/ex_stage_i/mult_i/_16218_/Y (sky130_fd_sc_hd__mux2i_2)
0.21 1.32 ^ core_i/ex_stage_i/mult_i/place12051/X (sky130_fd_sc_hd__buf_4)
0.21 1.54 ^ core_i/ex_stage_i/mult_i/place12050/X (sky130_fd_sc_hd__buf_4)
0.15 1.68 v core_i/ex_stage_i/mult_i/_18581_/Y (sky130_fd_sc_hd__o22ai_1)
0.33 2.01 ^ core_i/ex_stage_i/mult_i/_18583_/Y (sky130_fd_sc_hd__a21boi_0)
0.26 2.27 ^ core_i/ex_stage_i/mult_i/_18588_/X (sky130_fd_sc_hd__xor2_1)
0.19 2.46 ^ core_i/ex_stage_i/mult_i/_18695_/Y (sky130_fd_sc_hd__xnor2_1)
0.21 2.67 ^ core_i/ex_stage_i/mult_i/_18696_/X (sky130_fd_sc_hd__xor2_1)
0.20 2.86 ^ core_i/ex_stage_i/mult_i/_18800_/Y (sky130_fd_sc_hd__xnor2_1)
0.31 3.18 v core_i/ex_stage_i/mult_i/_18801_/X (sky130_fd_sc_hd__xor3_1)
0.15 3.32 v core_i/ex_stage_i/mult_i/place9367/X (sky130_fd_sc_hd__buf_4)
0.38 3.70 v core_i/ex_stage_i/mult_i/_18803_/X (sky130_fd_sc_hd__maj3_1)
0.25 3.95 ^ core_i/ex_stage_i/mult_i/_18841_/Y (sky130_fd_sc_hd__xnor2_1)
0.20 4.16 ^ core_i/ex_stage_i/mult_i/_18928_/Y (sky130_fd_sc_hd__xnor2_1)
0.20 4.35 ^ core_i/ex_stage_i/mult_i/_18937_/Y (sky130_fd_sc_hd__xnor2_1)
0.20 4.55 ^ core_i/ex_stage_i/mult_i/_19014_/Y (sky130_fd_sc_hd__xnor2_1)
0.25 4.80 ^ core_i/ex_stage_i/mult_i/_19016_/X (sky130_fd_sc_hd__xor2_1)
0.09 4.89 v core_i/ex_stage_i/mult_i/_19017_/Y (sky130_fd_sc_hd__nand2_1)
0.16 5.05 ^ core_i/ex_stage_i/mult_i/_19018_/Y (sky130_fd_sc_hd__o21ai_0)
0.25 5.30 ^ core_i/ex_stage_i/mult_i/_19335_/X (sky130_fd_sc_hd__xor2_1)
0.10 5.40 v core_i/ex_stage_i/mult_i/_19336_/Y (sky130_fd_sc_hd__o21ai_2)
0.20 5.60 v core_i/ex_stage_i/mult_i/_19338_/X (sky130_fd_sc_hd__a21o_1)
0.22 5.82 v core_i/ex_stage_i/mult_i/_19341_/X (sky130_fd_sc_hd__a21o_1)
0.21 6.03 ^ core_i/ex_stage_i/mult_i/_19343_/Y (sky130_fd_sc_hd__a21oi_2)
0.08 6.11 v core_i/ex_stage_i/mult_i/_19346_/Y (sky130_fd_sc_hd__o21ai_2)
0.31 6.42 v core_i/ex_stage_i/mult_i/_19348_/X (sky130_fd_sc_hd__maj3_2)
0.07 6.49 ^ core_i/ex_stage_i/mult_i/_19350_/Y (sky130_fd_sc_hd__nand2b_1)
0.08 6.57 v core_i/ex_stage_i/mult_i/_19351_/Y (sky130_fd_sc_hd__nand2_1)
0.18 6.75 ^ core_i/ex_stage_i/mult_i/_19353_/Y (sky130_fd_sc_hd__a21oi_2)
0.12 6.87 v core_i/ex_stage_i/mult_i/_19355_/Y (sky130_fd_sc_hd__o21bai_1)
0.35 7.21 v core_i/ex_stage_i/mult_i/_19357_/X (sky130_fd_sc_hd__maj3_2)
0.33 7.54 v core_i/ex_stage_i/mult_i/_19359_/X (sky130_fd_sc_hd__maj3_2)
0.24 7.78 ^ core_i/ex_stage_i/mult_i/_19361_/Y (sky130_fd_sc_hd__o211ai_1)
0.13 7.91 v core_i/ex_stage_i/mult_i/_19363_/Y (sky130_fd_sc_hd__a21oi_1)
0.22 8.14 ^ core_i/ex_stage_i/mult_i/_19366_/Y (sky130_fd_sc_hd__o21ai_1)
0.09 8.22 v core_i/ex_stage_i/mult_i/_19368_/Y (sky130_fd_sc_hd__a21oi_2)
0.19 8.41 ^ core_i/ex_stage_i/mult_i/_19370_/Y (sky130_fd_sc_hd__o21ai_1)
0.12 8.53 v core_i/ex_stage_i/mult_i/_19470_/Y (sky130_fd_sc_hd__a21oi_1)
0.16 8.68 ^ core_i/ex_stage_i/mult_i/_19473_/Y (sky130_fd_sc_hd__nor2_1)
0.20 8.88 ^ core_i/ex_stage_i/mult_i/_19568_/X (sky130_fd_sc_hd__maj3_1)
0.22 9.10 v core_i/ex_stage_i/mult_i/_19632_/Y (sky130_fd_sc_hd__xnor2_1)
0.23 9.33 v core_i/ex_stage_i/mult_i/place7610/X (sky130_fd_sc_hd__buf_4)
0.07 9.40 ^ core_i/ex_stage_i/mult_i/_19633_/Y (sky130_fd_sc_hd__nand2_1)

```

```

0.16 9.56 v core_i/ex_stage_i/mult_i/_22326_/Y (sky130_fd_sc_hd__o21ai_0)
0.44 10.00 ^ core_i/ex_stage_i/mult_i/_22328_/Y (sky130_fd_sc_hd__a221oi_2)
0.13 10.13 v core_i/ex_stage_i/mult_i/_22334_/Y (sky130_fd_sc_hd__o21ai_0)
0.33 10.46 v core_i/ex_stage_i/mult_i/_22335_/X (sky130_fd_sc_hd__a311o_1)
0.22 10.69 ^ core_i/ex_stage_i/_123_/Y (sky130_fd_sc_hd__a22oi_1)
0.14 10.83 v core_i/ex_stage_i/_125_/Y (sky130_fd_sc_hd__o21ai_2)
0.24 11.06 v core_i/id_stage_i/_2973_/X (sky130_fd_sc_hd__a22o_1)
0.17 11.24 ^ core_i/id_stage_i/_2974_/Y (sky130_fd_sc_hd__a21oi_2)
0.13 11.37 v core_i/id_stage_i/_2975_/Y (sky130_fd_sc_hd__o22ai_1)
0.22 11.59 ^ core_i/id_stage_i/_2976_/Y (sky130_fd_sc_hd__a211oi_2)
0.09 11.68 v core_i/id_stage_i/_2998_/Y (sky130_fd_sc_hd__nand2_1)
0.10 11.79 ^ core_i/id_stage_i/_2999_/Y (sky130_fd_sc_hd__nand2_1)
0.05 11.84 v core_i/id_stage_i/_3000_/Y (sky130_fd_sc_hd__a21oi_1)
0.38 12.22 ^ core_i/id_stage_i/_3779_/Y (sky130_fd_sc_hd__nor2_1)
0.17 12.39 v core_i/id_stage_i/_3784_/Y (sky130_fd_sc_hd__o21ai_0)
0.00 12.39 v core_i/id_stage_i/_4482_/D (sky130_fd_sc_hd__dftrtp_1)
    12.39 data arrival time

5.00 5.00 clock clk_i (rise edge)
0.00 5.00 clock network delay (ideal)
0.00 5.00 clock reconvergence pessimism
    5.00 ^ core_i/id_stage_i/_4482_/CLK (sky130_fd_sc_hd__dftrtp_1)
-0.15 4.85 library setup time
    4.85 data required time

-----
    4.85 data required time
   -12.39 data arrival time

-----

   -7.54 slack (VIOLATED)

wns max -7.54
tns max -11393.22
Clock clk_i
    1.79 source latency core_i/id_stage_i/register_file_i/_13410_/CLK ^
   -1.46 target latency core_i/if_stage_i/prefetch_buffer_i/instruction_obi_i/
        _251_/CLK ^
    0.00 CRPR
-----
    0.33 setup skew

```

Metric	Baseline	ALU-only modified
wns max	-10.03 ns	-7.17 ns
tns max	-13,541.69	-9,560.16
Setup skew	0.28 ns	-0.25 ns

Metric	ALU + mult modified
wns max	-7.54 ns (24.8% better than baseline)
tns max	-11,393.22 (15.9% better than baseline)
Setup skew	0.33 ns

Table 17.1: Post-global-route timing: baseline vs. ALU-only modified vs. ALU + multiplier modified.

The critical path runs through `ex_stage_i/mult_i` (startpoint `core_i/id_stage_i/_4450_`, endpoint `core_i/id_stage_i/_4482_`, data arrival 12.39 ns), the same module already identified as the worst-path location for both the ALU-only modified design (Chapter 12) and the post-CTS-repair stage of this design (Chapter 16).

17.3 Power

Group	Internal	Switching	Leakage	Total (W)	%
Sequential	2.53e-02	1.27e-03	3.49e-08	2.66e-02	47.5%
Combinational	4.41e-03	6.74e-03	9.86e-08	1.11e-02	19.9%
Clock	9.68e-03	8.60e-03	3.55e-09	1.83e-02	32.6%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.94e-02	1.66e-02	1.37e-07	5.60e-02	100.0%
	70.4%	29.6%	0.0%		

Table 17.2: Power breakdown, ALU + multiplier modified design, post-global-route stage.

Total power (56.0 mW) is below both the baseline (221 mW) and the ALU-only modified design (151 mW) at this stage; as with every power figure in this report, this is an OpenSTA default-activity estimate, not a validated result (Section 2.6).

17.4 Area

Design area 332533 μm^2 50% utilization.

Metric	Baseline	ALU-only modified	ALU + mult modified
Design area	314,907 μm^2	329,091 μm^2	332,533 μm^2
Utilization	51%	50%	50%
Change vs. baseline	—	+4.5%	+5.6%
Change vs. ALU-only	—	—	+1.05%

Table 17.3: Post-global-route area: baseline vs. ALU-only modified vs. ALU + multiplier modified.

17.5 Summary: post-global-route, all three designs

Metric	Baseline	ALU-only modified	ALU + mult modified
wns max	−10.03 ns	−7.17 ns	−7.54 ns
tns max	−13,541.69	−9,560.16	−11,393.22
Setup skew	0.28 ns	−0.25 ns	0.33 ns
Total power	221 mW	151 mW	56.0 mW
Design area	314,907 μm^2	329,091 μm^2	332,533 μm^2

Table 17.4: Complete post-global-route comparison, all three designs, same methodology applied to each.

Relative to the baseline, the final (ALU + mult) modified design improves wns by 24.8% and tns by 15.9%, at a design-area cost of +5.6% and roughly 4× lower reported power

(56.0 mW vs. 221 mW, power figures unconfirmed per Section 2.6). Relative to the ALU-only modified design specifically, the mult split’s incremental effect at this final stage is: wns 5.2% worse ($-7.17\text{ ns} \rightarrow -7.54\text{ ns}$), tns 19.2% worse ($-9,560.16 \rightarrow -11,393.22$), area +1.05% larger, and power roughly $2.7\times$ lower ($151\text{ mW} \rightarrow 56.0\text{ mW}$).

Why the multiplier split did not improve timing further. The critical path at both the post-CTS-repair stage and this final stage (Chapters 16 and 17) runs through a long `maj3/xnor/xor` chain inside `mult_i`, consistent with the multiplier’s partial-product/carry-save arithmetic tree — the core combinational depth of the multiplication itself. The RTL change applied (Section 6.3) split only the `MUL_MAC32/MUL_MSU32` datapath by inserting an EX1/EX2 pipeline register *after* this arithmetic tree, together with the control logic needed to select between the split and non-split (`MUL_H`) result paths. This register and its surrounding mux/control logic sit downstream of, or alongside, the arithmetic tree rather than inside it, so the tree’s own combinational depth — the actual worst-path bottleneck at this stage — was not shortened by the change. The additional cells introduced by the split (visible in the area increase, Section 17.4) instead added a small amount of extra logic on or near the same critical path and increased the total netlist size, which is consistent with wns and tns both moving slightly against the ALU-only modified design rather than improving further. In short: the split only reduces critical-path depth when the critical path runs through the specific logic that was cut in two, and by this stage of the flow it no longer did.

Chapter 18

Appendix: OpenROAD GUI Visualizations

The following screenshots, captured directly from the OpenROAD GUI, illustrate what the design physically looks like at three stages of the flow: floorplan, placement, and global routing.

18.1 Floorplan

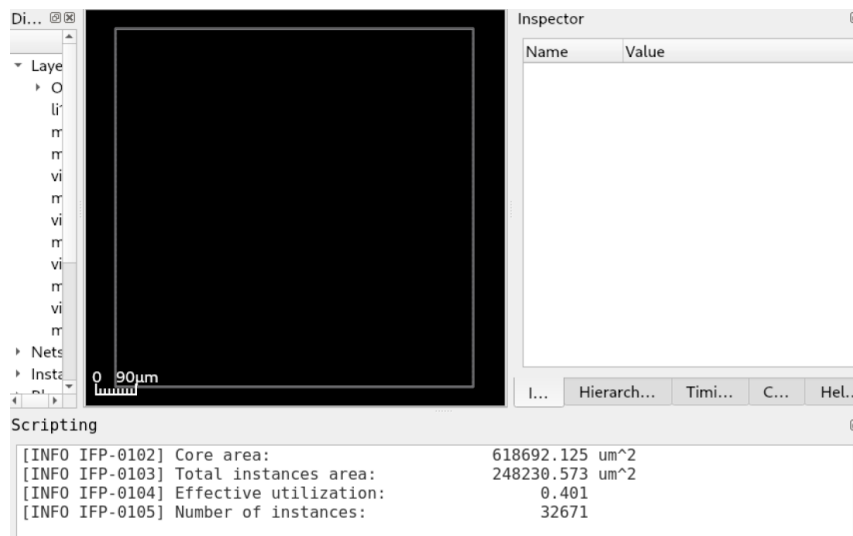


Figure 18.1: OpenROAD GUI immediately after `initialize_floorplan`, before PDN generation (core still empty). The Scripting panel confirms the baseline synthesis figures used throughout this report: core area $618,692.125 \mu\text{m}^2$, total instance area $248,230.573 \mu\text{m}^2$ (40.1% effective utilization), 32,671 instances.

18.2 Placement

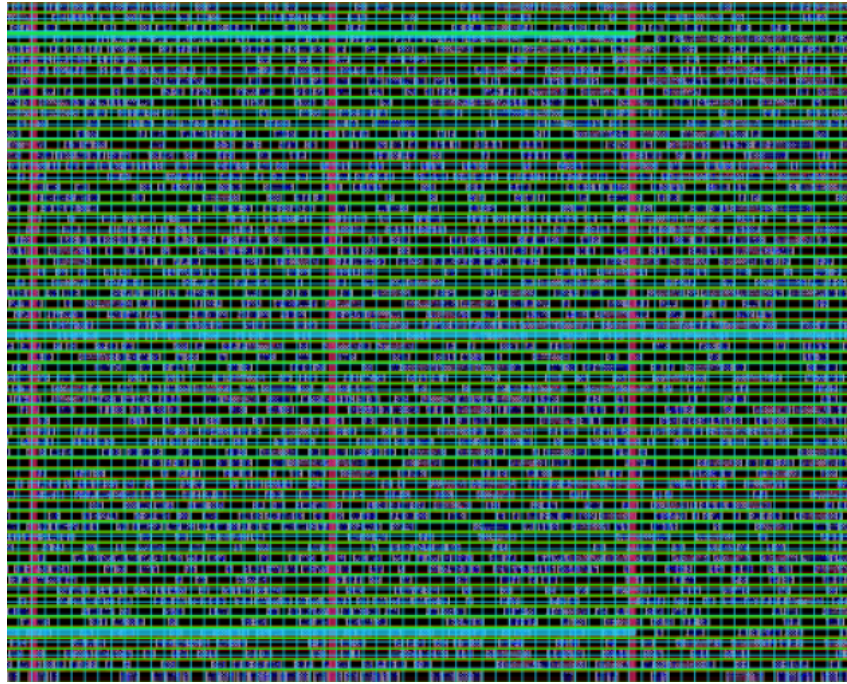


Figure 18.2: OpenROAD GUI view of the placed design. The dense grid of green/blue/red tracks shows the routing fabric (metal layers) with placed standard cells visible underneath; the bright green horizontal bands are wide power/ground straps from the PDN.

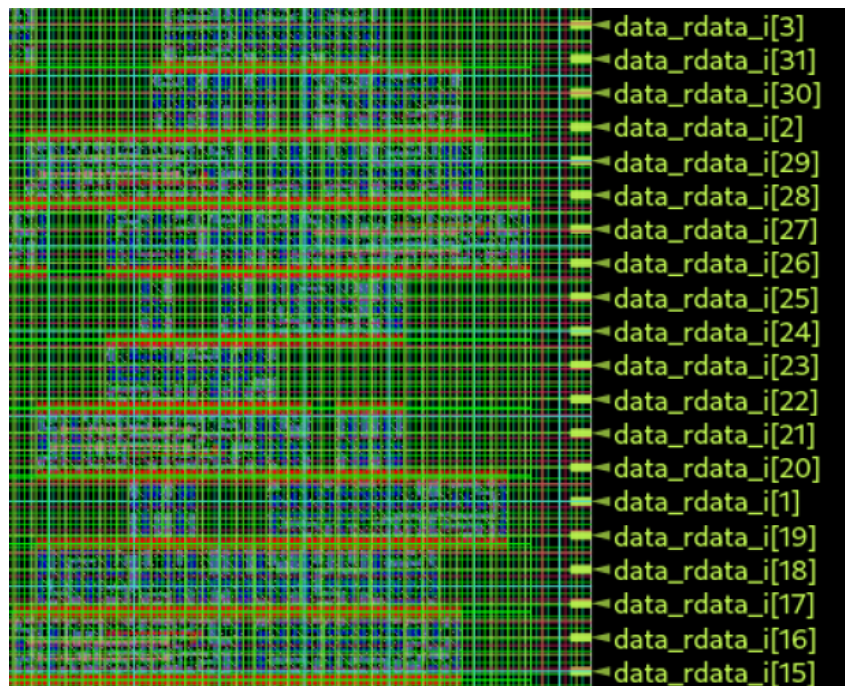


Figure 18.3: Zoomed-in placement view with individual net labels visible on the right-hand side (`data_rdata_i[*]`, the load-store unit's read-data bus). This level of zoom is useful for inspecting how a specific bus is placed and routed relative to its consuming logic.

18.3 Global routing

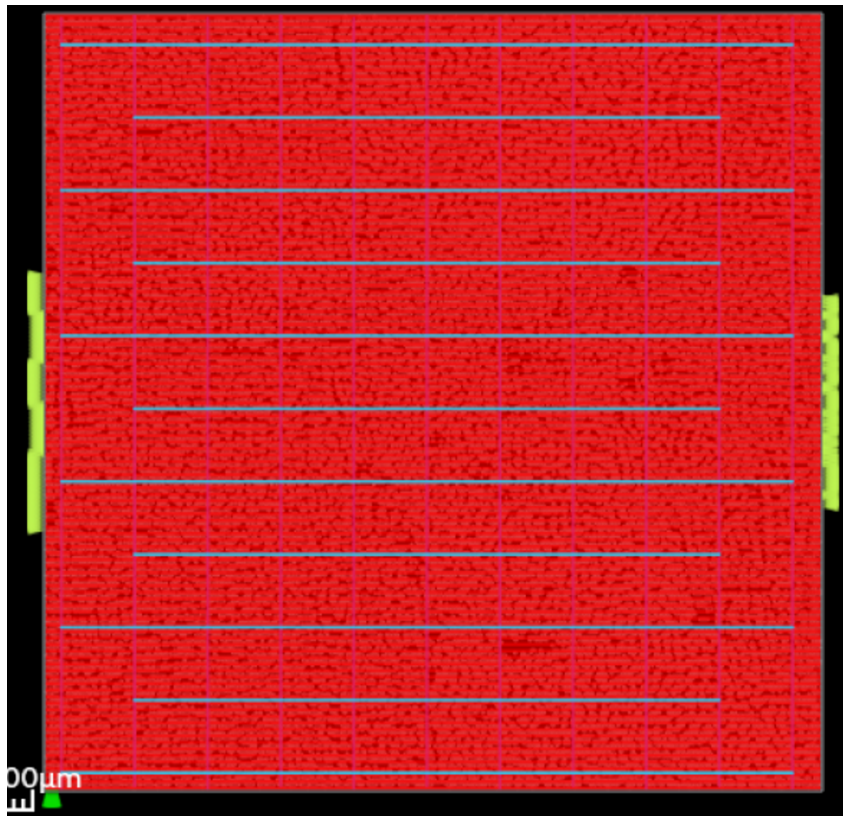


Figure 18.4: OpenROAD GUI view during the global routing stage. The dense red fill shows routing/cell density across the core, with horizontal blue lines marking PDN power straps and green marks on the left/right edges showing I/O pin locations.

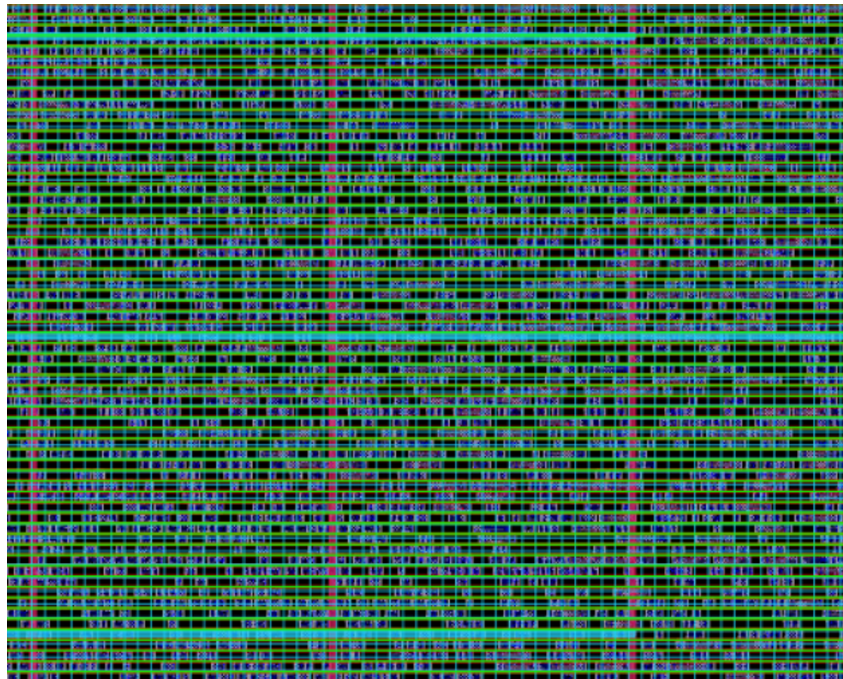


Figure 18.5: Global routing detail view, showing the resulting metal-layer track usage (cyan/green/blue horizontal and vertical segments) across a section of the die. This is the route-guide-level output of `global_route` — `detailed_route` did not converge for this design (Section 2.1), so no signoff-quality routed geometry is available.

References

External sources consulted to identify and resolve specific tooling issues encountered during this project:

1. sv2v (SystemVerilog-to-Verilog converter), used in Section 1.6:
<https://github.com/zachjs/sv2v>
2. CV32E40P RTL source (OpenHW Group), Section 1.1:
<https://github.com/openhwgroup/cv32e40p>
3. OpenROAD-flow-scripts, Section 1.2:
<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts>
4. micro2022tutorial, Section 1.3:
<https://github.com/The-OpenROAD-Project/micro2022tutorial>
5. OpenROAD-flow-scripts Flow Variables documentation (LATCH_MAP_FILE and other config.mk variables), Section 2.4:
<https://openroad-flow-scripts.readthedocs.io/en/latest/user/FlowVariables.html>
6. OpenROAD-flow-scripts sky130hd platform config.mk (IO_PLACER_H/IO_PLACER_V defaults), Section 3.1:
<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts/blob/master/flow/platforms/sky130hd/config.mk>
7. OpenROAD Initialize Floorplan documentation (initialize_floorplan, make_tracks), Sections 3.1, 3.6:
<https://openroad.readthedocs.io/en/latest/main/src/ifp/README.html>
8. OpenROAD-flow-scripts Platform Bring-Up guide (manual metal-track configuration reference), Section 3.6:
<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts/blob/master/docs/contrib/PlatformBringUp.md>
9. OpenROAD Pin Placer documentation (place_pins syntax), Section 3.7:
<https://openroad.readthedocs.io/en/latest/main/src/ppl/README.html>
10. OpenROAD Global Placement documentation (-timing_driven, -incremental), Sections 3.7, 5.1:
<https://openroad.readthedocs.io/en/latest/main/src/gpl/README.html>
11. OpenROAD Clock Tree Synthesis documentation (clock_tree_synthesis options), Chapter 4:
<https://openroad.readthedocs.io/en/latest/main/src/cts/README.html>

12. OpenROAD Global Routing documentation (`set_routing_layers`), Section 5.2:
<https://openroad.readthedocs.io/en/latest/main/src/grt/README.html>
13. OpenLane issue #1185 — “Add `insert_tiecells` features to handle special nets” (root cause and fix for the `zero_/GROUND` routing error), Section 5.1:
<https://github.com/The-OpenROAD-Project/OpenLane/issues/1185>
14. OpenROAD-flow-scripts Tutorial documentation (tie-cell insertion example and explanation), Section 5.1:
<https://openroad-flow-scripts.readthedocs.io/en/latest/tutorials/FlowTutorial.html>